

Disasters without diversification: Long-term labor responses to floods in Nigeria

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Abstract

Climate change is expected to decrease agricultural productivity across much of the world through both higher temperatures and more frequent extreme weather events. Migration and non-farm labor supply are potential coping and adaptation strategies, yet evidence on how weather shocks affect households' long-term labor allocation, migration, and livelihoods is limited. This paper studies the impacts of the catastrophic 2012 floods in Nigeria using four rounds of nationally-representative panel data on 5,000 households collected from 2010-2019 in a difference-in-differences framework exploiting spatial variation in community-level flooding. We first show that a common satellite imagery-based flood detection database disagrees significantly with survey flood reports: 73 of 495 communities are identified as flooded by satellite only compared to 86 by survey only and just 48 by both. The satellite measure also aligns poorly with administrative and media flood reports, and we argue that it is heavily limited by image resolution and high levels of cloud cover during the relevant period. Consistent with measurement error, we find largely null effects of exposure to this satellite-based flood measure, indicating a need for great care in the measurement and interpretation of "flood exposure" in applied work. In contrast, we find that survey-based flood exposure reduces household crop production value by 37% and consumption per capita by 13% on average, with fairly persistent magnitudes over seven years post-exposure, even after recentering around predicted flood exposure to isolate the exogenous component of the 2012 floods. Households do not change crop production decisions but experience persistently lower physical yields, suggesting lasting direct effects of flood damages. There is no evidence of agricultural exit or change in the supply of labor to different livelihood activities on average, though *ex ante* diversified households increase wage labor supply and are more resilient to the flood shock. Migration rises only after a delay of several years. These patterns suggest that household face constraints in their livelihood responses to climate shocks. Despite other studies showing short-term effects of floods on migration or non-farm labor, these results suggest that floods are more likely to entrench households in low-productivity agriculture, adding to concerns of how climate change may constrain structural transformation.

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1 Introduction

Climate change is predicted to reduce agricultural productivity in many parts of the world due to increases in temperatures and in precipitation variability, which are compounded by damages from increasingly severe and frequent extreme weather events. This has important implications for low- and lower-middle-income countries (LMICs) where a large share of the population depends on small-scale agricultural production for their livelihoods. The scope for on-farm adaptation and mitigation is limited (Hultgren et al., 2025; Kala et al., 2023; Pörtner et al., 2022), and there are also constraints to reallocating labor to non-farm alternatives (Albert et al., 2021; Alobo Loison, 2015; Desmet and Rossi-Hansberg, 2024), potentially increasing the costs of climate change (Cruz and Rossi-Hansberg, 2024). Weather shocks can have particularly severe effects on agriculture but it is unclear whether this encourages or hinders livelihood change or diversification over the long-term. A large literature finds short-term migration or non-farm labor responses to weather shocks in LMICs, but the literature on long-term impacts on livelihood outcomes is smaller and has shown mixed results. This paper analyzes the long-term livelihood impacts of exposure to flood shocks by analyzing the effects of a major flooding event in Nigeria on household livelihood activities up to seven years afterward.

Floods are a particularly important source of risk and damages, accounting for 44% of all disasters globally and 31% of all economic losses from 1970-2019 (World Meteorological Association et al., 2021). Over 170 million people in extreme poverty face high flood risk (Rentschler et al., 2022) and flood exposure is predicted to increase in most LMICs (Pörtner et al., 2022), with population growth driving large increases in exposure across much of sub-Saharan Africa (Rogers et al., 2025). Floods can completely destroy agricultural production, change perceptions of future production risk, and push households into new activities as coping strategies. But they may also decrease demand for non-farm goods and services and deplete assets needed to engage in non-farm activities. Consequently, it is unclear whether exposure to floods would result in a persistent reallocation of labor toward non-agricultural sectors. In 2012, excessive rainfall between September-November caused widespread flooding across much of Nigeria. The floods caused an estimated \$16.9 billion in damages and killed 431 people (Unah, 2021). We estimate effects of community-level exposure to these floods using four rounds of the nationally-representative General Household Survey-Panel (GHSP) from 2010-2019, and examine how the choice of how to measure flood incidence affects conclusions about flood impacts.

Existing work on the economic impacts of floods uses a variety of data sources including administrative data, satellite imagery or radar, survey or media reports, or some combination of these sources. Each source has different limitations which can lead to inconsistency in identifying flood incidence (Bangalore et al., 2026; Guiteras et al., 2015; Patel, 2024; Saunders et al., 2025).¹ We consider the implications of using a measure of community flood exposure based on household survey reports versus one based on 250 m daily MODIS satellite imagery from NASA’s Near Real

¹See Kar et al., 2024; Li et al., 2018; Tellman et al., 2021; Yan et al., 2015; Zhang et al., 2023 for examples of papers focusing on the question of flood detection.

Time Global Flood Mapping product (NASA, 2024).² These measures are not well-aligned: 48 communities are considered exposed to flooding in 2012 according to both measures compared to 86 using the survey measure only and 73 using the satellite measure only, with 288 not considered flooded by either.

We find limited evidence that the disagreement is due to survey measurement issues. Correlations with administrative reports of confirmed flooding and with media-based reports in the Dartmouth Flood Observatory archive are stronger with the survey measure than the satellite measure, indicating that the latter is both not capturing some true flooding and falsely detecting other flooding. The 250 m resolution of MODIS satellite imagery struggles to detect local variation in surface water, primarily identifying floods near large rivers where water is a larger share of pixel area. The median MODIS pixel is also cloudy for 71% of days during the period of the 2012 floods, preventing any flood detection. We also find that the satellite measure is less likely to identify floods in rural and agricultural communities that experienced more rainfall in 2012, while the survey measure does identify such cases of likely pluvial flooding. Given the limitations of the satellite measure in this context we focus our analysis on the survey-based measure. We test for sensitivity to different approaches to constructing the survey measure, and compare the main results to what we obtain using the satellite measure.

We estimate household effects of flood exposure in a difference-in-differences design comparing changes before and after the major flood events of 2012 for households that did and did not reside in communities exposed to those floods. To account for potential endogeneity in flood exposure, we balance the sample on recent flood history and underlying fluvial flood hazard, control for time-invariant correlates through household fixed effects, and recenter binary 2012 flood incidence around predicted incidence (as in Borusyak and Hull, 2023). We therefore estimate the effects of 2012 flood exposure from differences in flood incidence across communities in the same state with the same predicted flood incidence and histories of recent flood realizations. We show that recentering around predicted incidence is crucial for isolating effects of the 2012 floods from differential trends across areas with different levels of underlying flood risk – the ‘non-random’ component of flood shock exposure.

Community flood exposure causes large and significant immediate decreases in household consumption per capita, food security, asset value, and value of crop production. While household food security and asset values recover over time, consumption per capita does not, falling by 13% on average in survey rounds after 2012 relative to pre-flood levels. The apparent driver is a persistent decrease in the value of crop production, which falls by 37% on average following flood exposure with similar magnitude effects all the way through the 2018-19 survey round. These are economically large effects, particularly since they represent average effects of community-level flood exposure and less than half of households in communities flooded in 2012 report any direct adverse

²Higher resolution data from Sentinel-1 satellites, which carry synthetic aperture radar instruments which can see through clouds and at night, only became available in late 2014. These data address some limitations of satellite imagery data but have less frequent temporal coverage. A recent study combines the two data sources to build a more robust flood detection tool (Patel, 2024).

effects.

Decreased crop production value is driven by a decrease in average physical yields, both of which are largest in the year of the floods but persist over subsequent seasons. Crop portfolios and land, labor, and purchased inputs do not change significantly, indicating direct lasting effects of the floods on land productivity beyond the initial destruction. Farmers in flood-exposed communities also report selling their crops for lower prices: they do not benefit from the same price increases experienced over time in unexposed communities. Despite these effects we find no evidence of agricultural exit or reallocation of labor away from agriculture following flood exposure.

Nigerian households in the survey sample do not report commonly using migration or non-farm enterprises (NFEs) to cope with flood shocks; reducing consumption, selling assets, and both formal and informal assistance and loans are the most common coping strategies. In line with this, we find no average effects of flood exposure on household NFE engagement, labor hours, or income either immediately or over time. The probability that the household moves or that any member migrate does not change on average, though we do find significant increases 7 years after the floods (of 4 and 9 percentage points, respectively). Delayed increases in migration suggest this is potentially a desirable but constrained flood response. We also find no immediate or average effect of 2012 flood exposure on engagement in wage work, though effects on the count of household members working for a wage are significant for 2018-19 and average effects are significant in some robustness checks. These noisy increases in wage employment – suggesting some form of livelihood coping or adaptation – are concentrated among households previously active in wage work and are not associated with increases in wage income. In particular, only households engaged in *both* agricultural production and wage employment in the survey round before the floods increase their labor supply – to both activities – following flood exposure. Baseline livelihood activities are the only set of household or community characteristics along which we find consistent heterogeneity in the impacts of the 2012 floods. Consumption per capita does not decrease after the floods for these *ex ante* diversified households, but *ex post* diversification appears to be constrained. Household livelihoods in this context are highly concentrated and remain so after the major 2012 flood exposure shock: the shares of total household income from farm production value, NFE, and wage employment are unchanged.

The results are qualitatively similar across different approaches to defining survey-based community flood incidence, although the significance of some effects is lost when we consider specific subsets of flood-related survey questions, emphasizing the importance of how surveys ask about flood exposure. Differences in the results are much larger when using a satellite-based measure. An analysis of satellite-detected flood exposure would conclude that on average, floods decrease household farm production value but that this is more than offset by a large increase in NFE income, completely protecting households from adverse consumption or food security impacts. Strikingly, we find no significant effects of satellite-based 2012 flood exposure on household outcomes – including farm production value – in the 2012-2013 survey round, immediately after the flood. This is consistent with attenuation from measurement error, as this measure does not detect floods in a large

number of communities with survey flood reports and classifies a similar number of communities with no flood reports as flooded. These differences highlight the importance of decisions around flood incidence measurement for both interpretation and validity of economic analyses.

This paper contributes to our understanding of the dynamic impacts of weather-related disasters, particularly floods, on labor supply and livelihoods in sub-Saharan Africa, adding to a rich literature on the impact of climate shocks in LMICs. Many studies have analyzed the effects of floods on household outcomes, though studies of its impacts on migration and labor supply have focused primarily on South Asia and on short-term effects (see Section 2 for an overview). In general, few studies of natural disasters in LMICs have analyzed long-term effects on measures of livelihood or labor supply.³ Using measures of community exposure to a major flood event in Nigeria, we find persistent decreases in household consumption per capita driven by lower crop yields and farm production value. Wage employment increases only among households with members already working for a wage before the floods and NFE activity does not change while migration does not increase until many years after the floods, indicating that access to non-farm opportunities is limited. These findings align with some prior studies of long-term livelihood effects of floods in South Asian contexts (Sajid, 2023) but not others (Mueller and Quisumbing, 2011; Patel, 2024), indicating that effects of floods may depend on the specific shocks and populations studied.

Our analysis of long-term labor allocation across activities following a weather shock also builds on the literature on factors encouraging or impeding structural transformation in LMICs (Alobo Loison, 2015; Gollin and Kaboski, 2023). Global reductions in agricultural productivity growth over the past few decades due to climate change have important implications for structural transformation.⁴ While a large empirical literature studies the consequences of climate change and many studies analyze short-term labor supply responses to extreme weather events in LMICs, there is limited evidence on the relationship between extreme weather events and livelihood diversification or structural change over the long-term in sub-Saharan Africa (Barrett et al., 2021). We show no evidence that flood adaptation or coping strategies push reallocation of labor from farm to non-farm activities following Nigeria’s 2012 floods. Household livelihoods in this context are highly concentrated and remain so after the floods. Though we do not find significant increase in agricultural labor, decreased consumption and agricultural productivity suggest floods may slow down processes of structural transformation. This has important implications in a context of increasing flood risk, particularly as we show that more diversified households are more resilient to floods.

Finally, we add to a growing literature on measurement concerns in evaluating the impacts of flood exposure, and to a broader literature on using remote sensing in economic analysis and particularly the limitations of such measures.⁵ We find large differences in the communities identified as affected by the 2012 floods in Nigeria according to survey reports and a NASA database using

³Some examples include Alfani et al. (2024), Efobi (2022), and Mueller and Osgood (2009b) on droughts, Kirchner (2017) on earthquakes, Van den Berg (2010) on hurricanes.

⁴See Auffhammer and Kahn (2018) and Kala et al. (2023) for discussions of potential responses for agricultural households in LMICs.

⁵See Donaldson and Storeygard, 2016; Fowlie et al., 2019; Gibson et al., 2021; Jain, 2020; Josephson et al., 2026.

MODIS satellite imagery (building on previous work in Bangalore et al., 2026). Differences in vulnerability across communities play a role in what variations in surface water are experienced as flood shocks by humans, but we show that much of the disagreement is due to constraints in what floods can be detected by the satellite data available for 2012. While the limitations of different flood incidence measures are known, we highlight how they influence what types of flood events are captured and analyzed and how analyses of the effects of floods should be interpreted. The vast majority of studies on the impacts of floods consider a single source of flood incidence data, and many on rely satellite imagery data similar to what we use.⁶ The striking differences in estimated impacts under different measurement approaches in this paper – we find largely null effects of exposure to floods detected by satellite, even in the short-term – suggest a need for more careful consideration of how to approach measurement⁷ as well as transparency and sensitivity checks in future research on floods.

2 Conceptual framework and literature

The structural transformation of economies is a key aspect of modern economic development. Historically, structural transformation has involved decreasing employment in agriculture, hump-shaped employment shares in manufacturing, and increasing employment in services as a function of the level of economic development, though not all countries follow this trajectory (Sen, 2019). Models of structural transformation emphasize the role of technological progress in driving reallocation across sectors (Herrendorf et al., 2014). In agriculture, innovation-driven productivity growth tends to *push* labor out, as decreasing amounts of labor are needed to meet finite demand for agricultural products. In manufacturing and services, productivity growth *pulls* labor in by increasing the marginal product of labor and therefore wages while lower prices and increased incomes drive increasing demand for manufactured goods and services.

A large literature studies the drivers and constraints to structural transformation (see Gollin and Kaboski (2023) for a recent overview), but it is clear that differences across contexts in productivity across sectors can affect whether and how structural change occurs. These considerations make it important to consider how climate-related productivity shocks could affect structural transformation, particularly in LMICs with high shares of employment still engaged in agriculture. Climate change is decreasing agricultural productivity in most regions of the world. Recent research suggests that current adaptive measures are insufficient to mitigate these effects (Hultgren et al., 2025; Kala et al., 2023; Pörtner et al., 2022) and that constraints to migration – an important adaptation pathway (see e.g., Ibáñez et al., 2026) – increase the losses from climate change (Cruz and Rossi-Hansberg, 2024).

⁶This is the most common data source for studies of the labor supply or migration effects of floods, with different sources of satellite imagery used in Akter (2021), Balboni et al. (2023), Chen and Mueller (2019), Chen et al. (2017), Sajid (2023), and Vitellozzi and Giannelli (2023).

⁷For example, some recent studies propose machine learning algorithms to help address some limitations of satellite imagery (Patel, 2024; Saunders et al., 2025).

Studies on the impacts of climate change on structural transformation have focused on the effects of increasing temperatures and report mixed results. Two global analyses find that decreased agricultural productivity increases agriculture’s share of employment as consumption expenditure falls and shifts towards subsistence goods (Cruz, 2024; Nath, 2025). At the national and sub-national level, there is evidence of both increases (Albert et al., 2021; Colmer, 2021; Pham, 2025) and decreases (Liu et al., 2023; Pham, 2025) in non-agricultural employment. What is clear from the literature is that reallocating labor to a more productive non-farm activity would reduce losses from permanent or transitory decreases in agricultural productivity.

In addition to increasing temperatures, climate change is also affecting the frequency and severity of extreme weather events. Many studies find that agricultural households respond to adverse weather events by increasing labor supply to non-agricultural activities in the same year in order to offset lost agricultural income and smooth household consumption. A growing body of evidence evaluates whether exposure to such shocks has persistent effects on household livelihood strategies,⁸ but the results are mixed so the implications for structural transformation are unclear.

If structural transformation is most commonly the result of productivity *increases* in the agricultural and non-agricultural sectors, how might an agricultural productivity *decrease* from a weather-related shock such as a flood promote labor reallocation? Several mechanisms may push households to exit agriculture or diversify their livelihood strategies. First, households may update their beliefs about risks and returns to agricultural production, though only if the shock is seen as an indicator of a changing risk profile and not as an extreme realization from an unchanged risk distribution. Second, the short-term non-farm labor and migration coping strategies documented in many studies to smooth consumption needs following shocks may help households learn about non-farm livelihood opportunities that are more productive than agriculture. Third, for weather disasters causing physical damages such as floods or cyclones, damage or loss of productive assets may change perceptions about sunk costs of remaining in agriculture. Forced displacement may similarly decrease the opportunity cost of allocating labor away from agriculture.

On the other hand, a variety of factors can constrain households wanting to respond to a shock by diversifying their livelihood activities or exiting agriculture. In developing contexts which often have high trade costs, low agricultural productivity can create a ‘food problem’ increasing the need to allocate land and labor to produce food and meet subsistence needs (Gollin et al., 2007; Nath, 2025). As documented in studies on climate change and structural transformation, reduced agricultural productivity also decreases local incomes. This will on average reduce demand for non-agricultural goods and services and associated non-agricultural labor (Liu et al., 2023), limiting local non-farm opportunities which may already be low in rural agricultural areas with weak infrastructure (Barrett et al., 2021). Migration is a possible response, but resource constraints following a productivity shock can interact with other barriers to prevent migration in search of non-agricultural work. Even successful migrants may face high costs of living and lower wages in

⁸E.g., Mueller and Osgood (2009a, 2009b) on drought in Brazil, Efobi (2022) on drought in Nigeria, Alfani et al. (2024) on drought in Morocco, Kirchberger (2017) on earthquakes in Indonesia, and Van den Berg (2010) on hurricanes in Nicaragua. We discuss studies on the impacts of floods in more detail below.

urban areas due to increased market competition (Auffhammer and Kahn, 2018), and many rural households and individuals prefer not to migrate away permanently (Lagakos et al., 2023). Even where there are non-farm opportunities, household strategies to smooth consumption after a shock can deplete household assets (Carter et al., 2007), reducing resources available to invest in new skills, activities, and technologies that could be needed to engage in new livelihood activities.

This conceptual framework highlights how adverse shocks to agricultural productivity could have different effects on household livelihood strategies depending on the context and household characteristics. In well-integrated areas with more developed labor markets, agricultural households with sufficient resources may reallocate labor to non-farm activities while in less-integrated areas and for poorer households, reduced incomes and limited alternatives may increase engagement in agriculture. For example, Mueller et al. (2020) find that temperature and rainfall anomalies increase short-term out-migration in some African countries but decrease it in others and point to differences in non-farm opportunities and the need for subsistence production labor as potential mechanisms. Pham (2025) finds that extreme heat in Vietnam only leads to reallocation of labor away from agriculture in more globally-integrated regions.

Most weather-related shocks do not only affect agricultural productivity. This paper focuses on floods, which can destroy or damage both agricultural and non-agricultural property and assets as well as local infrastructure. These other effects may affect livelihood responses to the shock. Local destruction can decrease non-agricultural work opportunities by disrupting infrastructure, supply chains, and demand, though reconstruction may increase demand at least temporarily. Livelihood effects of a flood or other weather shock may therefore differ from those of a pure shock to agricultural productivity that does not have broader economic effects (e.g., a crop disease). However, analyzing the livelihood effects of floods in an LMIC through the lens of agriculture remains relevant its continued importance for livelihoods and its vulnerability to weather shocks.

The evidence on the impacts of flood exposure on livelihoods, labor, and migration in LMICs is mixed and has overwhelmingly focused on South Asian countries.⁹ For example, four studies report increases in short-term out-migration following flood exposure (Balboni et al., 2023; Berlemann and Tran, 2020; Patel, 2024; Pavel et al., 2023), two report decreases (Chen and Mueller, 2019; Chen et al., 2017), two report null effects (Gray and Mueller, 2012; Mueller et al., 2014), and one reports mixed results (Maystadt et al., 2016). While many studies find that flood exposure increases short-term non-farm labor activity (e.g., Maystadt et al., 2016; Mueller and Quisumbing, 2011; Vitellozzi and Giannelli, 2023), the small literature on long-term effects of floods on livelihood activities is less consistent. Mueller and Quisumbing (2011) find that wages in both farm and non-farm activities fall and that workers move into non-farm casual labor in the short-term following flood exposure in Bangladesh, but that these effects do not persist 5 years later. Sajid (2023) finds that flood exposure in the past decade in rural areas of India decreases household wealth, non-agricultural work, and migration to urban areas while increasing engagement in agriculture. In contrast, Patel

⁹Studies in other contexts are limited by include Arnall et al., 2013 on resettlement and livelihoods in Mozambique and Berlemann and Tran, 2020 on migration in Vietnam.

(2024) finds that floods in Bangladesh persistently increase migration and push employment out of agriculture, with effects mitigated by prior adaptation to past flooding exposure.

The different results across these studies motivate further evaluation of the long-term household labor supply and livelihood impacts of flood exposure in a new context (without the monsoon-related flooding prevalent in South Asia), and a need to identify factors which could lead to heterogeneous effects.

3 Context and data

Floods are one of the most common types of natural disasters globally and the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report finds that there is high confidence that climate change is increasing the risk and severity of flooding, due largely to increases in extreme precipitation events (Caretta et al., 2022). Tellman et al. (2021) estimate that the proportion of the global population exposed to floods has increased by 20-24 percent from 2000 to 2015, and find that this will increase further by 2030 under current climate change projections. The largest projected increases are concentrated in South and Southeast Asia and in Sub-Saharan Africa (Caretta et al., 2022). Rogers et al. (2025) estimate that by 2100, 63% of global flood exposure will be in the areas with the lowest GDP levels.

Nigeria is one of the countries most affected by increasing flood exposure. Tellman et al. (2021) estimate that the population exposed to flooding nearly doubled from 2000 to 2015. Rogers et al. (2025) project that it will nearly double again from 2020 to 2050, from 34 to 64 million at risk of 100-year floods of at least 10 cm. The Nigeria Hydrological Services Agency (NIHSA) reports that the frequency and severity of flooding have increased in recent years, disrupting agricultural production and livelihoods but also causing death, population displacement, and destruction of housing and other infrastructure (NIHSA 2021). In 2012, the most severe floods in Nigeria’s recorded history affected 30 of 36 states between September and November as a result of excessive rainfall. The floods caused an estimated \$16.9 billion in damages, displaced over 2.1 million people, and killed 431 people (Unah, 2021).

Increasing flood risk and a recent catastrophic national flood disaster make Nigeria a relevant context for studying the economic impacts of flood exposure. Nigeria is also an important setting to study labor reallocation and structural transformation, with the largest population and economy in Africa. While 76% percent of Nigeria’s GDP comes from non-agricultural sectors, around 70% of Nigeria’s population is engaged in agriculture (Statista, 2023), primarily at low productivity for subsistence purposes (FAO, 2022).

3.1 Survey data

The main data source for the analysis is Nigeria’s General Household Survey Panel (GHSP).¹⁰ The GHSP, part of the World Bank’s Living Standards Measurement Study - Integrated Surveys on

¹⁰See [Appendix C](#) for additional details.

Agriculture (LSMS-ISA), is a nationally-representative panel survey including approximately 5,000 households conducted by the Nigeria National Bureau of Statistics. In each round, households are surveyed twice: once post-planting (around August-September) and once post-harvest (around February-March). Four survey rounds were been completed between 2010 and 2019, with two-thirds of the sample being replaced for the 2018-19 round. The analysis sample excludes all new households surveyed for the first time in 2018-19. [Figure A1](#) shows a timeline of GHSP data collection. Efforts to track households even if they move result in quite low levels of attrition: just 7.7% of households drop out of the sample between 2010 and 2016.¹¹ These efforts also allow us to track household-level migration over time.

The household survey includes detailed modules on household agricultural production, labor supply, assets, consumption, shocks, and socio-demographics. The primary household-level outcomes are measures of total annual income or production value by activity and engagement of members in different types of work. We consider three broad livelihood activities: household farm (crop and livestock) production, household non-farm enterprise, and wage employment. In addition, we consider effects of flood exposure on measures of household consumption, food security, and assets, and on a variety of crop production outcomes. We convert all monetary values to constant 2016 PPP USD, accounting for inflation and changes in purchasing power, and winsorize continuous variables at the 99th percentile.

Incomes are measured at the annual level over the 12 months prior to the post-harvest survey. We assign a value of 0 for households not engaged in particular activities. In 2010, 40% of households reported at least one member engaged in wage employment at some point in the year. Household wage income is the sum of individual members' earnings for all wage employment. Similarly, income from household non-farm enterprise (76% of households) is the sum of revenues from all enterprises. For household farm activities (69% of households), we estimate the value of all crop and livestock production. The value of crop production is based on crop sales revenue plus the value of unsold production, calculated as the quantity not sold times the median price for sales of that crop in the community. We follow the same approach to value livestock products (e.g., animal labor, eggs, milk, meat, live animals). We also calculate total income from other sources, such as investments, remittances, and pensions.

We measure migration using information on household-level moves and changes in composition of household members. Household-level migration is defined by whether the household has moved to a different community between survey rounds. These movers remain in the survey sample, and 9.7% of the initial 2010-11 sample of households moved at some point over the following survey rounds. This measure will not capture temporary displacement if the household returns to its original community by the time of the next survey. Individual-level migration is defined using changes in household rosters. In each round, respondents are asked whether any previous members have left the household and whether any new members have joined. We construct measures of general member departures (for reasons other than death), and of member migration for work or

¹¹For the panel sample of the 2018-19 survey round, 95% of targeted households were tracked and included.

better land and for marriage or other family-related reasons. We also measure whether any new adult member joined the household.

Each household is associated with community-level data based on surveys with a group of community informants conducted at the same time as the household surveys.¹² The public survey data also include spatial characteristics calculated by the World Bank at the community level such as distances to markets and administrative centers, local climate and weather, and land cover. We complement this with data on fluvial flood hazard at a 90 meter resolution from the European Commission Joint Research Centre (JRC; Baugh et al., 2024) and data on rivers and bodies of water from the Digital Chart of the World. We calculate the distance between each survey community and the nearest body of water and the mean and maximum depth of 100-year floods within a 5 km radius as a measure of local flood hazard.

3.2 Flood data

An important challenge in analyzing floods is determining how to measure flood incidence – where floods have occurred. The main raw data sources include government records, media reports, survey reports, precipitation data, river or other water gauges, and satellite imagery or radar. All of these sources have been used in applied research studying the impacts of floods. Each source has different advantages and limitations.¹³ The most common flood incidence measures in recent years draw on high-frequency and high-resolution satellite imagery or radar data, applying various algorithms to first identify surface water and then determine when and where to classify this as flooding.

Most of the economics literature on the household effects of flood exposure uses a single flood incidence measure and does not engage critically with flood remote sensing literature, a criticism that has also been applied to economists’ use of nighttime lights data (Gibson et al., 2025). Guiteras et al. (2015) were among the first (economists) to highlight that different flood measures were poorly correlated, considering rainfall, satellite imagery, and survey reports in Bangladesh, but did not test flood impacts. Chen et al. (2017) find higher correlations between a more refined rainfall-

¹²Communities are referred to as ‘enumeration areas’ (EAs) in the GHSP.

¹³Biscaye et al. (2026) provide a summary and overview of current approaches to flood measurement in empirical research. Government and media reports can provide validated information on the timing and location of floods and are incorporated in many public databases of flood events such as the Dartmouth Flood Observatory (DFO, Brakenridge, 2023) and the EM-DAT International Disaster Database (Guha-Sapir et al., 2023). But media sources typically only approximate the spatial extent of flooding and are not comprehensive, with particularly limited coverage outside more populated and developed areas (Jones et al., 2023; Patel, 2024), while the level of detail in government flood records is highly variable. Survey reports provide direct evidence of areas affected by floods but are limited in their spatial and temporal coverage and subject to concerns around measurement error and reference dependence or adaptation. Local precipitation may be correlated with flooding, though the quality of the proxy is questionable (Chen et al., 2017; Guiteras et al., 2015; James and Schumacher, 2024), as most flood events result from the overflow of rivers and other bodies of water (fluvial floods) rather than local rainfall accumulation that is not absorbed or drained away (pluvial floods). Precipitation data may also be combined with hydrological models to predict flood incidence. Water gauges are limited in spatial extent and are more commonly used for validation of other flood measures than for measuring flood incidence. Satellite data sources have the advantages of high spatial and temporal coverage but may fail to capture floods that are of short duration, limited in extent, or masked by clouds (an issue for imagery but not radar), buildings, trees, or topography. They are also sensitive to algorithm parameters selected to identify surface water and classify flood events.

based measure and a measure based on satellite imagery in Bangladesh, and estimate similar qualitative effects of flood exposure on out-migration. Akter (2021) considers survey reports and satellite imagery detection of floods in Pakistan and finds similar qualitative effects on men’s and women’s time use. In contrast, Saunders et al. (2025) report that choosing between measures of incidence based on satellite imagery (including state-of-practice and state-of-the-art models), a physical inundation model, rainfall deviations, and river gauges significantly affects the accuracy and timeliness of simulated flood index insurance payouts in Bangladesh.

In closely related work, Bangalore et al. (2026) compare different approaches to measuring incidence of Nigeria’s 2012 floods. They evaluate three GHSP survey-based measures, two measures based on satellite imagery, one based on media and government reports (the Dartmouth Flood Observatory (DFO) Archive; Brakenridge, 2023), and one combining DFO flood events with satellite imagery (the Global Flood Database; Tellman et al., 2021). The study finds significant disagreement across sources in which GHSP communities are identified as experiencing floods in 2012, and this leads to inconsistent estimates of the short-term effects of these floods on household agricultural production. We build on Bangalore et al. (2026) and other papers considering approaches to flood measurement by analyzing the drivers of disagreement between survey and satellite measures of community flood incidence in Nigeria in 2012, and explore the implications of flood measurement approaches to estimating long-term effects of the 2012 floods.

First, we use a *survey-based* measure of flood incidence using household reports in the 2012-13 GHSP.¹⁴ We combine reports from separate questions in different parts of the survey on whether floods in 2012 caused harvest failures, loss of harvest, loss of property, and food insecurity. We count the number of households in each community reporting any type of flood exposure in 2012 (Figure A2), and define community flood incidence as having at least two households reporting flood exposure. This approach has the advantage of capturing flood events that affect households in different ways, though will not capture floods that do not affect households through these particular dimensions. Requiring more than one household flood report effectively treats isolated reports as measurement error, as a ‘non-flood’ shock (some households may report damage from heavy precipitation as a ‘flood’), or as a highly-localized shock. This requirement could bias estimated effects downward due to some flooding among control observations, or upward by identifying effects of potentially more severe floods. We test the sensitivity of the results to different survey-based flood incidence measures, including measures that incorporate reports of flood events in the community survey or that exclude different forms of household flood reports. Appendix C provides additional details on the specific survey questions used and correlations in flood reports across survey questions.

Second, we follow a large number of economics papers (Akter, 2021; Chen and Mueller, 2019; Chen et al., 2017; Guiteras et al., 2015; Sajid and Bevis, 2021; Vitellozzi and Giannelli, 2023) in using a *satellite-based* flood incidence measure using MODIS (Moderate Resolution Imaging Spectroradiometer) satellite imagery. In particular, we use NASA’s Near Real-Time (NRT) 2-day

¹⁴See e.g., Freudenreich and Kebede (2022), Gray and Mueller (2012), Mueller and Quisumbing (2011), and Stein and Weisser (2022) for examples of research on the household impacts of floods measured using survey reports.

250 m resolution Global Flood Mapping product (NASA, 2024), taking their approach to identifying flood events as given.¹⁵ We define pixels as having been exposed to floods if the NASA NRT product identifies any flood incidence in that pixel in 2012 (Figure A3 Panel G presents a map). We match GHSP communities to satellite flood incidence based on the community coordinates. These coordinates are randomly offset from the original community centroids in the public data, by 0-2 km in urban areas and by 0-5 km in rural areas (and up to 10 km for a random 1% of rural communities). We therefore define a community as exposed to a satellite flood incidence if its offset coordinates are within 5 km of any flooded pixel, and test robustness to using a 10 km radius. Moving forward we refer to this as the ‘MODIS’ or ‘satellite’ flood measure.

Recent studies using remotely-sensed flood measures increasingly rely on satellites with radar sensors, such as Sentinel-1. The Sentinel-1 radar data are higher resolution than the MODIS data and the radar has the ability to see through clouds and at night, but this satellite was launched in late 2014, after the 2012 flood event we study. We do not include a precipitation-based proxy of flood incidence due to the coarseness of publicly-available databases and limitations in using local precipitation deviations to identify flooding.

4 Analyzing flood detection across sources

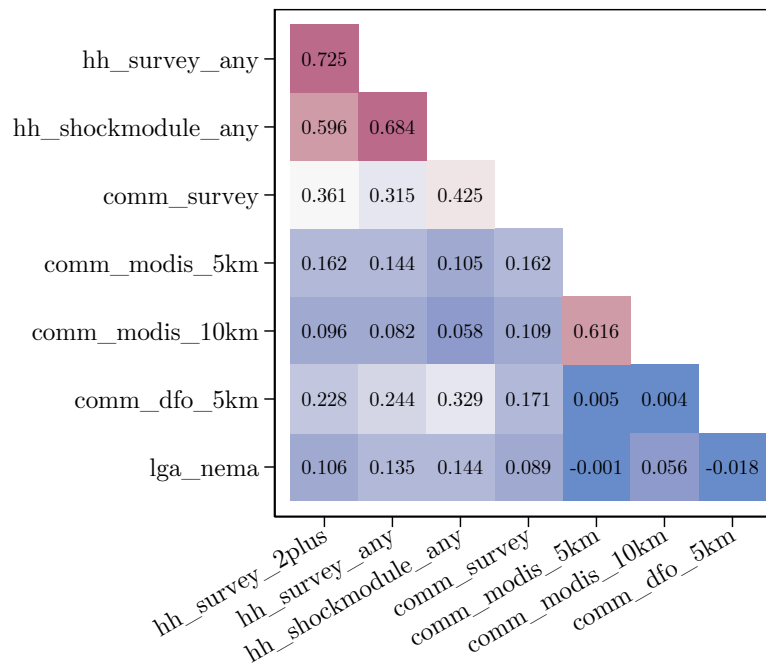
The set of communities identified as exposed to floods in 2012 differs significantly depending on the survey measure. Figure 1 shows a heatmap of pairwise correlations between flood measures (Figure A3 maps exposed communities by definition). Measures based on the household survey are highly correlated with one another (values above 0.6), though measures based on household reports versus one based on informant reports in the community survey are only moderately correlated (values between 0.3-0.4). Correlations between survey measures and the satellite-based measure using MODIS imagery are weak, with none greater than 0.16. The correlation is slightly stronger when using a 5 km buffer compared to 10 km to link communities to MODIS-detected floods, and when using the main survey-based flood definition that considers all flood-related questions and requires at least two separate household reports.

Potential validation data are limited, but we analyze the correlations between these floods measures and two sources of flood records. The media report-based DFO Archive identifies five broad polygons of flooded areas for Nigeria in 2012 (Figure A3 Panel I).¹⁶ We do not directly use this DFO measure in the analysis because of the coarseness of the flood identification, but to the extent it captures zones where some flooding definitely occurred it can be used to compare the survey and satellite measures. As an additional source of validation data, we directly obtained a list of local

¹⁵The algorithm identifies surface water using the Normalized Difference Water Index (NDWI), applies a permanent water mask to determine when this water is not expected, and then applies various corrections to deal with cloud and terrain shadow and other known issues in surface water detection. The NASA NRT product transitioned to a different format in 2022 and access to historical flood data has been restricted and is no longer possible through the online platform.

¹⁶The EM-DAT International Disaster Database, which draws on media and government reports, only records a subset of the states in which floods were reported in 2012.

Figure 1: Correlations between different measures of community-level 2012 flood incidence



Note: This heatmap shows pairwise correlations between different measures of flood incidence in 2012 at the level of communities in the Nigeria GHSP. ‘hh_survey_2plus’ is the main survey-based measure, indicating at least two households reporting a flood in any survey module. ‘hh_survey_any’ is the same but only requires one household report. ‘hh_shockmodule.any’ indicates any household reporting a flood but only considering reports in the household shocks module. ‘comm_survey’ indicates a flood report by informants in the community survey. ‘comm_modis_5km’ and ‘comm_modis_10km’ indicate any flooded pixel identified in the NASA MODIS flood database within 5 and 10 km of the community centroid, respectively. ‘comm_dfo_5km’ indicates whether the community is within 5 km of a polygon of flooding recorded in the Dartmouth Flood Observatory archive. ‘lga_nema’ indicates floods reported by NEMA in the local government area (LGA) containing the community.

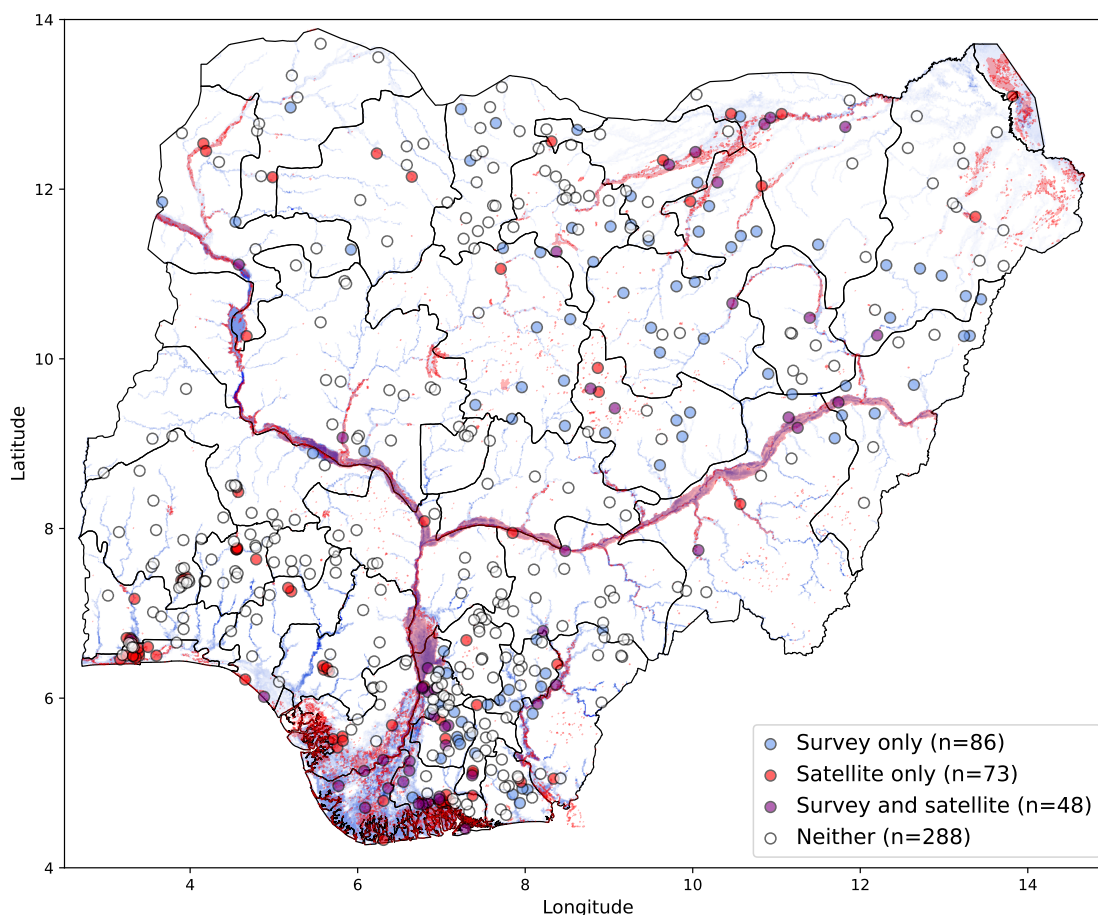
government areas (LGAs) – the administrative level below the state – where floods were reported in 2012 from the Nigerian National Emergency Management Agency (NEMA), and match this information to communities based on their location. NEMA is responsible for overseeing disaster management in Nigeria, and collects information on flood reports from State Emergency Management Agencies, other government agencies, and media sources.¹⁷ For both validation sources, we find weak to moderate correlations with the survey-based flood measure, as expected based on the inability of the DFO and NEMA data to reflect local variation in flooding, but essentially zero correlation with the satellite-based measure. This pattern indicates that the survey measure better captures the reality of flooding experienced on the ground than the satellite measure.

Figure 2 maps the locations of GHSP communities and their 2012 flood exposure status based on the main survey measure and the satellite measure. The figure illustrates how widespread flooding was across the country as well as the local variation in exposure, and highlights differences in community flood exposure between the two measures. Of 495 GHSP communities in the sample, 134 have at least two household flood reports and 121 are within 5 km of a pixel identified as

¹⁷This was the first year where NEMA attempted to collect information on flooded areas across the country, as the extent of the flooding and damages in 2012 was unprecedented. Resources and tracking efforts were therefore limited but were greatly increased in subsequent years. The 2012 NEMA records list the LGAs for only 21% of communities identified as flooded by satellite and 28% of those identified by survey.

flooded by the satellite measure, but just 48 communities are considered flooded according to both measures.

Figure 2: 2012 community flood exposure, by flood measure



Note: The map shows the locations of communities in the Nigeria GHSP sample, colored by their flood exposure in 2012. Flood incidence is based on either at least two survey reports of being affected by floods in the community or on the community being within 5 km of a pixel where flooding was identified in the NASA MODIS-based flood tracking product. Pixels identified as flooded by the NASA product are shown in red. The blue shading in the background corresponds to the depth of 100-year fluvial floods, which largely tracks major rivers and bodies of water.

Survey measurement error does not seem to drive the difference, as alignment between the survey and satellite measures only increases slightly when we require at least two households to have reported floods for the survey measure compared to one household, and the correlation is similar for community survey flood reports. Strategic misreporting also does not seem an important factor, as on average 34% of households (10 households are sampled in most communities) report flooding in the 86 communities identified as flooded by survey only, compared to 50% on average in the 48 communities identified as flooded by both measures. Part of the mismatch may be due to the noise in the publicly-provided GHSP community coordinates,¹⁸ but the correlation between the two measures is worse if we extend the radius for the satellite measure to 10 km (Figure A4 shows the distributions of distance to a MODIS-flooded pixel by survey flood status). This aligns

¹⁸We were unable to obtain a dataset linking the true community locations to the MODIS flood incidence data from the World Bank LSMS team.

with less than 1% of community coordinates being offset by more than 5 km. Twenty-four of the 86 communities classified as flooded by survey only are within 10 km of a flooded pixel, and their true location – or those of their more remote households – may in fact overlap with the area where floods are detected. But the other 62 are more than 10 km from any satellite-identified flooding, indicating other reasons for the mismatch in flood classification.

Different types of errors may explain communities identified as flooded by only one measure. The accuracy of the MODIS measure is affected by three important limitations: missing data due to cloud cover and gaps in temporal coverage, relatively low imagery resolution, and an inaccurate masking of permanent or seasonal water.¹⁹ Most flooding in Nigeria is associated with periods of heavy rainfall so may be masked by clouds. During the period of the 2012 floods from August through November, the median (mean) MODIS pixel in Nigeria was cloudy for 71% (73%) of images, while the median across GHSP communities for the share of pixels within 5 km masked by clouds was 100% for nearly all of August-October (Figure A9). The satellites also only take snapshots at particular points in time every 1-2 days so will miss flash floods where standing water does not persist. The 250 m resolution of the MODIS imagery also make it difficult to detect localized floods, since detection is based on pixel-level reflection of light at different wavelengths. Imagery at this resolution are best suited to detect large contiguous bodies of water that make up a large share of pixel area (see Figure A10 for some examples). These two limitations likely account for most of the 86 communities identified as flooded by the survey only. A flood measure based on higher-resolution satellite radar such as Sentinel-1 data, or a combination of radar and imagery, would help overcome this limitation (see Patel (2024) for a thoughtful example of how this can be done), but public satellite radar data are not available for 2012. Another possibility is that survey respondents may report water-related damages from heavy rainfall that did not involve inundation as floods, as the surveys do not allow for separate reporting of such damages. Such events would not be captured by satellites, but requiring multiple household flood reports in our main survey-based definition should reduce the risk of this type of misclassification.

There are 73 communities within 5 km of MODIS-detected flooded pixels but with no survey-reported floods. In 17 of these communities there is exactly one household flood report, but using this less conservative survey-based flood measure would lead to an additional 54 communities with survey-reported floods but no satellite-detected floods. The correlation between survey reports and the MODIS measure is weaker if we only consider household reports in the shock module, showing the importance of including respondent reports in other parts of the survey. Including community survey reports, which include a broad definition of “affected by flooding,” does not strengthen the correlation. The mismatch is therefore not driven mainly by our choice of how to construct the main survey-based flood measure.

Several other possibilities could explain satellite flood detection without survey reports. The detected floods may have been sufficiently far away not to affect households, or floods near these communities may not have affected households in the specific ways they are asked about (harvest

¹⁹ Appendix D presents more detail and examples of the limitations of MODIS imagery for flood detection.

loss or failure, property loss, food insecurity). Reference dependence and adaptation may also affect whether communities experience and report adverse effects of surface water changes (Guiteras et al., 2015). Finally, challenges in masking out permanent or seasonal water in the satellite-based measure may lead to false detection of floods. The NASA MODIS-based flood product primarily identifies flooding near rivers and other bodies of water where the imagery resolution is sufficient to detect surface water (Figure 2). The flood detection algorithm is a “best guess” as to the flood status of detected water (Slayback, 2024), and may misclassify some expected or seasonal variation in surface water as flooding.²⁰

To test whether there are systematic differences in the types of communities identified by the different flood measures, we estimate a series of regressions of community characteristics on dummies for flood incidence under different combinations of the two measures, and present the results in Table 1. Communities where 2012 floods are identified in both survey reports and MODIS imagery are primarily rural (72%), within 10 km of a body of water, have average 100-year flood inundation depth within 5 km of 1.5 m, and experienced 331 mm less precipitation in 2012 than their historical average (first column). Areas that are not classified as flooded according to either flood measure are 32 km farther from a body of water and have much lower underlying flood risk, with average 100-year flood inundation depth within 5 km of 0.1 m, higher elevation, and greater average terrain slope. They had similar precipitation in 2012 as usual, but are otherwise similar in terms of rural location and household characteristics (column β_3). This aligns with floods being primarily fluvial rather than pluvial and concentrated in areas with high underlying flood hazard.

The significant differences from historical annual precipitation in communities identified by satellite highlight the importance of cloud cover to this measure: floods are primarily identified in areas with less rainfall – and therefore less cloud cover – than usual. Relative to communities identified as flooded according to both survey and satellite, communities classified as flooded according to satellite only are 39 percentage points (pp) less likely to be rural, are 3.6 km closer to a major road, are 11 km farther from a body of water, and have half the average level of flood hazard (column β_1). Households in these communities have 78% less agricultural land, are 35 pp less likely to be engaged in crop production, and are 13 pp less likely to be below the poverty line. The satellite-only measure is therefore more likely to identify floods in more urban, less agricultural settings with lower flood hazard. Households in these areas may be less likely to experience property and harvest destruction from floods due to more robust housing materials and less agricultural engagement and have access to more resources to cope with food insecurity, and so be less likely to report a flood shock. The lower flood hazard also suggests possible false positives from the NASA MODIS flood detection algorithm.

Communities identified as flooded according to the survey only are more likely to be rural, are 16 km farther on average from a body of water, have higher elevation and terrain slope, face

²⁰Slayback (2024) notes that the NASA Global Flood Mapping product algorithm includes time compositing to deal with clouds and shadows, terrain and cloud shadow masking, Height Above Nearest Drainage masking, and a reference surface water mask based on water detection over previous years. It is not clear if the water mask accounts for seasonal water fluctuations.

Table 1: Differences in GHSP community characteristics by 2012 flooding status

	Mean: Both definitions N=48	β_1 : Satellite only N=73	β_2 : Survey only N=86	β_3 : No flood N=289	$\beta_1 = \beta_2$ (<i>p-val</i>)	$\beta_1 = \beta_3$ (<i>p-val</i>)	$\beta_2 = \beta_3$ (<i>p-val</i>)
Rural	0.72 (0.45)	-0.39*** (0.08)	0.16** (0.08)	-0.02 (0.07)	0.00***	0.00***	0.00***
Distance to nearest major road (km)	7.68 (10.60)	-3.62** (1.50)	0.49 (1.45)	-2.02 (1.26)	0.00***	0.11	0.01***
Elevation (m)	176.81 (263.03)	10.07 (38.42)	167.38*** (37.16)	110.61*** (32.21)	0.00***	0.00***	0.01**
Slope (percent)	2.15 (1.16)	0.40 (0.45)	0.73* (0.43)	1.35*** (0.37)	0.28	0.00***	0.05*
Distance to nearest water area (km)	9.72 (14.06)	11.34** (5.27)	15.85*** (5.10)	31.22*** (4.42)	0.23	0.00***	0.00***
Avg 100-year flood depth within 5 km (m)	1.48 (1.54)	-0.75*** (0.12)	-1.32*** (0.12)	-1.37*** (0.10)	0.00***	0.00***	0.20
Diff. in annual precipitation from hist. avg. (mm)	-330.68 (571.96)	61.68 (70.81)	358.02*** (68.50)	265.36*** (59.38)	0.00***	0.00***	0.02**
Baseline share HHs w/ >5 members	0.48 (0.27)	-0.08* (0.04)	0.12*** (0.04)	-0.03 (0.04)	0.00***	0.08*	0.00***
Baseline share HHs below 1.90 USD/cap poverty line	0.34 (0.31)	-0.13** (0.05)	0.17*** (0.05)	0.02 (0.04)	0.00***	0.00***	0.00***
Baseline share HHs w/ above-median asset value	0.52 (0.25)	0.05 (0.05)	-0.08* (0.05)	-0.03 (0.04)	0.00***	0.02**	0.14
Baseline mean HH agricultural land (ha)	0.99 (1.81)	-0.77** (0.36)	0.34 (0.35)	-0.20 (0.31)	0.00***	0.01**	0.04**
Baseline share HHs w/ crop production	0.69 (0.40)	-0.35*** (0.06)	0.25*** (0.06)	0.02 (0.05)	0.00***	0.00***	0.00***
Baseline share HHs w/ non-farm enterprise	0.78 (0.16)	0.00 (0.04)	0.00 (0.04)	-0.02 (0.03)	0.93	0.34	0.37
Baseline share HHs w/ wage employment	0.39 (0.25)	0.08* (0.05)	-0.06 (0.05)	0.01 (0.04)	0.00***	0.03**	0.01**

Note: This table presents the results from regressions of community characteristics (in each row) on dummies for flood incidence in 2012 according to different measures. The reference group is communities considered flooded under both the survey-based and satellite-based measures. Baseline household data are from the GHSP 2010-11 survey round. The difference in annual precipitation from the historical average is the total precipitation for 2012 minus the annual average from 1960-1990.

limited fluvial flood hazard, and experienced significantly more precipitation in 2012 (column β_2). These characteristics suggest that flood reports in these communities are more likely to represent pluvial floods or heavy rain-related damages that MODIS imagery will often miss. Households in these communities are also larger, more likely to be poor, and more likely to be engaged in crop production, meaning they are more vulnerable to the types of flood damages asked about in the survey.

In summary, the two flood measures identify different sets of flooded communities. While Guiteras et al. (2015) argue that survey reports of flooding in Bangladesh have little value relative to “objective” satellite-based measures because households “perceive exposure relative to their average environment” (p. 235), if we are interested in analyzing the effect of floods as an economic shock it is not obvious that a satellite measure should be preferred. In addition, we show that a measure based on MODIS satellite imagery misses a large share of reported floods, and discuss how it may also include false positives of flood incidence. The survey measure only captures floods that adversely affect households in specific ways, but it is arguably these types of events that are most relevant to study.

In this paper we focus our analysis on the impacts of survey-reported floods and test how the main effects differ by the choice of flood measure, because of the limitations in the MODIS measure. The effects we identify are of residing in a community where floods were severe enough to damage crops, property, or food security of at least two of ten surveyed households, relative to residing in a community with no survey-reported floods or floods that were less severe.

5 Empirical approach

We analyze the impacts of community exposure to floods in 2012 on household outcomes in subsequent years in a difference-in-differences framework. Specifically, we estimate two-way fixed effects regressions

$$Y_{icst} = \alpha + \beta Flood_c \times Post_t + \mu_i + \gamma_{st} + \lambda_{it} + \epsilon_{icst} \quad (1)$$

where outcome Y varies across households i , communities c , states s , and years t .²¹ *Flood* is an indicator for having been a resident in 2012 of a community exposed to floods in 2012. Defining flood exposure at the community level avoids concerns about endogeneity in which households are directly affected. Floods, perhaps more so than drought or temperature shocks, have a strong idiosyncratic element where vulnerability varies across households even within a community, creating variation in the extent to which different households in flood-exposed communities are affected. On average, 40% of households report being affected by floods in communities with at least two household flood reports (of ten total sample households per community). Analyzing floods at the community level in an ‘intent to treat’ specification captures both direct effects from households that were themselves flooded as well as any indirect impacts from proximity to flooding, though we cannot distinguish between these types of effects.²²

Post is an indicator for being observed in a survey round after 2011. The major flood events in 2012 largely occurred between August and November, at the same time as the 2012 GHSP post-planting survey.²³ As the primary outcomes we consider are reported in the post-harvest survey conducted in February-April 2013, we consider the 2012-13 round (along with the 2015-16 and 2018-19 rounds) to be post-flood. We use data from 2010-11 survey round to serve as a pre-flood comparison period and to test for baseline balance in household and community characteristics. In addition to the average effects over all post-2012 survey rounds estimated in the above specification, we also estimate dynamic effects by interacting *Flood* with indicators for each survey round. We

²¹As we consider many outcomes, we calculate Anderson (2008) adjusted q -values to correct for family-wise false discovery rates (FDR) with multiple hypothesis tests. These results are available upon request. In general, treatment effects that are significant at more than a 95% confidence level remain statistically significant after this adjustment, while other estimated effects lose statistical significance.

²²Household outcomes may be affected by proximity to flooding even without reporting direct physical effects through local infrastructure damage or expectations of flood risk, for example. Stein and Weisser (2022) find that households living in close proximity to a flood event have lower subjective well-being even if they do not report being directly affected by the flood. We therefore do not use community-level flooding to instrument for household-level flooding due to concerns about violations of the exclusion restriction.

²³One-third of sample households in the 2012-13 post-planting round were surveyed in September and 60% were surveyed in October, with the few remaining households surveyed just before or after.

have less power to estimate effects in 2018-19 after the panel sample was refreshed, so standard errors are larger on average and any significant effects represent fairly strong effects.

We include household fixed effects (μ) to control for time-invariant household characteristics which might affect livelihood decisions and 2012 flood exposure, and state-by-round fixed effects (γ) to control for common changes over time across broad geographic areas. λ_{it} are household baseline characteristic-by-round controls, where we include characteristics that are unbalanced between flooded and non-flooded communities at baseline. We test the robustness of the results to varying the set of fixed effects and controls. We cluster standard errors at the level of the community of residence in 2012 since this is the level at which the flood treatment is assigned.

This is a ‘canonical’ difference-in-differences model where all units receive the treatment – being a resident in 2012 of a community that flooded – at the same time and the control group never receives this treatment. The key assumption of this model is the parallel trends assumption, that households that resided in communities which flooded in 2012 would have experienced similar trends over time as households in communities which did not flood, if it had not been for the floods. If the occurrence of floods in 2012 were random over space, this assumption would hold in expectation, but while the timing of floods is exogenous their incidence in space is endogenous to local characteristics such as proximity to a river, elevation, slope, soil type, and other hydrological features. The household fixed effects will absorb these characteristics to the extent that they are largely time-invariant, but the analysis may still fail to recover the effects of the 2012 floods if trends in outcomes over time differ between areas at higher and lower risk of experiencing these floods.

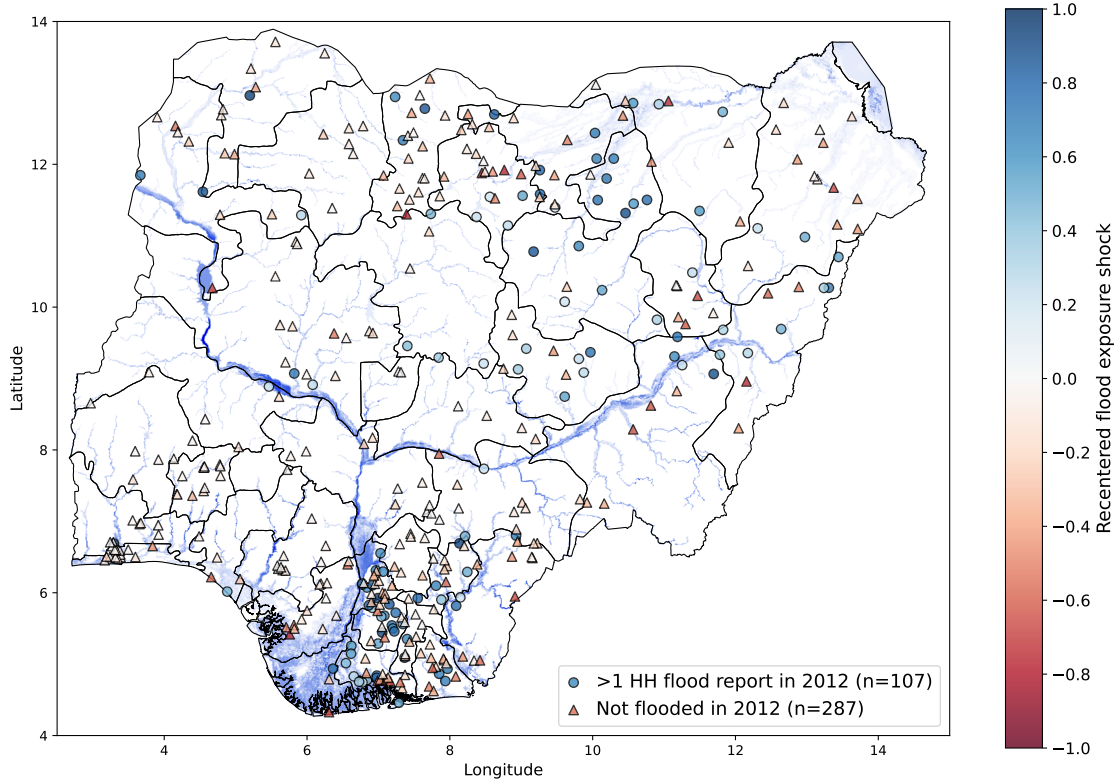
We use two approaches to address this concern. First, we balance the sample on recent reported flood incidence and underlying flood hazard (from the JRC data). We count the number of years each community was exposed to floods from 2007-2011 using household recall data, and for each strata of years we trim the sample to only include communities within the common support of the distribution of underlying flood hazard by 2012 survey-reported flood incidence. This excludes from the sample 13 communities for which there is no close 2012 flood counterfactual in terms of flood risk. For each flood measure, we also predict 2012 community flood incidence as a function of geographic, weather, and mean household characteristics, using a Lasso regression with 10-fold cross-validation (Table A1 shows the results).²⁴ We again drop communities with predicted values outside the common support of the distribution by 2012 flood incidence from the analysis.

Second, we recenter our 2012 flood treatment variable around an estimate of its expectation, following the Borusyak and Hull (2023) approach to dealing with non-random exposure to exogenous shocks. As we do not observe the distribution of potential 2012 flood realizations, we proxy the

²⁴The predicted 2012 flood likelihood does not come from a full hydrological model and may depend on the selection of predictors. The most common important predictors are distance to the nearest body of water, average 100-year flood depth, household flood reports in 2009, and the share of households with any crop activity in 2010. Interactions between 2012 rainfall deviations and both elevation and agricultural area, as well as average wettest quarter rainfall, are also selected in most predictions. Other rainfall predictors including interactions of 2012 rainfall deviations with several other geophysical characteristics are not commonly selected, consistent with prior evidence on the weak relationship between local rainfall and flood incidence.

expected realization with our predicted flood incidence. We subtract the predicted flood probability from actual flood exposure to obtain the recentered (or residualized) treatment variable. This method directly controls for observable treatment correlates, isolating the plausibly exogenous component of 2012 flood exposure to identify causal impacts of flooding. Figure 3 presents the communities in the analysis sample colored by the recentered 2012 flood shock variable for the main survey definition. A value of 1 for the recentered flood exposure exposure indicates that a community was flooded in 2012 despite a predicted flood propensity of 0, while a value of -1 indicates that a community was not flooded despite a flood propensity of 1. A comparison with Figure 2 shows which communities are dropped from the analysis sample from the balancing on recent flood history, underlying flood hazard, and predicted 2012 flood incidence.

Figure 3: Recentered 2012 flood shock by survey-reported community flood incidence



Note: The figure plots the locations of GHSP communities in the analysis sample by 2012 flood incidence and recentered flood treatment. Flood incidence is defined as at least two survey households reporting being affected by floods within a community. The recentered flood treatment subtracts the estimated 2012 flood propensity from the binary incidence. A value of 1 indicates a flooded community with an estimated flood propensity of 0, and a value of -1 indicates a non-flooded community with an estimated flood propensity of 1. The blue shading in the background corresponds to the depth of 100-year fluvial floods.

The combination of sample restrictions and recentering in a difference-in-differences framework generates doubly-robust estimates of the impacts of flood exposure. We identify effects of the 2012 floods from exogenous differences in flood incidence across communities in the same state with the same predicted flood incidence and histories of recent flood realizations. The different combined approaches may control for similar types of endogeneity in flood exposure. For example, the

household fixed effects already account for the time-invariant community characteristics included in the flood incidence prediction and the recent flooding histories. In [Subsection 6.5](#) we show that the main results are not sensitive to different sample restrictions and approaches to controlling for correlates of 2012 flood incidence. Estimated magnitudes are slightly larger with no sample restrictions and with no recentering but are not significantly different and maintain similar levels of statistical significance.

The parallel trends assumption cannot be tested but we assess its plausibility by testing for baseline balance in community and household characteristics by treatment status ([Table A2](#)). In line with the fact that flood incidence is not exogenous across space and with [Table 1](#), we find that characteristics of communities flooded in 2012 differ from those not flooded along many margins. These differences largely disappear after recentering treatment around predicted flood incidence, and we cannot reject the joint hypothesis that all characteristics are the same ($p = 0.987$). Our main specifications include baseline characteristic-by-year controls for the few characteristics where we find significant differences. In addition to balance in baseline household and community characteristics, recentering the treatment variable also leads to balance in survey-reported flood incidence across the full survey period, whereas the binary treatment variable is correlated with other flood events ([Table A3](#)). These results support the assumption that the recentered treatment variable identifies exogenous variation in 2012 flood incidence.

The final analysis sample after balancing the sample of communities and recentering the flood exposure variable includes 3,724 unique households across 389 communities. We only include households that did not leave their original survey community prior to the 2012-13 survey round. Although movers were tracked in the GHSP surveys, we cannot construct the same community-level flood measures for households outside the main sample communities in 2012. There is no significant difference in moving or attrition between 2010-11 and 2012-13 by 2012 flood exposure. Overall, 6.6% of analysis sample households – observed in both 2010-11 and 2012-13 – dropped out of the survey over the following two rounds. Attrition is slightly higher in non-flooded communities but the difference is not statistically significant ($p = 0.212$). Of the analysis sample households, only 1,105 are observed in the 2018-19 round because of the partial GHSP sample refresh. The lower sample size for 2018-19 leads to larger standard errors for estimated dynamic effects in this period.

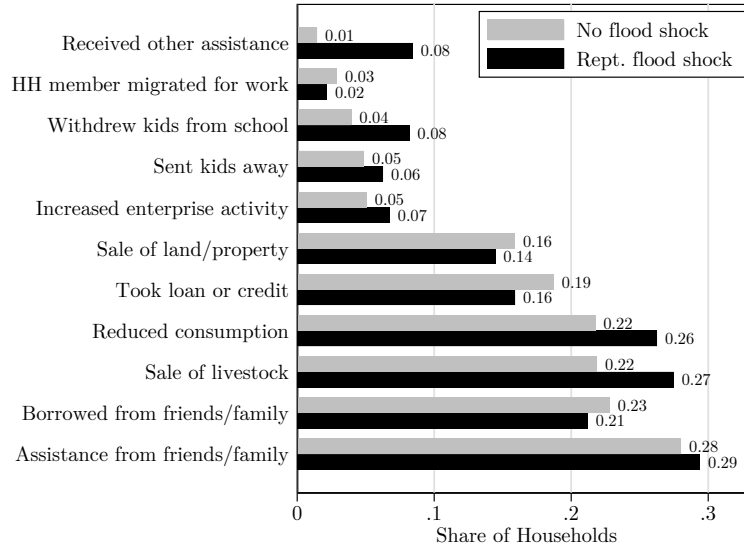
6 Results: Impacts of flood exposure

6.1 Immediate flood coping strategies

Before estimating impacts of 2012 flood exposure on household outcomes, we first explore households’ reported coping strategies. Just 2.9% of sample households have any formal insurance of any kind in 2012, meaning they must use a variety of strategies to cope with shocks. Information on these strategies comes from the household survey shocks module in the 2013 post-harvest survey, where households are asked to report what adverse shocks they have faced and to list the coping strategies they have used. Death or disability of an adult household member is the most com-

monly declared shock, followed by floods (Figure A6). Figure 4 compares the strategies reported by households that declare a flood shock against those reported by households declaring only non-flood shocks. For both groups, we consider all the coping strategies households report across all shocks they face, as many shocks are interrelated.²⁵

Figure 4: Household shock coping strategies, by reported flood exposure



Note: The figure shows the share of households reporting each type of coping strategy in 2012-2013, separately for households declaring experiencing a flood shock and for households declaring only non-flood shocks. Coping strategies are aggregated across all shocks faced by the household as many shocks are interrelated.

We find that the coping strategies reported by flooded and non-flooded households are broadly similar in terms of relative frequency.²⁶ Households declaring flood shocks are, however, significantly more likely to report receiving assistance from outside of family and friends, withdrawing children from school, and reducing their consumption, after controlling for state fixed effects. These suggest that flood shocks may be both more severe and attract more support than other types of household shocks, though even among flooded households just 8% report receiving assistance from outside their immediate network, such as government and non-profit relief. The most common coping strategies – borrowing and selling assets – may reduce household wealth in the long term and could drive persistent effects of flood exposure. We also observe that livelihood responses are not a frequent coping strategy. Few households increase enterprise activity and even fewer report any member migrating to find work in response to a shock. These results suggest long-term livelihood effects of flood exposure may be limited in this setting.

²⁵For example, households declaring a flood shock are more likely to also declare non-flood harvest destruction, death of livestock to illness, and adverse price shocks than households declaring shocks other than floods, even after including state fixed effects (Table A4).

²⁶We set aside responses of ‘did nothing’ which are the most common for both groups, at 40% and 36% of households, respectively.

6.2 Household consumption and assets

We now consider impacts of 2012 community flood exposure on a series of household outcomes. Table 2 shows that households in communities with at least two household reports of floods in 2012 are 5 percentage points (pp) more likely to report experiencing any food insecurity over the last 12 months on average over the three post-flood survey rounds than households in non-flooded communities, a 28% increase. In line with this and with reported decreases in consumption in Figure 4, daily household consumption per capita falls by 0.68 USD (13%). This decrease is driven largely by decreased consumption on food items which falls by 0.66 USD (15%), potentially explaining the increase in food insecurity, though consumption on non-food items also falls by 0.18 USD (10%). The decrease in consumption per capita does not increase the probability that households are below the 1.90 USD PPP daily consumption per capita poverty line. Although sales of property and livestock are commonly-reported shock coping strategies, we find no significant average effects of 2012 flood exposure on the total value of household assets or on livestock holdings.

Table 2: Impacts of 2012 community flood incidence on household consumption and assets

	N	Control Mean (SD)	2012 flood recentered × Post (SE)	2012 flood recentered × 2013 round (SE)	2012 flood recentered × 2016 round (SE)	2012 flood recentered × 2019 round (SE)
Any food insecurity in last 12 months	11951	0.18 [0.38]	0.05** (0.02)	0.07*** (0.02)	0.02 (0.03)	0.01 (0.04)
Daily HH consumption per capita (USD)	11951	5.28 [4.23]	-0.68*** (0.21)	-0.88*** (0.27)	-0.45* (0.26)	-0.74** (0.31)
Daily HH food cons. per capita (USD)	11951	4.35 [4.50]	-0.66*** (0.18)	-0.54*** (0.20)	-0.83*** (0.20)	-0.43 (0.29)
Daily HH non-food cons. per capita (USD)	11951	1.82 [2.63]	-0.18* (0.10)	-0.14 (0.11)	-0.21* (0.13)	-0.21 (0.13)
HH under 1.90 USD PPP per capita daily poverty line	11951	0.37 [0.48]	0.01 (0.02)	0.02 (0.02)	-0.00 (0.03)	0.04 (0.04)
Value of HH assets (USD)	11951	1966.60 [4708.85]	-109.17 (147.78)	-322.42* (186.71)	119.46 (170.38)	-105.50 (296.95)
Livestock holdings (TLU)	11951	2.20 [36.37]	0.95 (1.01)	0.68 (1.06)	1.28 (1.03)	0.71 (0.74)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. Control means are for communities with no flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. Flood exposure is defined as at least two households in the community reporting being affected by floods in 2012, and is recentered around predicted community incidence. Estimates therefore represent the effect of residing in a community that was exposed to floods 2012 for a given predicted propensity of incidence. All regressions include household and state-by-round fixed effects as well as controls for baseline characteristics-by-round for characteristics not balanced in Table A2. Standard errors are clustered at the level of the community of residence in 2012. Each row represents an outcome. Daily household consumption outcomes are calculated on the basis of reported consumption across different goods and services, and are provided directly in the publicly-available GHSP data. Poverty is calculated based on aggregated consumption. The value of household assets is the sum of declared values across all assets owned. Livestock are measured in tropical livestock units (TLU) in order to aggregate across species. Monetary values are in 2016 PPP USD. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Turning to dynamic impacts, we find that the value of household assets does fall significantly in 2013 (by 16%) but recovers by 2016, indicating that households find ways to replenish their assets. Household food insecurity is also only significantly larger in 2013, when the fall in consumption per capita is largest. This pattern is consistent with floods represent a severe immediate shock and households using coping strategies that deplete available resources. These results build on previous work on household flood impacts in African countries showing short-term decreases in food security and consumption (Amolegbe et al., 2023; Baez et al., 2020; Devereux, 2007; Reed et al., 2022), and similar results for an analysis of effects of natural disasters in Nigeria (Escalante et al., 2025).

Over the longer term, although the decreases in food security and value of assets do not persist, the decrease in daily consumption per capita remains statistically significant in all post-flood survey rounds. This indicates that community flood exposure makes households significantly and persistently worse off on average, even accounting for the fact that over half of households sampled in flooded communities do not report any direct adverse impacts of floods. These households may be indirectly affected by local flooding, or affected in ways not captured by the survey questions around flood damages.

6.3 Household composition and livelihood activities

Prior studies have found increases (Balboni et al., 2023; Berlemann et al., 2023; Patel, 2024; Pavel et al., 2023), decreases (Chen and Mueller, 2019; Chen et al., 2017), and null or mixed (Gray and Mueller, 2012; Maystadt et al., 2016; Mueller et al., 2014) effects of flood exposure on short-term out-migration. Consistent with few households reporting migration as a shock coping strategy in this sample (Figure 4), Table 3 shows no average effects of 2012 flood exposure on the probability that any individual member left the household. This aligns with a study of the impacts of temperature shocks on migration in Northern Nigeria (Dillon et al., 2011), which finds null average effects but that frequent heat shocks increase the probability of male migration, suggesting migration may be a response to particular types of shocks. We also find no effect on the likelihood any adult joined the household, on the count of household members, or on the count of household members ages 0-4, the latter of which implies no effects on household fertility decisions.

By construction, the sample only includes households that remained in the same community as in 2010-11 at the start of the 2012-13 survey. A marginally significant but effectively 0 effect on the probability that the household moved in 2012-13 therefore implies no impact of floods on the probability that the household moved between the 2012-13 post-planting and post-harvest survey rounds.²⁷ By the time of the 2018-19 survey round six years after the floods, however, households in exposed communities are 4 pp more likely to have moved. This is a large effect relative to the fact that just 15% of households retained in the panel by the 2018-19 round had moved since the initial 2010-11 survey.

Though there are no significant effects on the counts of household members in any round, we also find some delayed effects of flood exposure on the probability of households moving and members leaving or joining the household. New adults are 3 pp more likely to have joined households exposed to floods by the time of the 2015-16 round, and flood exposure increase the probability that any household member left for work by 9 pp by the time of the 2018-19 round. The latter effect is large compared to the 12% of all panel households had any member leave in search of work by the 2018-19 round, though it does not drive an increase in the overall probability that any member left the household.

²⁷Although initial displacement is a common consequence of flooding, fieldwork we conducted in Nigeria’s Jigawa State after the 2022 floods showed that nearly all households return to the same community if not the same dwelling within 6 months of being displaced.

These results indicate that individual migration is not a common short-term response to floods in Nigeria, but the positive long term effects – also reported in Patel (2024) – suggest that this could be a result of individual and household migration constraints. In particular, the short-term decreases in household consumption and assets following flood exposure may reduce resources available to support migration.

Table 3: Impacts of 2012 community flood incidence on household migration, composition, and livelihood activities

	N	Control Mean (SD)	2012 flood recentered × Post (SE)	2012 flood recentered × 2013 round (SE)	2012 flood recentered × 2016 round (SE)	2012 flood recentered × 2019 round (SE)
HH moved since first survey round	11951	0.00 [0.00]	-0.00 (0.00)	-0.00* (0.00)	-0.01 (0.01)	0.04* (0.02)
Any member left HH during/since last round	11951	0.08 [0.27]	0.01 (0.02)	-0.02 (0.02)	0.03 (0.02)	-0.01 (0.04)
Any member left HH for work during/since last round	11951	0.02 [0.13]	-0.00 (0.01)	-0.02 (0.01)	-0.00 (0.02)	0.09** (0.04)
Any member left HH for family during/since last round	11951	0.03 [0.17]	-0.02 (0.02)	-0.02 (0.02)	-0.02 (0.03)	-0.04 (0.05)
Any adult member joined HH during/since last round	11951	0.07 [0.25]	0.01 (0.01)	-0.00 (0.01)	0.03** (0.01)	-0.00 (0.03)
Count of HH members	11951	5.58 [2.97]	0.01 (0.06)	-0.06 (0.06)	0.07 (0.08)	0.10 (0.24)
Count of HH members ages 0-4	11951	0.93 [1.12]	-0.02 (0.04)	-0.01 (0.04)	-0.05 (0.05)	0.17 (0.12)
Any HH farm activity	11951	0.73 [0.44]	0.03* (0.01)	0.02 (0.02)	0.03* (0.02)	0.03 (0.04)
Any HH non-farm enterprise activity	11951	0.75 [0.43]	0.02 (0.02)	-0.01 (0.02)	0.03 (0.02)	0.05 (0.04)
Any wage employment activity	11951	0.41 [0.49]	0.00 (0.02)	0.02 (0.03)	-0.01 (0.02)	0.01 (0.04)
Count of HH members wkg. in HH farm	11951	1.87 [2.14]	0.08 (0.10)	-0.06 (0.10)	0.19 (0.13)	0.34* (0.20)
Count of HH members wkg. in HH non-farm enterprise	11951	1.22 [1.37]	0.00 (0.06)	0.01 (0.06)	-0.02 (0.07)	0.06 (0.15)
Count of HH members wkg. in wage employment	11951	0.78 [1.40]	0.09 (0.06)	0.11 (0.08)	0.06 (0.06)	0.22** (0.10)
Count of HH members wkg. in any work	11951	2.72 [2.02]	0.03 (0.10)	-0.07 (0.09)	0.08 (0.13)	0.33* (0.19)

Note: This table presents effects of 2012 community flood exposure on household outcomes in subsequent survey rounds, following Table 2. Each row represents an outcome. Household moving is measured in reference to their location of residence during the initial survey in 2010, and includes moves both within and outside of the baseline community. Changes in household composition are measured in reference to the previous survey round, as the household roster is updated in each survey and respondents can indicate which members have left and if any new members have joined. Counts of household members working in different activities are based on the household labor and agricultural labor modules in the post-planting and post-harvest surveys where respondents indicate which members are engaged in different activities. We code each member as engaged in a given activity for a survey round if they were engaged in it during either the post-planting or post-harvest round, then take the sum of members in each activity for the household. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The literature on floods fairly consistently finds that flood exposure increases short-term engagement in non-farm labor (Akter, 2021; Chen et al., 2017; Gray and Mueller, 2012; Maystadt et al., 2016; Patel, 2024; Vitellozzi and Giannelli, 2023), with the exception of Mueller and Quisumbing (2011) and Sajid (2023). We do not find clear evidence of such effects in this context. There is no average or immediate effect of 2012 flood exposure on household engagement in either household non-farm enterprise (NFE) or wage employment on the extensive margin. There is also no intensive margin effect on the count of household members working in household NFEs in any round (in line with Figure 4). The average effect on the count of member in wage employment is close to being

statistically significant at conventional levels ($p = 0.120$), and is large and significant in 2018-19. We find no significant effects of flood exposure on either community survey or household reports of average wages paid for agricultural labor (Table A13). Reported wages earned by household wage workers are marginally significantly lower during the post-planting period but not during the post-harvest period. These results on average wages suggest that any increase in the supply of wage workers following flood exposure is not sufficient to bring down equilibrium wages.

We also find no evidence of households exiting agriculture. If anything, households in flooded communities are 3 pp (4%) *more* likely to engage in crop or livestock production on average after flood exposure, though this is only marginally statistically significant on average and in 2015-16. The average effect on the count of household members working in household farm activities is not statistically significant, but this masks a negative point estimate for 2012-13 followed by positive estimates in 2015-16 ($p = 0.166$) and 2018-19 ($p = 0.086$). Around 1 in every 4 flood-exposed households has an additional member engaged in farm work over those later two survey rounds, an 11% increase, but these effects do not surprise adjustments for multiple hypothesis testing. Smaller magnitude effects on the count of household members engaged in any work activity and no significant decrease in the count engaged in household NFE imply that to the extent engagement in farm and wage work increases it is in large part among members already engaged in other activities, indicating an intensification of household labor supply. This intensification implies a reduction in leisure time, another way in which flood exposure may decrease household well-being.

To relate these results to the few prior studies of long-term effects of floods on labor supply, the positive but largely non-significant effects on wage work are closest to Mueller and Quisumbing (2011) who find null effects of floods on non-agricultural employment in Bangladesh and to Patel (2024) who finds positive effects in Bangladesh, and contrast with Sajid (2023) who reports negative effects in rural areas of India. On the other hand, the null or delayed effects on agricultural labor hours are more in line with Sajid (2023) who shows increased engagement in agriculture in India, in contrast to Patel (2024) who finds that floods push employment out of agriculture in Bangladesh. In general, the results do not align well with any prior study, with weak evidence of any changes in labor supply following flood exposure, even in the short term.

6.4 Income sources and diversification

The central question of this paper is whether a weather-related natural disaster – in this case, floods – can promote sustained shifts of livelihood activities away from more vulnerable household agriculture. We find that community exposure to Nigeria’s 2012 floods did not increase individual or household migration (at least in the short-term) or consistently affect household member labor supply over the following seven years, with only suggestive evidence of increased household farm production and wage employment activity (Table 3). The largely null effects of flood exposure on labor supply and livelihood activities suggests we should not expect meaningful diversification of household income sources.

Table 4 presents the results from directly testing the effects of community flood exposure in

2012 on household incomes or value of production from different sources and on relative income shares. We drop 807 sample observations with missing data for income or production value for all activities they are engaged in, though results are qualitatively similar if they are included.²⁸ Although flood exposure persistently decreases household daily per capita consumption, we find no significant effect on total household income and production value on average or in any survey round. This is largely because estimated effects on non-farm income are very noisy, though the point estimates are positive except for the effect in the 2015-16 round. The total value of household farm – crop and livestock – production decreases significantly and persistently (though the effect in 2018-19 is imprecise, $p = 0.142$). On average, income from household farm activities decreases by 485 USD (26%) per year. The largest decrease comes immediately after the floods, and is likely explained by crop destruction, a key input into the survey definition of flood exposure. But the decreases remain similar in magnitude in the following years, indicating sustained effects beyond the initial destruction.

Table 4: Impacts of 2012 community flood incidence on incomes and income shares by livelihood activity

	N	Control Mean (SD)	2012 flood recentered × Post (SE)	2012 flood recentered × 2013 round (SE)	2012 flood recentered × 2016 round (SE)	2012 flood recentered × 2019 round (SE)
Total HH income and production value (USD 100s)	11144	82.86 [177.76]	-4.00 (7.73)	-0.75 (8.97)	-8.48 (7.85)	2.48 (10.70)
Total value of HH farm production (USD 100s)	11144	18.79 [37.48]	-4.85*** (1.80)	-5.50*** (2.13)	-4.26** (1.97)	-4.32 (3.20)
Total income from non-farm activities (USD 100s)	11144	64.07 [176.24]	0.84 (7.57)	4.75 (8.75)	-4.22 (7.77)	6.80 (10.24)
Total crop production value (USD 100s)	11144	16.43 [33.15]	-6.01*** (1.67)	-6.13*** (2.06)	-6.09*** (1.72)	-4.87 (3.11)
Total non-farm HH enterprise income (USD 100s)	11144	26.59 [81.05]	0.55 (3.28)	3.32 (3.51)	-3.55 (3.77)	7.72 (4.83)
Total wage employment income (USD 100s)	11144	34.60 [154.85]	0.19 (6.56)	0.64 (7.55)	-0.12 (6.73)	-0.67 (9.34)
Total HH income from other activities (USD 100s)	11144	2.88 [19.57]	0.10 (1.10)	0.79 (1.15)	-0.54 (1.26)	-0.25 (2.12)
HH farm share of total HH income and prod value	11144	0.47 [0.43]	-0.02 (0.02)	-0.04** (0.02)	-0.00 (0.02)	-0.02 (0.03)
NFE share of total HH income and prod value	11144	0.32 [0.39]	0.03 (0.02)	0.04* (0.02)	0.01 (0.02)	0.01 (0.04)
Wage share of total HH income and prod value	11144	0.17 [0.34]	-0.00 (0.01)	0.01 (0.02)	-0.01 (0.01)	-0.00 (0.02)
Other share of total HH income and prod value	11144	0.03 [0.15]	-0.00 (0.01)	-0.00 (0.01)	-0.00 (0.01)	0.00 (0.03)
HH income Herfindahl-Hirschman Index	11144	0.84 [0.20]	0.00 (0.01)	0.00 (0.01)	0.00 (0.01)	-0.01 (0.02)

Note: This table presents effects of 2012 community flood exposure on household outcomes in subsequent survey rounds, following Table 2. Each row represents an outcome. Section 3.1 describes how the income and value of production variables are constructed. Farm production includes crop and livestock activities. ‘Other activities’ includes income from pensions, remittances, investments, transfers, etc. Households not active in a given activity are assigned an income of zero for that activity. We drop households with zero total income and production value from the analysis as these likely represent missing data rather than true zeros. Table A5 shows the results including these households. The household income Herfindahl-Hirschman-Index (HHI) is the sum of squared shares of income from farm production, NFE, wage work, and other sources. Values closer to one indicate a greater concentration of income sources. Monetary values are in 2016 PPP USD. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

²⁸We describe our approach to calculating value of production for household farm activities and incomes from all other activities in Section 3.1. Across all survey rounds we find 807 sample household observations (6.7%) with zero total income and value of farm production, with no significant difference by 2012 flood exposure. These households do report being engaged in farm production (43%), NFE (51%), and wage employment (23%). Most of these zeros are therefore likely the result of missing values from refusal to answer, inability to estimate, or coding errors.

In line with crop destruction being the key driver, we find that the value of crop production falls by 601 USD (37%) per year on average after flood exposure. Smaller decreases in total farm income suggest that reductions in crop production value are slightly offset by increases in livestock production value. Looking at specific sources of non-farm income, there are no significant effects of flood exposure on income from NFE, wage employment, or other non-wage and non-enterprises sources on average or in any period. The point estimates for effects on NFE income are noisy but positive and fairly large in 2012-13 and 2018-19, which explains the null effects of flood exposure on total household income.

If households whose crop production value falls after flood exposure are active in other livelihood activities, this could lead to shifts in the income shares of different activities even if the households are not increasing labor supply to other activities to cope with or adapt to flood exposure. The median and mean sample household is active in two out of farm production, NFE, and wage employment, and [Table 3](#) indicates that this does not change following flood exposure. On average, household income shares appear quite diversified, with 47% to farm production, 32% to NFEs, 17% to wage employment, and 3% to other sources. But the mean household Herfindahl-Hirschman Index value of 0.84 – where 1 indicates all income is from a single activity – reveals much greater concentration of incomes at the household level. The mean of the maximum income share across all activities is 87% for sample households in 2010-11, but the median is 96%. While income shares are diversified across households on average, most individual households are highly specialized with very concentrated incomes even if engaging in multiple livelihood activities is common.

Consistent with highly-concentrated household livelihoods, [Table 4](#) shows no effects of the 2012 floods on household income concentration on average or in any survey round. This could appear to contrast with a study of the impacts of harmful degree days in Nigeria using the same survey data, which finds that same-year decreases in agricultural productivity decrease the household crop income share and increase the share from livestock and wage employment (Amare and Balana, 2023). But that study only considers immediate impacts, and we also find a marginally significant decrease in the agricultural income share in the year the floods occurred, which is offset by an increase in the NFE income share. In addition, we combine crop and livestock production value, though results are similar when consider crop income separately. The farm share of income is not significantly lower in subsequent rounds even as the decrease in the value of household farm production remains persistent. This indicates that the most affected households are not offsetting these production decreases.

6.5 Robustness

The main results are robust to a variety of alternative specifications. As we consider a large number of outcomes, we focus our robustness checks on a subset of key well-being, income, labor supply, and crop production outcomes. First, we show that statistical significance is largely not sensitive to the choice of standard error clustering ([Table A6](#)). SEs are nearly identical when clustering at the LGA level and typically slightly smaller with two-way community and year clustering and with

Conley (1999) SEs allowing for spatial autocorrelation within 50 or 100 km. Wild bootstrapped SEs clustered at the state level also generate similar patterns of statistical significance.

Next, we consider changes to the controls and fixed effects (Table A7). Estimates are nearly unchanged when controlling for annual precipitation deviations from historical means at the community level, indicating we are indeed capturing the effects of flood shocks and not of any associated precipitation shocks. Dropping the baseline characteristic-by-round controls and shifting from state-by-round fixed effects to round or zone-by-round fixed effects does not qualitatively affect the results. Some results change when using LGA-by-round fixed effects but these impacts are identified from just nine LGAs with variation in flood incidence across communities.

We also consider different sample restrictions (Table A8). The main analysis sample stratifies communities by years of survey-reported flood incidence from 2007-2011 and drops communities outside the range of common support for average 100-year flood depth within 5 km. Results are nearly unchanged if we include all communities and if we instead drop communities more than 50 km from any community with a 2012 flood report. This is likely because the construction of the recentered flood shock variable already drops communities outside of the common support of estimated 2012 flood propensity by actual flood incidence, making the additional sample restrictions less meaningful. The results are largely similar if we exclude control communities within 20 km of or within the same LGA as any flooded community, which could potentially have been indirectly affected by nearby flooding. The effect on the probability of any farm activity loses its marginal statistical significance, while an increase in the count of household members working in wage employment becomes statistically significant. These results indicate that the effects are driven by direct flood exposure and not by localized spillovers.

Estimated flood impacts are similar focusing just on the sample of households with any farm activity during the survey (78% of households). The decrease in farm production value is larger in magnitude while the decrease in consumption per capita is smaller, and we find a significant increase in the count of members engaged in wage work. The magnitudes of adverse effects of flood exposure on food insecurity, consumption per capita, and farm production value are larger in the sample of communities with non-0 fluvial flood hazard, but we do not find any change in which variables are significantly affected by flood exposure. We analyze heterogeneity in flood impacts by community and household characteristics in more detail below.

We next compare alternative approaches to controlling for differences in 2012 flood propensity besides recentering the flood incidence treatment variable (Table A9). We first replace the recentering with 2012 flood propensity-by-round or flood hazard-by-round and local 2012 rainfall deviation-by-round fixed effects, and then consider stripped down models with fewer controls and sample restrictions. The increase in food insecurity loses statistical significance in some specifications, while the magnitudes of the decrease in consumption of capita and the increase in the probability of any farm activity are larger across all alternative specifications. While some specifications show significant increases in the count of household members working on the household farm, all alternative specifications show significant increases in the count of members engaged in

wage employment, in contrast to non-significant effects in the main specification. Another notable difference is that when we do not control for or restrict the sample based on the predicted 2012 flood propensity, we find large and significant increases in the total family crop labor hours, in line with the extensive margin effects on household farm engagement. These tests show that accounting for 2012 flood propensity – the non-random component of the flood shock – meaningfully changes some of the results relative to a naive difference-in-difference analysis treating the 2012 floods as completely exogenous.

Finally, we conduct two tests to assess whether the results identify the effects of realized flood exposure separately from underlying flood propensity. We first estimate the effects of estimated 2012 flood propensity using the same empirical specification but looking at the interaction of flood propensity and *Post* rather than the interaction of the recentered flood exposure shock and *Post*, among non-flooded communities. We find no significant effects of predicted 2012 flood propensity on food insecurity, consumption, or farm production value after 2012, but some outcomes – such as farm engagement and crop area – do exhibit differential trends associated with flood propensity (Table A10). This indicates that flood-prone areas may experience distinct trends even absent realized flooding, underscoring the importance of isolating the unexpected component of flood exposure. Consistent with this interpretation, specifications that do not directly account for estimated flood propensity yield larger changes in farm activity (Table A9), highlighting the role of our identification strategy which recenters treatment around flood propensity.

We also test whether the results are driven by the allocation of treatment across flood-prone communities by implementing a permutation test that reassigns flood exposure within bins of predicted flood propensity while preserving the number of treated communities in each bin. Focusing on daily per capita consumption and farm production value, we find that the resulting placebo distributions are centered near zero (Figure A5). The actual estimated effects lie in the extreme tails of these distributions and would be very unlikely under random reassignment of treatment within flood risk bins: we estimate randomization p values of 0.004 for daily per capita consumption and 0.013 for the value of farm production. These results suggest that the observed effects are unlikely to be driven primarily by baseline differences across flood-prone areas but rather reflect realized flood exposure, consistent with a framework in which identification comes from realized exposure net of predicted flood risk.

In summary, the robustness tests show the importance of our shock recentering approach and that the main significant results are otherwise fairly robust to alternative specification and sample choices. Increases in family farm labor workers are significant only in models that do not control in any way for 2012 flood propensity, highlighting the importance of directly controlling for this given differential trends over time by flood propensity. The main outcome that is sensitive to the choice of specification is the increase in the count of household members engaged in wage employment, where $p = 0.120$ in the main specification but we find larger and statistically significant effects in certain models.

7 Implications of flood measurement decisions

Figure 1 and Figure A3 show that different approaches to measuring flood incidence identify different communities as being exposed to floods in 2012. Table 5 therefore presents the impacts of 2012 flood exposure using different flood incidence definitions but retaining the same sample restrictions and recentering approach after estimating definition-specific 2012 flood propensities. We do not include any baseline characteristic-by-round controls in these regressions, as the set of baseline characteristics that differ by flood exposure varies with the flood measure used, but Table A7 shows that not including controls does not greatly affect the main results.

7.1 Comparing results across survey-based definitions

Moving from requiring two household survey flood reports in a community (column 1) to any household report (column 2) to define a community as exposed generally weakens the results. Decreases in consumption per capita and value of farm production become smaller in magnitude and less statistically significant, while the increases in food insecurity the count of household wage workers lose statistical significance. Additionally including communities with any community survey flood reported as exposed to floods (column 3) weakens the effect on household farm engagement but strengthens effects on consumption per capita, value of farm production, and household members working for a wage relative to a definition based on household reports along (column 2). This broadest possible survey-based flood definition also results in a marginally significant decrease in household NFE income. These differences imply that the main survey flood definition captures more severe floods, though the less restrictive definitions likely also include more measurement error. The decrease in NFE income captured when including community survey flood reports may reflect a limitation of the main measure based on household survey reports where only the questions of flood-related property loss and food insecurity may indirectly capture effects of floods on NFE.

We also test how defining flood exposure based only on specific household survey questions – dropping communities with floods reported in other survey questions – affects the estimated impacts. A common approach to studying household shocks using survey data is to rely on household reports from specific shocks modules. The GHSP shocks module asks respondents if they experienced floods that caused harvest failure or floods that causes property loss. If we define community flood exposure based only on household reports of floods in the shocks module (column 4), the effects of the 2012 floods on the value of farm production and farm engagement are no longer statistically significant and we find a larger increase in the count of household wage workers, relative to a definition using all household survey questions (column 2). Null effects on food insecurity and consumption may stem from no longer considering household reports of flood-related food insecurity in the flood measure, while null effects on farm production value may be due to not considering reports of flood-related crop losses (short of complete failures) in the agricultural modules. These differences show that the choice of how survey respondents are asked about floods matters. Focusing just on a few specific ways that floods can affect households leads to different conclusions

Table 5: Sensitivity to alternative ways of measuring 2012 flood incidence

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	>1 HH flood rept all modules	Any HH flood rept, all modules	Any HH or comm flood rept	Any HH flood rept, shocks module	HH non-crop or comm. flood rept	<5 km from MODIS flood	<10 km from MODIS flood
Any food insecurity in last 12 months	0.04** (0.02)	0.02 (0.02)	-0.01 (0.02)	0.03 (0.02)	0.00 (0.02)	0.02 (0.03)	-0.01 (0.02)
Daily HH consumption per capita (USD)	-0.69*** (0.22)	-0.56** (0.23)	-0.48*** (0.18)	-0.58* (0.35)	-0.42** (0.20)	0.23 (0.30)	-0.37* (0.21)
Total value of HH farm production (USD 100s)	-4.63*** (1.74)	-3.25* (1.80)	-4.00** (1.77)	-1.87 (2.27)	-3.70 (2.41)	-3.85** (1.82)	-0.60 (1.79)
Total non-farm HH enterprise income (USD 100s)	0.06 (3.24)	-4.07 (3.20)	-5.04* (2.78)	-3.64 (3.94)	-1.71 (2.82)	6.92* (3.99)	2.44 (3.73)
Total wage employment income (USD 100s)	-0.38 (6.53)	-1.93 (5.05)	2.02 (5.01)	-0.28 (5.00)	0.87 (5.54)	-1.72 (7.80)	9.73* (5.70)
Any HH farm activity	0.05*** (0.01)	0.03*** (0.01)	0.02* (0.01)	0.02 (0.02)	0.01 (0.01)	-0.01 (0.02)	0.01 (0.01)
Any HH non-farm enterprise activity	0.02 (0.02)	0.01 (0.02)	0.02 (0.02)	-0.03 (0.02)	0.03 (0.02)	0.01 (0.02)	-0.01 (0.02)
Any wage employment activity	0.01 (0.02)	0.01 (0.02)	0.02 (0.02)	0.03 (0.03)	0.01 (0.02)	0.00 (0.02)	-0.01 (0.02)
Count of HH members wkg. in HH farm	0.11 (0.09)	0.01 (0.08)	0.07 (0.08)	0.04 (0.12)	0.06 (0.07)	-0.03 (0.11)	-0.08 (0.08)
Count of HH members wkg. in HH non-farm enterprise	0.01 (0.06)	0.01 (0.05)	0.03 (0.05)	-0.07 (0.06)	0.02 (0.05)	0.05 (0.05)	0.02 (0.05)
Count of HH members wkg. in wage employment	0.12** (0.06)	0.09 (0.05)	0.14*** (0.05)	0.16** (0.07)	0.11** (0.05)	-0.01 (0.07)	-0.12* (0.07)
Total area planted (ha)	-0.07 (0.20)	-0.03 (0.18)	0.10 (0.17)	0.02 (0.24)	0.11 (0.14)	0.28 (0.21)	0.30* (0.17)
Total family crop labor hours in past year	98.00 (140.14)	99.84 (130.43)	152.26 (130.05)	220.30 (185.75)	222.83* (134.11)	82.37 (145.03)	170.91 (113.85)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. Each row represents an outcome; see the main analysis tables for explanations of the variables. Monetary values are in 2016 PPP USD. Each column indicates how 2012 community flood incidence is defined. Figure A3 shows maps of flood incidence according to the different definitions. For each flood measure, we estimate the 2012 flood propensity following the same approach as for the main flood definition and use this to recenter the community flood exposure treatment variable. All regressions include household and state-by-round fixed effects and the main analysis sample that is balanced on underlying flood hazard by 2007-2011 survey flood report strata, but do not include any baseline characteristic-by-round fixed effects. This explains differences between the results in column (1) and those in the preceding tables. Column (1) presents the results using the main definition of at least 2 household flood reports for 2012 within a community. In column (2) we only require 1 household flood report to define community flood incidence. In column (3) we define a community as flooded if there is any household flood report in any module or any flood report in the community survey, the broadest possible survey-based definition. In column (4) we only consider flood reports in the household shocks survey module, ignoring reports in the agriculture and food security modules. Column (5) follows column (3) but excludes communities where floods are only reported in the household agriculture module or crop failure shock questions. In both columns (4) and (5), we drop from the analysis communities with other survey flood reports so that they are not included among the control communities. Columns (6) and (7) use MODIS satellite imagery from the NASA NRT Global Flood Mapping product rather than survey reports to define flood incidence. Communities are considered flooded if there is any pixel classified as flooded at any point in 2012 within 5 and 10 km of the community centroid, respectively. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

about the impacts of flood exposure when relying on survey reports to define flood incidence, even if many estimated effects are similar. This has important implications for studies of other shocks measured based on specific survey questions, and particularly for interpreting the estimated effects of such shocks.

We note that the large decreases in farm production value we estimate may be partly mechanical due to the definition of flood exposure based on household reports on crop failure or losses. We therefore construct another survey flood measure based only on household questions about flood-related food insecurity and property loss and on community survey flood reports (column 5). As with the main flood measure, we find that floods significantly decrease consumption per capita and increase the count of members working for a wage. Effects on food insecurity or engagement in

farm work are no longer significant, but we find a marginally significant increase in engagement in household NFE. Importantly, the decrease in farm production value is also no longer statistically significant at conventional levels ($p = 0.126$) but remains fairly large in magnitude. These differences indicate that floods adversely affect households even where they do not result in crop failure, but that the worst effects appear to be associated with floods that cause severe crop losses.

7.2 Comparing results between survey- and satellite-based definitions

While there are some differences in results across survey-based flood exposure definitions, they also frequently align which is not surprising considering the strong correlations between these definitions. But survey- and satellite-based measures are poorly correlated, and [Table 1](#) shows that these measures identify flood incidence in communities with different characteristics. In line with this, estimated effects of the 2012 floods as identified by MODIS satellite imagery ([Table 5](#) columns 6 and 7) are considerably different from those using a survey-based definition. With a 5 km buffer we find only a decrease in the value of farm production and a marginally significant but large increase in NFE income. The point estimate on consumption per capita becomes positive and non-significant, and we no longer find increases in farm activity or the count of wage workers. The estimates are almost completely different with a 10 km buffer. Here we do observe a decrease in consumption per capita but find no effects on farm production value or NFE income and instead a large and marginally significant increase in wage income, despite a marginally significant decrease in the count of household wage workers. It is difficult to interpret what effects this 10 km buffer definition is identifying given its very low correlation with all survey and other measures of 2012 flood exposure ([Figure 1](#)), and none of these effects survive adjustments for multiple hypothesis testing.

Focusing on the 5 km buffer, we strikingly find no significant impacts of being in a community with flooding identified within 5 km by MODIS imagery in 2012 on any of the main household outcomes in 2013, immediately after the floods ([Table 6](#)). Significant average effects of this flood measure on value of farm production and NFE income are driven by significant effects in the 2015-16 survey round. This strongly indicates that this measure of flood incidence is not accurately identifying adverse flood events as experienced by survey households and communities, and validates our focus on a survey-based measure. Null immediate effects are consistent with measurement error from MODIS imagery’s inability to detect flooding covered by clouds or narrow in geographic extent and potential false positives from seasonal water fluctuations.

The difference in results between survey- and satellite-based definitions, if taken at face value, would lead to very different conclusions about the effects of flood exposure. Survey-based measures generally agree that flood exposure makes households persistently worse off in terms of consumption per capita on average over the following years, and that this is driven by reductions in the value of farm production that are not offset by returns to increased labor supply in wage employment. In contrast, the 5 km satellite buffer measure would suggest that households are largely resilient to flood shocks and offset decreases in farm production value by increasing income from household

Table 6: Impacts of satellite-detected 2012 community flood incidence

	N	Control Mean (SD)	2012 MODIS recentered × Post (SE)	2012 MODIS recentered × 2013 round (SE)	2012 MODIS recentered × 2016 round (SE)	2012 MODIS recentered × 2019 round (SE)
Any food insecurity in last 12 months	10304	0.19 [0.39]	0.02 (0.03)	0.02 (0.03)	0.02 (0.03)	0.07 (0.07)
Daily HH consumption per capita (USD)	10304	5.32 [4.48]	0.23 (0.30)	0.52 (0.39)	-0.13 (0.32)	0.49 (0.44)
Total value of HH farm production (USD 100s)	9541	18.77 [37.62]	-3.85** (1.82)	-2.73 (2.06)	-5.05** (2.05)	-3.44 (2.90)
Total non-farm HH enterprise income (USD 100s)	9541	26.32 [79.14]	6.92* (3.99)	5.19 (4.12)	9.66* (4.95)	1.78 (5.44)
Total wage employment income (USD 100s)	9541	36.75 [173.45]	-1.72 (7.80)	1.83 (8.89)	-4.07 (9.87)	-8.01 (18.56)
Any HH farm activity	10304	0.75 [0.43]	-0.01 (0.02)	-0.02 (0.02)	-0.00 (0.02)	0.06 (0.06)
Any HH non-farm enterprise activity	10304	0.75 [0.43]	0.01 (0.02)	-0.00 (0.03)	0.01 (0.02)	0.07 (0.05)
Any wage employment activity	10304	0.40 [0.49]	0.00 (0.02)	0.02 (0.03)	-0.01 (0.03)	-0.04 (0.04)
Count of HH members wkg. in HH farm	10304	1.95 [2.19]	-0.03 (0.11)	-0.06 (0.10)	-0.07 (0.14)	0.42** (0.18)
Count of HH members wkg. in HH non-farm enterprise	10304	1.29 [1.37]	0.05 (0.05)	0.06 (0.06)	0.01 (0.06)	0.26* (0.14)
Count of HH members wkg. in wage employment	10304	0.76 [1.37]	-0.01 (0.07)	0.01 (0.09)	-0.04 (0.07)	0.06 (0.08)
Total area planted (ha)	10304	1.76 [3.43]	0.28 (0.21)	0.17 (0.22)	0.36* (0.21)	0.52** (0.24)
Total family crop labor hours in past year	10304	900.28 [1796.90]	82.37 (145.03)	128.32 (176.59)	38.93 (181.20)	41.23 (176.49)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure, as defined using satellite data, interacted with being observed after 2012. Each row represents an outcome; see the main analysis tables for explanations of the variables. Monetary values are in 2016 PPP USD. Control means are for communities with no flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. The main difference between this table and the estimates in Table 2 and the other main results is in how flood exposure is defined. Instead of using survey reports, in this table flood exposure is defined as a community being within 5 km of any pixel identified as flooded at any point in 2012 by MODIS satellite imagery in the NASA NRT Global Flood Mapping product. As in the main analyses, 2012 flood incidence is recentered around predicted incidence. Estimates therefore represent the effect of residing in a community that was exposed to floods 2012 according to MODIS satellite imagery for a given predicted propensity of satellite-based incidence. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

NFEs, with the largest effects occurring several years after the floods rather than in the same year. For the reasons we have given, we do not believe the satellite-based measure accurately captures 2012 flood exposure, whereas the survey-based measures directly identify where households were adversely affected by these floods. We next turn to exploring the mechanisms behind the effects of flood exposure under the main survey definition.

8 Mechanisms and discussion

We find no evidence that Nigerian households respond to flood exposure by either exiting agriculture or diversifying their income sources. Households also are persistently worse off in terms of consumption per capita. In this section, we first explore the drivers of the persistent decrease in farm production value. We then test for heterogeneity in the impacts of flood exposure by baseline household and community characteristics to evaluate what factors influence vulnerability and livelihood responses.

8.1 Explaining crop production value effects

Table 4 shows that the value of crop production is persistently and significantly lower for households in communities exposed to the 2012 floods. The average effect is economically very large, representing a 37% decrease relative to the mean in non-flooded communities. Many studies report that floods can decrease agricultural output, with a focus on direct damages. The large decrease in 2013, when households reported on production during the season affected by the 2012 floods, would be consistent with direct damage and destruction of crops by floodwaters. Indeed, such direct effects on crops are an important source of the household flood reports we use to define the flood exposure measure.

In addition to direct damage from inundation, significant effects in subsequent survey rounds imply persistent *indirect* effects of flood exposure on crop production. The effect in 2015-2016 is almost identical in magnitude to the effect in 2012-13, and while the effect in 2018-19 is not significant at conventional levels ($p = 0.119$), the magnitude of the point estimate remains large.

Decreases in crop production value are not due to agricultural exit. Table 4 shows that households are if anything more likely to engage in farm activities after flood exposure, and point estimates for effects on the count of members engaged in farm work are noisy but positive. Decreased production value could be explained by reallocation of farm labor away from crops toward livestock, decreases in other crop production inputs, decreases in land productivity, or decreases in prices of cultivated crops. Table 7 presents estimates of the impact of flood exposures on household crop production outcomes to test these potential explanations, focusing on households that cultivated crops in 2010-11 or any subsequent round (79% of the sample).²⁹ Among these households, annual crop production value falls by 656 USD (32%) on average after flood exposure, and there is a more noticeable difference over time with the largest effects in 2013 and non-significant but still economically large effects in 2019.

We find no effects of flood exposure on crop production labor inputs. Point estimates for effects on total annual hours of household crop production labor and total family crop labor days are positive but non-significant, as with the effects on the count of household members engaged in farm work. The average effect for hired or exchange labor days is negative but not significant and is more than offset by the positive estimates for family labor days. We also find that households do not significantly change their crop production practices in response to the 2012 floods, with no significant effects on the number of crops cultivated, the diversity of area planted across crops, total area planted, inorganic fertilizer use, or spending on crop production.³⁰ Therefore, production decisions cannot explain the persistence of the fall in crop production value to 2015-16 and, more

²⁹The number of observations is smaller for the household crop area planted HHI and the proportion of crop production value sold because these are only defined for households with non-zero area planted in a given round. Households with zero area planted are included in the sample for Table 7 if they cultivated crops in any survey round, and assigned values of zero for rounds in which they did not plant crops.

³⁰Only in 2018-19 for households in the long panel do we see some evidence of changes in practices, with significant decreases in hired/exchange labor and the probability of inorganic fertilizer use and a suggestive decrease in area planted ($p = 0.148$) and increase in family crop labor hours ($p = 0.134$). It is possible that households change varieties of crops but we do not observe variety choice in the data.

Table 7: Impacts of 2012 community flood incidence on household crop production outcomes

	N	Control Mean (SD)	2012 flood recentered × Post (SE)	2012 flood recentered × 2013 round (SE)	2012 flood recentered × 2016 round (SE)	2012 flood recentered × 2019 round (SE)
Number of crops cultivated	9275	2.74 [1.72]	-0.03 (0.12)	0.02 (0.12)	-0.09 (0.13)	-0.03 (0.28)
Household crop area planted HHI	6448	0.54 [0.27]	0.02 (0.02)	0.03 (0.02)	0.01 (0.02)	0.01 (0.03)
Total area planted (ha)	9275	2.03 [3.40]	-0.11 (0.22)	-0.17 (0.23)	-0.01 (0.23)	-0.27 (0.19)
Total hired/exch. crop labor days in past year	9275	12.77 [28.55]	-3.99 (3.04)	-4.74 (3.12)	-0.81 (4.39)	-18.29** (8.61)
Total family crop labor days in past year	9275	157.74 [362.79]	14.79 (34.21)	20.81 (45.84)	4.72 (32.90)	36.27 (34.47)
Total family crop labor hours in past year	9275	1225.87 [1989.80]	140.77 (157.62)	152.44 (202.76)	93.47 (183.53)	350.44 (233.57)
Any inorganic fertilizer use	9275	0.52 [0.50]	-0.02 (0.03)	-0.05 (0.04)	0.03 (0.03)	-0.10** (0.05)
Total spending on crop production (USD 100s)	9275	4.44 [10.35]	0.03 (0.62)	0.11 (0.73)	0.19 (0.81)	-1.45 (1.26)
Yield by area planted of cereals (kgs/ha)	4478	2235.54 [4338.81]	-962.19** (426.07)	-1046.75* (553.73)	-907.38** (424.99)	-743.92 (605.83)
Average cereal price (USD/kg)	4464	0.55 [0.92]	-0.27*** (0.08)	-0.27* (0.15)	-0.31*** (0.11)	-0.11 (0.15)
Total crop production value (USD 100s)	9275	20.56 [35.90]	-6.56*** (1.87)	-7.23*** (2.27)	-6.15*** (1.94)	-4.57 (3.49)
Total crop sales (USD 100s)	9275	5.91 [15.30]	-1.93*** (0.69)	-1.86** (0.81)	-2.11*** (0.73)	-1.29 (0.95)
Proportion of value of crop production sold	6448	0.26 [0.33]	-0.04* (0.02)	-0.05** (0.02)	-0.04 (0.03)	-0.01 (0.04)

Note: This table presents effects of 2012 community flood exposure on household outcomes in subsequent survey rounds, following [Table 2](#). The sample includes all households that cultivated crops in 2010-11 and any household cultivating in a subsequent round. Each row represents an outcome. All outcomes refer to the year including the most recent growing season. Outcomes for households not active in crop production in a given round are assigned values of 0, except for the crop area HHI and proportion of crop value sold which are undefined for households with 0 area planted and the cereal yield and price values which are undefined for households not cultivate cereals. The number of crops cultivated is across all household plots. The crop area Herfindahl-Hirschman-Index (HHI) is the sum of squared shares of total planted area allocated to different crops. Values closer to 1 indicate less diversification across crops. Total labor days and family labor hours are summed across all crop production activities reported in the post-planting and post-harvest surveys. Crop production costs are the sum of expenditures on purchased inputs such as seeds, fertilizer, labor, land, etc. Cereal yield is the total quantity harvested divided by the area planted. Average cereal price is the quantity-weighted mean price at which different cereals produced by the household are valued. Crop sales are the sum across crops of quantity sold times sales price. Crop production value is the sum across crops of crop sales plus the value of unsold crops, measured as quantity harvested but not sold times the median sales price for that crop in the community. The proportion of value of crop production sold is the ratio of the two above variables. Monetary values are in 2016 PPP USD. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

noisily, to 2018-19.

Instead, [Table 7](#) indicates that the effects are due to decreased physical crop yields and average prices. The table shows results for cereal yields and prices as most crop-producing households grow at least one cereal. Among households producing cereals ([Table A12](#)), crop production value falls by 813 USD (26%). Average household cereal yields falls by 962 kg/ha (43%) and average household cereal price falls by 0.27 USD/kg (49%), on average. These are large effects. The fact that these decreases are larger than the decrease in total crop production value suggests households may be somewhat protected by production of less affected non-cereal crops. We show results for yields and prices of individual cereal crops and for other commonly-cultivated crops in [Table A11](#). Individual estimated impacts of floods are noisy but the overall patterns consistently show negative and sometimes statistically significant point estimates.

Cereal yield is calculated as total quantity harvested divided by the area planted across all

cereal crops. As we find no significant changes in observed crop production decisions, this implies that effects on physical crop yields are driven by changes in land productivity. The immediate decrease in cereal yield can be explained by flooding and destruction of cultivated plots. Over the long-term, flooding could decrease productivity through soil erosion, siltation, or contamination, and increased salinity (in coastal areas), or increase it by depositing nutrients and sediments. Though we cannot identify the specific mechanism, the persistent negative impacts on crop yields indicate that adverse effects of floods dominate. Further, households do not appear to respond to this reduced agricultural potential by changing complementary inputs. No reduction in inputs may suggest constraints in non-farm livelihood alternatives. No increase in inputs, which could potentially offset reduced agricultural potential, may be a rational response to reduced soil fertility or represent tighter budget constraints following flood exposure.

The other main factor in valuing crop production is crop (sales) prices. The average household cereal price is the quantity-weighted mean price at which different cereals produced by the household are valued, based on household or community sales prices. Crop sales prices would generally be expected to rise in the short-term following flood exposure due to destruction of local output and potential supply disruptions. Instead, while effects on sales prices are not statistically significant for commonly-cultivated non-cereal crops in Nigeria, point estimates are all negative ([Table A11](#)). The negative effects do not represent falling prices but rather smaller increases over time in flooded communities. Average household cereal prices in communities not exposed to floods rise from 0.55 USD on average in 2011 to 1.16 USD in 2013, 0.95 USD in 2016, and 0.89 USD in 2019 (all prices are in 2016 USD adjusted for national inflation in Nigeria). The rise in price from 2011 to 2013 is 11% smaller in flood-exposed communities and the average prices in subsequent rounds are also 10-20% lower. These differences imply that crop producers in unexposed communities may have benefited from higher prices of their crops. But [Table A8](#) shows that the estimated effects of floods on crop production value are if anything larger if we exclude nearby communities and those in the same LGA – more likely to experience spillovers – from the sample of controls, suggesting limited upward bias in our estimates.

Turning to market price data collected in the community surveys, we find that crop prices in the post-harvest period are not in general significantly different after flood exposure ([Table A13](#)). This implies that flood exposure primarily affects prices received by producers at the point of sale, rather than broader consumer prices. Crop sales decrease by 193 USD (33%) annually on average after flood exposure, and the proportion of crop production value that is sold decreases by 4 percentage points (15%). Households are therefore selling less of their reduced output and receiving lower prices for it. Since we value crop production using household sales prices were available, if they sold part of their crop at a low price this would reduce the estimated value of the entire output. One possibility is that households exposed to floods may be selling at less opportune times, such as right after harvest. Kakpo et al. (2022) finds that weather shocks in Niger lead to reduced millet market prices immediately after harvest, but increases 6 months later. Differences in sales timing could explain part of the difference we observe in average crop sale prices between flood exposed

and unexposed communities, but we cannot test this as the surveys do not ask about the timing of crop sales.

We test the relative importance of yield and price changes by estimating impacts of flood exposure on crop production value controlling for each of these mediators (Table A12). Controlling for average cereal price does not meaningfully change the estimated average effect of flood exposure on crop production value among cereal producers. Bias from spillovers in price effects may therefore be limited. In contrast, controlling for average cereal yield reduces the estimated effect by around a third. Effects on physical yield therefore appear to be a more important driver of reduced production value. In the short term, this is consistent with a key role of crop damage and destruction. In the longer term, given no significant changes in crop production decisions on average, this suggests direct and persistent effects on soil fertility.

8.2 Heterogeneity in responses to flood exposure

We next examine heterogeneity in the effects of flood exposure to better understand which households are more vulnerable to flood shocks and how adjustment occurs across livelihood activities. Specifically, we estimate versions of our baseline specification that allow the treatment effect to vary across baseline household and community characteristics, including geography, market access, prior flood exposure, and household size, assets, and livelihood activities. We focus on effects of flood exposure on consumption, food insecurity, and incomes and member engagement in the three types of livelihood activities we consider. In each regression, we estimate a triple-differences model fully interacting a baseline household or community characteristic H with the Post and Flood dummies, the state-by-round fixed effects, and all controls in the main specification (following Feigenberg et al., 2025).

Across most dimensions, we do not find consistent or statistically distinguishable differences in treatment effects across outcomes. In particular, heterogeneity by baseline household size, asset holdings, proximity to a major road, market, or town, proximity to water, fluvial flood risk, and recent community flood exposure yields mixed or imprecise patterns (Table A14). While some of these characteristics are associated with a greater risk of food insecurity or larger decreases in farm production value, they do not consistently predict differential adjustment in labor allocation or income outcomes. This suggests that vulnerability to flood shocks does not map cleanly into a broad ranking based on demographics, wealth, community centrality, or general geographic exposure.

The only dimensions along which there appears to be informative heterogeneity are baseline livelihood activities – agriculture and wage work – and rural location (Table 8). As would be expected mechanically, Panel A shows that decreases in the value of agricultural production are concentrated among households engaged in agriculture at baseline. But these households are not significantly worse off in terms of consumption and food insecurity. Positive but non-significant point estimates for effects on wage and NFE incomes suggest these households are able to offset farm production decreases with income from other activities. Only agricultural households increase the count of members engaged in wage work, though the difference is not significant at conventional

levels ($p = 0.11$). Somewhat less intuitively, Panel B shows that the fall in agricultural production is only significant in households engaged in wage work at baseline, though the difference is not significant ($p = 0.18$). But flooding does not decrease consumption or increase food insecurity for these households, and the results suggest this may be because they respond with much larger labor allocations to household agriculture ($p = 0.02$) and wage work ($p = 0.00$), while also earning significantly more NFE income ($p = 0.07$) to offset the decrease in farm production value. We do not observe a clear pattern of differences in effects by baseline household NFE engagement (Table A15).

Panel C shows that the patterns in panels A and B are driven by the 28% of households engaged in *both* agriculture and wage work at baseline. Households not engaged in either activity do not increase the count of members engaged or income from any livelihood activity and experience significantly larger decreases in consumption per capita and food security. The results could be interpreted as showing that intensification of existing activities, particularly among those engaged in wage work, increases resilience of households to the flood shock. However, increases in the count of wage workers are in no case associated with a significant increase in wage income, and are not consistently associated with mitigated impacts on consumption or food insecurity. This suggests that labor supply adjustments along existing margins may be incomplete or low-productivity, limiting contributions to household resilience and recovery.

Why don't more households respond to the flood shock by participating more in wage work? One possibility is that agricultural households – those experiencing the largest income losses due to flood exposure – live in areas with limited opportunities for wage work, and face constraints to having members migrate to work. But we do not find that being closer to a major market, major road, or town with a population of at least 20,000 is associated with a larger effect on the count of household wage workers (Table A14). Further, Panel D of Table 8 find that flood exposure increases the count of wage workers by significantly *more* in rural than in urban areas. Rural areas include a greater share of agricultural households and the decrease in the value of farm production is only significant in rural areas, though we cannot rule out equality of impacts ($p = 0.26$). But the pattern follows the other panels of the table: households experiencing significant decreases in farm production value increase their allocation of household labor to wage work and (less consistently) to farm work.

Taken together, these patterns point to an important constraint on household adjustment following large shocks. While households appear able to reallocate labor within existing activities, they do not readily enter new activities in response to flood exposure. This is consistent with the presence of short-run barriers to occupational switching, such as liquidity constraints, skill requirements, or limited local demand for labor. The fact that wage work responses are not larger for households in communities where we would expect larger labor demand suggests the main constraints to livelihood responses are at the household level. In particular, resilience to flood shocks is closely tied to pre-existing livelihood diversification, particularly access to wage employment, rather than to broad measures of wealth or demographic characteristics. These findings help explain why

Table 8: Impacts of 2012 community flood incidence by baseline household activities and rural location

	Consumption		HH agriculture		Wage work		HH NFE	
	(1) Daily HH cons./cap.	(2) Any food insec.	(3) Value prod.	(4) Count mems.	(5) Total inc.	(6) Count mems.	(7) Total inc.	(8) Count mems.
A. H = Any baseline HH agriculture (76%)								
Flood $\times$ Post, H=0	-1.13 (0.70)	0.06 (0.05)	-1.21 (1.00)	-0.03 (0.08)	1.40 (28.91)	-0.06 (0.09)	-10.34 (9.31)	-0.06 (0.08)
Flood $\times$ Post, H=1	-0.50** (0.21)	0.04* (0.03)	-5.34*** (2.04)	0.08 (0.12)	1.60 (4.93)	0.13* (0.07)	4.26 (3.06)	-0.02 (0.07)
<i>p</i> , equality of effects	0.39	0.73	0.06	0.47	0.99	0.11	0.14	0.72
B. H = Any baseline wage work (39%)								
Flood $\times$ Post, H=0	-0.81*** (0.26)	0.07** (0.03)	-2.36 (2.13)	-0.07 (0.10)	-1.40 (1.77)	-0.05 (0.03)	-2.99 (4.24)	-0.06 (0.06)
Flood $\times$ Post, H=1	-0.52 (0.34)	0.01 (0.03)	-7.32** (2.94)	0.32** (0.16)	10.59 (14.42)	0.33*** (0.12)	9.64* (4.93)	0.07 (0.10)
<i>p</i> , equality of effects	0.49	0.13	0.18	0.02	0.40	0.00	0.07	0.24
C. H = Both HH agriculture and wage work at baseline (28%)								
Flood $\times$ Post, H=0	-0.98*** (0.27)	0.07*** (0.02)	-2.16 (1.89)	-0.01 (0.09)	-0.55 (6.65)	-0.05 (0.04)	-2.42 (3.81)	-0.04 (0.06)
Flood $\times$ Post, H=1	-0.02 (0.26)	-0.01 (0.04)	-9.47** (3.69)	0.31 (0.20)	4.74 (12.77)	0.38** (0.16)	10.81** (5.43)	0.04 (0.13)
<i>p</i> , equality of effects	0.01	0.04	0.09	0.12	0.71	0.01	0.06	0.53
D. H = Rural community (74%)								
Flood $\times$ Post, H=0	-0.51 (0.45)	0.07* (0.04)	-2.10 (1.40)	-0.29** (0.14)	-14.86 (19.96)	-0.18* (0.10)	-6.07 (9.06)	-0.08 (0.09)
Flood $\times$ Post, H=1	-0.56** (0.24)	0.04* (0.03)	-4.91** (2.07)	0.09 (0.11)	5.86 (5.16)	0.15** (0.06)	3.37 (3.04)	-0.02 (0.07)
<i>p</i> , equality of effects	0.92	0.54	0.26	0.04	0.32	0.01	0.32	0.58

Note: This table presents average effects of 2012 community flood exposure on household outcomes in subsequent survey rounds, by baseline household characteristics H as indicated in the panel headings. For each H , we show the share of households for which $H = 1$. Each column represents an outcome. Column 1 shows daily consumption per capita and column 2 shows whether any food insecurity was reported over the previous 12 months. For each category of livelihood activities (HH agriculture, HH NFE, and wage work), we show effects on income or production value from that activity and count of household members engaged. The results for each column in each panel are from a single triple-differences regression fully interacting the baseline household characteristic H with the Post and Flood dummies, state-by-round fixed effects, and all controls from the main specification. We calculate the average effect in the group with $H = 1$ by taking the sum of the $Flood \times Post$ and $Flood \times Post \times H$ coefficients and estimate SEs using the *xlincom* function in Stata. We also report the p -value for the triple interaction term, which tests for equality of effects between the two groups. Monetary values are in 2016 PPP USD. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

we do not observe widespread shifts in livelihood structure in response to flooding: adjustment occurs primarily along margins that are already available to households.

9 Conclusion

Floods are among the most common and destructive natural disasters, and climate change is increasing flood risk. This has implications for the well-being and livelihoods of households in areas at risk of flooding. Increased flood risk and damages in agricultural communities in developing countries may also affect household labor market decisions and broader structural transformation by affecting both expected returns to different livelihood strategies and resources available to invest in these activities.

We find that, controlling for predicted risk of flooding, community-level exposure to floods during Nigeria’s 2012 flood disaster persistently decreases agricultural production and household food consumption. In line with studies finding that climate change is likely to trap many workers in low-productivity agriculture (Cruz, 2024; Nath, 2025), we find no impacts of flooding on agricultural exit, though farm labor inputs also do not increase significantly on average. Persistent decreases in agricultural productivity in flood-exposed communities imply that floods are more likely to constrain the process of structural transformation in the absence of any policy interventions.

In contrast with much work showing short-term increases in wage employment following agricultural shocks, we find no consistent evidence of short- or long-term changes in household labor allocations to different activities or of livelihood diversification. Non-farm enterprise engagement does not change, consistent with few Nigerian households reporting using such work as a shock coping strategy. Only households already engaged in both agriculture and wage employment before the floods increase their labor supply in both activities. As community characteristics are not consistently associated with differences in labor supply effects of the floods, these results suggest that household-level experience, skills, or access are the main constraints to livelihood responses to flood exposure. Future work analyzing constraints to and drivers of such responses, building on broader work on climate adaptation in sub-Saharan Africa (Di Falco et al., 2012; Gbetibouo et al., 2010), is needed to support greater resilience to flooding.

Migration is highlighted as a key potential response in the literature on climate change and structural transformation or labor supply, but in many contexts migration decisions are constrained (Cruz, 2024; Ibáñez et al., 2026). We find that migration is not affected by the 2012 floods on average, but this masks differences over time. Flood-exposed households are less likely to move or have any members migrate in the short-term, but this reverses 7 years after flood exposure suggesting households need time to overcome migration constraints. A longer-term analysis would be needed to evaluate whether this eventual migration reverses the persistent consumption decreases following flood exposure.

Another important conclusion from this study is that decisions of how to measure flood incidence matter, echoing some previous work. We show that the identification of flooded areas in Nigeria in 2012 differs depending on whether survey reports or MODIS satellite imagery is used, with limited overlap between the two measures. We argue that much of the difference can be attributed to limitations in satellite imagery and flood detection algorithms, and highlight the importance of understanding what flood measures capture for interpretation. Floods reported in household

surveys by definition affected households, and therefore represent cases where floods were not expected, not adequately prepared for or defended against, or more severe than expected. MODIS satellite imagery on the other hand can capture fluctuations in surface water, including anticipated seasonal fluctuations and other cases that did not adversely affect households, but misses floods masked by clouds – an important factor when floods are driven by heavy precipitation – and floods that are not broadly spread over space. These differences affect conclusions about the impacts of floods. In contrast to the effects we document for survey-reported floods in Nigeria in 2012, we find no consistent effects of floods detected by MODIS imagery and included in the NASA NRT Global Flood Mapping database. This may not be surprising given that this database primarily identifies floods in communities where none are reported while missing a large share of communities where floods are reported.

In contrast to previous work dismissing survey data in favor of more ‘objective’ satellite data, we argue that survey reports have great value as ground-truth measures of situations where floods caused damages, though it is important for interpretation to be clear that such reports will only capture a subset of all floods depending on how survey questions are worded. New methods incorporating satellite-based radar and cross-validating measures of flooding from multiple data sources may help increase accuracy of remote flood detection but will still be subject to data limitations and to decisions of how to define what constitutes a ‘flood.’ Researchers must carefully consider both definition and measurement issues in empirical studies of the effects of flood exposure, as these affect how the effects should be interpreted. They should also test the sensitivity of estimated impacts of flood exposure to data and definition decisions, in line with recent calls to do so for other remotely-sensed data.

One limitation of the study is that we study a single flood event. If part of the mismatch in flood identification across data sources is due to adaptation of communities to floods in certain areas, this would imply that analyses of a single major flood event will fail to capture longer-run effects of changes in flood risk and resulting adaptation responses of households, unless prior floods were limited in frequency and severity. We find no consistent differences in impacts of exposure to the 2012 floods in Nigeria by recent community flooding, but future work on the long-term economic impacts of flood exposure should consider effects of changes in flood hazard and cumulative flood exposure over longer time frames, as in the recent working papers by Patel (2024) and Sajid (2023). Satellite-based measures are likely be the best source of flood incidence data for such analysis due to the lack of high-frequency survey data in vulnerable communities, but researchers should seek additional data sources of ground truth to test against what is detected by satellite.

The results are also relevant to policymakers in developing countries. The failure of satellite imagery to identify a large share of survey communities where flood damages are reported has implications for emerging recommendations to use satellite data for policy targeting. An understanding of the limitations of what satellites can detect can inform decisions about how to allocate monitoring resources. The findings indicate challenges for households attempting to cope with or adapt to flood exposure, which could motivate policies to provide support for adaptation such as agricultural

extension and training, access to inputs, and support to access non-farm wage employment.

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A Additional Figures

Figure A1: GHSP data collection timeline

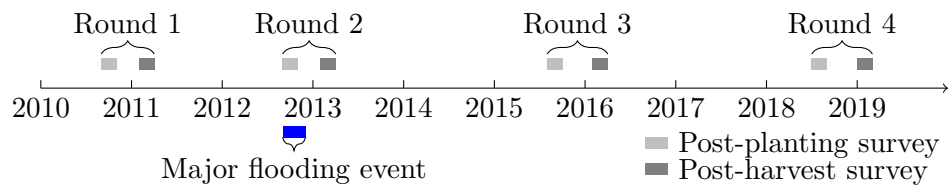
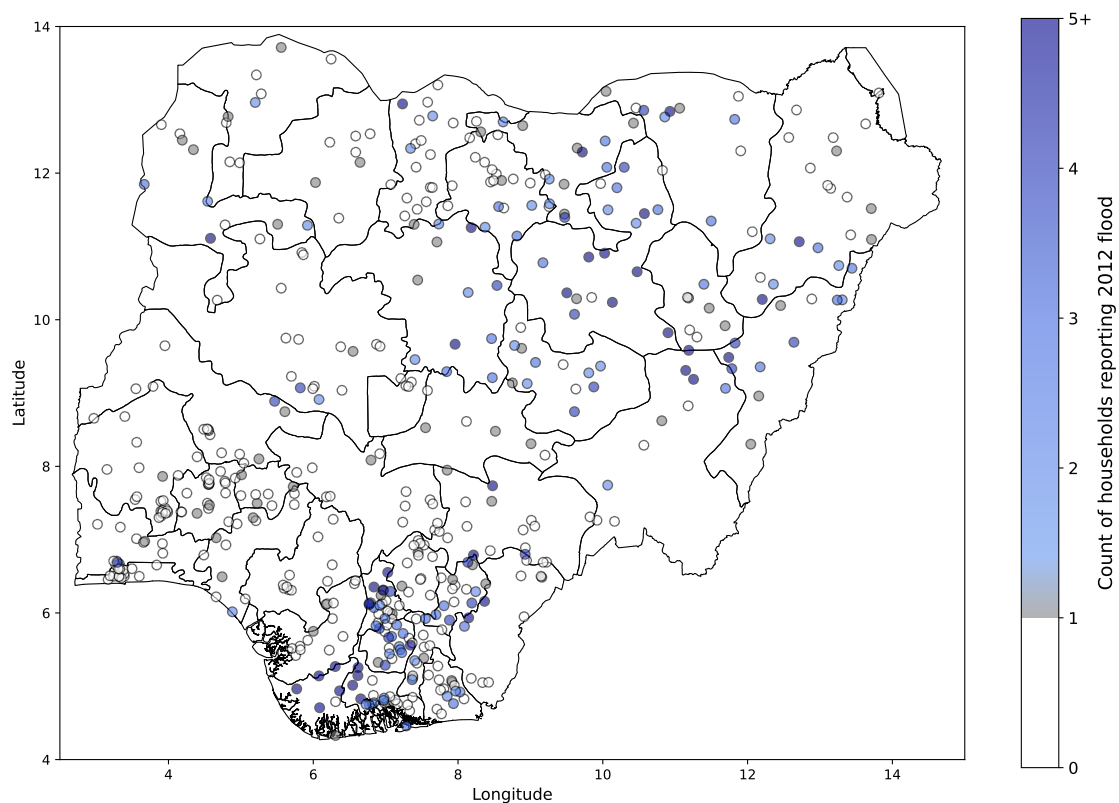
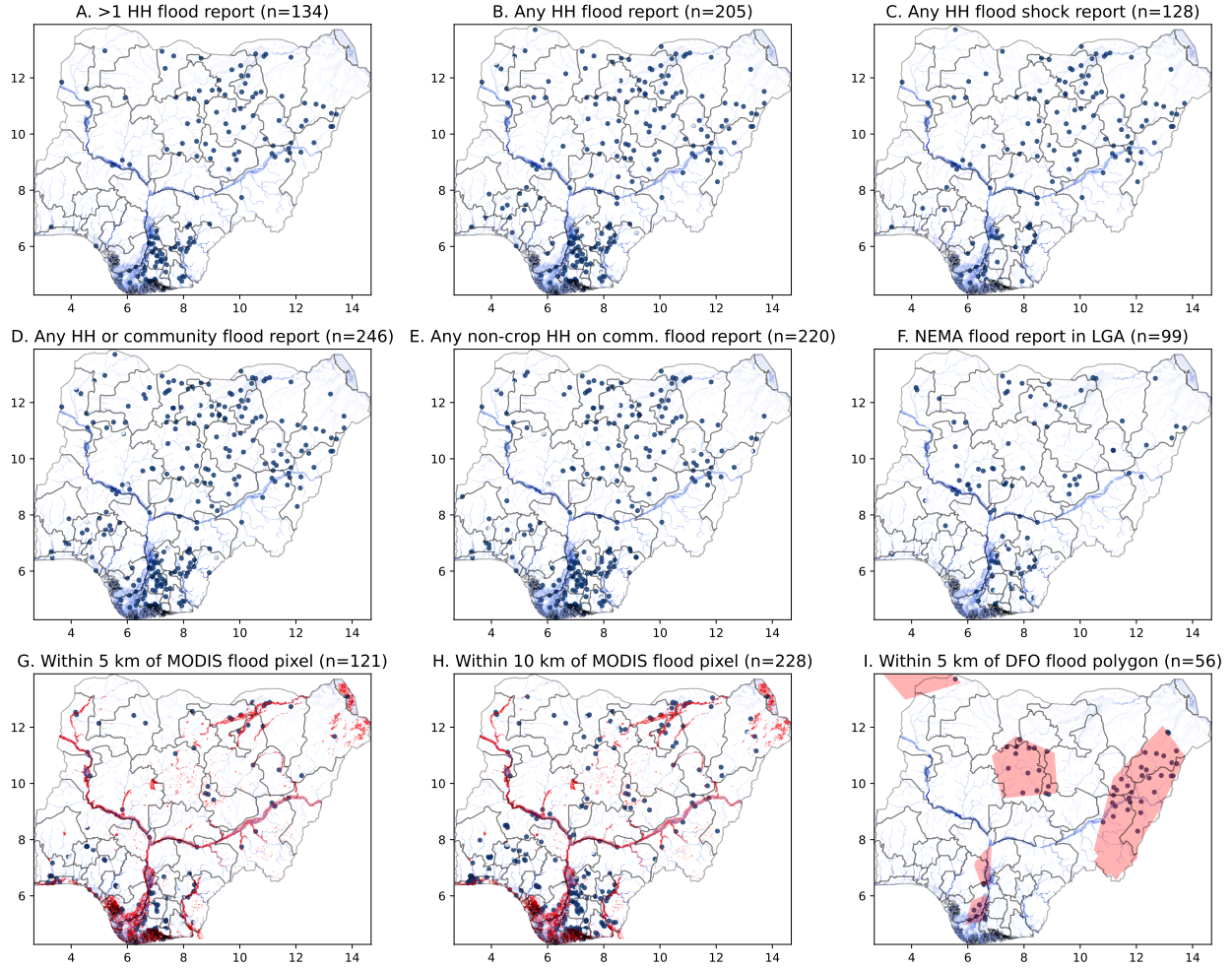


Figure A2: Counts of households reporting flood exposure in 2012 by community



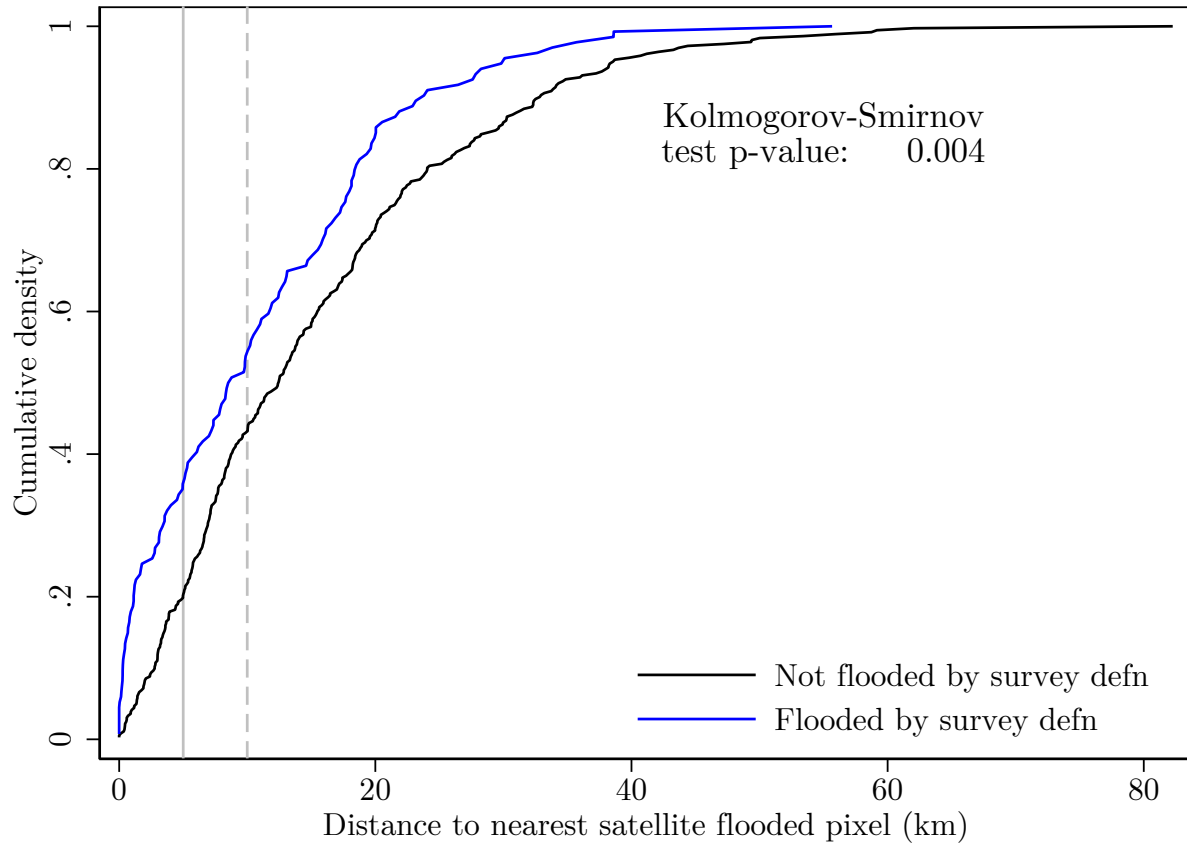
Note: Nigeria GHSP household flood reports are aggregated across questions from the household shocks, food security, and crop production modules. For each community, we calculate the number of households reporting being affected by floods in 2012 in one of these modules. White circles indicating communities with no household flood reports. Ten households are sampled for each survey community.

Figure A3: Community flood detection by flooding measure



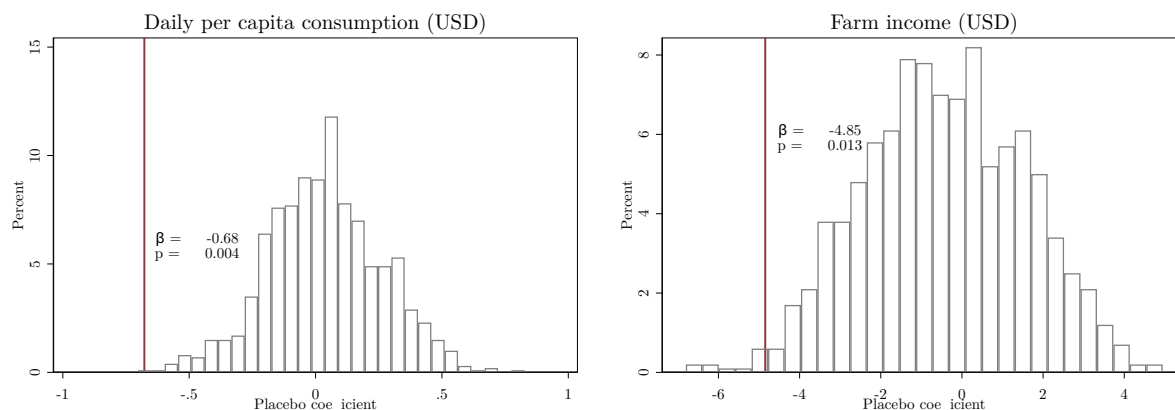
Note: Each panel shows the set of GHSP communities identified as exposed to floods in 2012 according to a different definition of flood incidence. The blue shading in the background of each panel corresponds to the depth of 100-year fluvial floods. Panels A-E rely on survey reports of floods in 2012. Panel A shows communities with at least two household reports of floods in any module. Panel B shows communities with any such household reports. Panel C shows communities with any household reports of floods in the household shocks module only. Panel D shows communities with any household or community survey flood report. Panel E shows communities with any household non-crop-related flood or community flood report. Panel F shows communities in local government areas (LGAs) where NEMA reports any floods in 2012. Panels G and H show communities within 5 and 10 km of any pixel identified as flooded by MODIS satellite imagery at any point in 2012 in the NASA NRT Global Flood Mapping product. Panel I shows communities within 5 km of polygons where flooding is reported in the Dartmouth Flood Observatory archive. The flooded pixels and polygons in panels G-I are colored red.

Figure A4: Distance to nearest flooded pixel by survey flood definition



Note: The figure plots the cumulative distribution of GHSP community distance to the nearest pixel identified as flooded at any point in 2012 in the NASA MODIS flood database by whether the community was exposed to the 2012 floods according to our main survey-based definition. For each community, we calculate straight-line distances between the jittered community centroid and the centroids of 250 m MODIS flooded pixels. The solid vertical line represents the 5 km cutoff used in the main satellite-based flood exposure definition, and the dashed vertical line represents a 10 km cutoff. We also show the p -value for the Kolmogorov-Smirnov test of equality of the two distributions.

Figure A5: Distribution of impacts of permutated 2012 flood exposure within flood propensity bins



Note: The two panels show the results of 1000 simulations of the impacts of 2012 recentered flood exposure permutating actual exposure within bins of estimated flood propensity to keep the same number of flooded communities in each bin. For each outcome, we plot the distribution of the estimated coefficients from each simulation. The vertical red line shows the coefficient for the effect of actual realized flood exposure. The text alongside the red line shows the coefficient estimate and the randomization p -value, calculated as the share of simulation estimates more extreme (in absolute terms) than the observed estimate. A small p -value indicates a very low likelihood that we would observe our estimate under the null of random allocation of flood exposure within flood propensity bins.

B Additional Tables

Table A1: Predictors of 2012 flood incidence

	(1) Any HH flood rept, shocks module	(2) Any HH flood rept, all modules	(3) > 1 HH flood rept, all modules	(4) HH non-crop or comm flood rept	(5) Any HH or comm flood rept	(6) Any MODIS flood pixel w/in 5 km	(7) Any MODIS flood pixel w/in 10 km
Distance to nearest water area (km)	-0.016*** (0.006)	-0.014*** (0.005)	-0.023*** (0.007)	-0.003 (0.004)	-0.012** (0.005)	-0.020** (0.008)	-0.025*** (0.005)
Avg 100-year flood depth within 5 km	0.826*** (0.205)	0.664*** (0.192)	0.385 (0.276)	0.456*** (0.167)	0.580** (0.286)	2.905*** (0.703)	3.333*** (0.786)
Area equipped for irrigation in cell in 2005 (ha)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000** (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
>15% broadleaved evergreen or semi-deciduous forest (>5m)	-1.805** (0.890)	-0.494 (0.459)			-0.801* (0.450)	-1.583 (0.996)	-0.529 (0.574)
>15% closed to open herbaceous vegetation	1.398** (0.679)	0.682 (0.681)	-0.012 (0.818)		0.183 (0.647)	1.429 (1.106)	
>50% artificial surfaces and associated urban areas		-0.354 (0.588)	-0.879 (0.868)		-0.241 (0.482)	0.029 (0.869)	0.641 (0.582)
Percent ag land w/in approx 1 km	0.009 (0.006)	0.011* (0.006)		0.004 (0.005)	0.013** (0.006)	-0.041*** (0.015)	-0.025*** (0.006)
Slope (percent)		0.031 (0.066)	-0.141* (0.076)	-0.055 (0.057)	-0.036 (0.063)	-0.274** (0.131)	-0.029 (0.066)
Avg. precipitation of wettest quarter		0.003*** (0.001)	0.004*** (0.001)	0.002 (0.001)	0.002*** (0.001)	0.001 (0.002)	
Percent ag land X 2012 deviation from avg. annual precipitation	0.002 (0.002)	0.001 (0.002)	0.003 (0.002)	0.003* (0.002)	0.002 (0.002)	0.015*** (0.003)	0.007*** (0.002)
Elevation (m) X 2012 deviation from avg. annual precipitation	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.001 (0.001)	-0.000 (0.000)
Any HH rept. flood shock in 2008	0.741 (0.632)	1.063 (0.661)	0.616 (0.630)	0.981* (0.554)		-1.409 (1.123)	-1.254* (0.730)
Any HH rept. flood shock in 2009	1.528*** (0.391)	1.243*** (0.419)	1.488*** (0.399)	0.896** (0.361)	1.253*** (0.443)	1.297** (0.617)	
Any HH rept. flood shock in 2011	0.399 (0.424)	0.405 (0.418)	0.164 (0.434)			0.264 (0.572)	1.268*** (0.464)
Mean HH area planted with crops in 2010 (ha)	0.310*** (0.107)	0.140 (0.094)	0.263*** (0.093)		0.089 (0.094)	0.327*** (0.119)	0.159 (0.101)
Share of HHs w/ any crop activity in 2010	1.556*** (0.551)	2.627*** (0.569)	2.254*** (0.562)	0.790** (0.326)	1.077*** (0.394)	-0.539 (0.731)	
Observations	486	494	486	495	491	483	490

Note: This table shows the results of logit regressions predicting 2012 flood incidence under different measurement approaches indicated in the column headings. See [Figure A3](#) for a description of the different measures. The predictors are selected from a broader set of geographic, weather, and mean household characteristics in a first stage using a Lasso regression with 10-fold cross-validation. Differences in the number of community observations result from certain selected land cover classes predicting failure or success perfectly, and observations in those landcover classes dropping out from the estimation. For simplicity, we only show characteristics selected for at least five flood measures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A2: Balance in baseline characteristics by 2012 flood incidence

	No flood report Mean (SD)	Flood report difference (SE)	Recentered flood shock difference (SE)
<i>Community characteristics</i>			
Rural	0.62 (0.49)	0.10** (0.05)	-0.08 (0.05)
Distance to nearest major road (km)	13.54 (18.34)	1.41 (2.11)	-0.55 (2.20)
Distance to nearest market (km)	67.12 (46.67)	4.37 (3.70)	3.02 (3.99)
Distance to nearest pop. center w/ >20k pop. (km)	17.77 (19.30)	2.11 (2.19)	-1.89 (2.17)
Percent ag land w/in approx 1km	24.53 (26.00)	6.92*** (2.54)	1.02 (2.74)
Slope (percent)	3.32 (2.65)	-0.29 (0.25)	-0.04 (0.25)
Elevation (m)	268.30 (193.65)	-33.96** (14.81)	-14.24 (13.53)
Avg. annual precipitation (mm)	1439.11 (606.20)	8.14 (21.00)	-44.24* (23.86)
Distance to nearest water area (km)	36.58 (31.60)	-4.76** (2.39)	2.75 (2.24)
Avg 100-year flood depth within 5 km	0.23 (0.53)	0.28*** (0.08)	0.01 (0.08)
Any comm. HH flood shock rept. from 2007-2011	0.20 (0.40)	0.13*** (0.05)	-0.10* (0.05)
<i>Household characteristics</i>			
Female-headed HH	0.16 (0.37)	-0.02 (0.01)	-0.02 (0.02)
Count of HH members	5.25 (2.95)	0.36** (0.15)	0.13 (0.17)
Any food insecurity in last 12 months	0.20 (0.40)	0.02 (0.02)	0.00 (0.02)
HH under 1.90 USD PPP per capita daily poverty line	0.32 (0.47)	0.05** (0.02)	-0.02 (0.03)
Household asset index	0.07 (1.07)	-0.20*** (0.06)	-0.01 (0.06)
Any HH farm activity	0.64 (0.48)	0.09*** (0.03)	-0.07** (0.03)
Any HH non-farm enterprise activity	0.76 (0.43)	0.04** (0.02)	0.02 (0.02)
Any wage employment activity	0.42 (0.49)	-0.10*** (0.03)	0.02 (0.03)
Test of joint significance		$F = 9.00$	$F = 0.42$
p -value		$p < 0.001$	$p = 0.987$

Note: This table shows the baseline (2010-11) control mean and difference by 2012 flood exposure status for selected community and household characteristics from a series of separate regressions. All regressions include state fixed effects, and standard errors are clustered at the 2012 community level ($N = 497$). 2012 flood incidence is defined as at least two households in a community reporting being affected by floods. The first column of differences uses the simple binary flood treatment measure while the second recenters the treatment around predicted flood incidence. The bottom of the table shows results for the test of the hypothesis that the relationship between a given 2012 flood exposure measure and all variables is jointly 0. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Balance in flood risk by 2012 flood incidence

	(1)	(2)
2012 reported flood exposure	0.046*** (0.016)	
2012 recentered flood exposure		0.005 (0.016)
Observations	3830	3830
Mean, not flooded in 2012	0.060	0.060
State and Year FE	Yes	Yes

Note: This table shows the results from regressing a dummy for any community flood incidence in a given year according to the survey measure on dummies for flood exposure in 2012. The first row uses the simple binary flood treatment measure while the second recenters the treatment around predicted flood incidence. Observations are at the community-year level for 2007-2018, and 2012 is omitted. Fixed effects control for state and year. Standard errors are clustered by community, but are similar if we cluster by year. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Differences in probability of reporting a given shock in 2013 by household report of any flood shock

	No flood shock Mean (SD)	Any flood shock difference (SE)
Death/disability of adult member	0.15 (0.36)	-0.02 (0.02)
Death of remittance sender	0.04 (0.20)	-0.00 (0.01)
Departure of income earner	0.01 (0.10)	-0.00 (0.01)
Non-farm business failure	0.05 (0.21)	0.02 (0.02)
Theft of crops/cash/property	0.03 (0.16)	-0.00 (0.01)
Other harv. failure/destruction	0.04 (0.19)	0.04** (0.02)
Dwelling damaged/demolished	0.02 (0.14)	0.01 (0.01)
Death of livestock to illness	0.02 (0.13)	0.02* (0.01)
Input/output price shock	0.03 (0.17)	0.02* (0.01)
Increase of food prices	0.06 (0.24)	0.02 (0.02)
Other shock	0.06 (0.24)	-0.04*** (0.01)

Note: This table shows the results from separate regressions of indicators that households declare different shocks in the 2013 post-harvest survey on an indicator for whether they declare any flood shock. All regressions include state fixed effects. Standard errors are clustered by enumeration area. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Impacts of 2012 community flood incidence on incomes, including households with zero estimated total income

	N	Average impact		Dynamic impacts		
		Control Mean (SD)	2012 flood recentered × Post (SE)	2012 flood recentered × 2013 round (SE)	2012 flood recentered × 2016 round (SE)	2012 flood recentered × 2019 round (SE)
Total HH income and production value (USD 100s)	12218	77.18 [173.48]	1.39 (5.59)	0.54 (6.42)	0.42 (5.99)	13.49 (11.17)
Total value of HH farm production (USD 100s)	12218	17.42 [37.48]	-4.11*** (1.55)	-4.26** (1.69)	-4.09** (1.83)	-3.19 (3.13)
Total income from non-farm activities (USD 100s)	12218	59.76 [171.26]	5.50 (5.33)	4.79 (6.17)	4.51 (5.71)	16.68 (10.41)
Total non-farm HH enterprise income (USD 100s)	12218	25.29 [82.52]	2.03 (2.89)	4.89* (2.96)	-2.29 (3.28)	8.78* (4.52)
Total wage employment income (USD 100s)	12218	31.78 [147.98]	3.07 (4.57)	-1.02 (5.50)	6.91 (5.02)	7.97 (8.79)
Total HH income from other activities (USD 100s)	12218	2.70 [20.05]	0.40 (1.03)	0.92 (1.10)	-0.11 (1.18)	-0.08 (1.92)

Note: This table reproduces the first rows of Table 4 but including households with zero total income and production value. The results for income shares are unchanged because the shares can only be calculated for households with non-zero income, and therefore are not shown. Monetary values are in 2016 PPP USD. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Sensitivity of estimated impacts to different standard error clustering

	(1) Any food insec.	(2) Daily HH cons./cap.	(3) HH farm prod. val.	(4) Any HH farm act.	(5) Ct. wage wkrs.	(6) Area planted
Flood Exposure X Post	0.045** (0.022)	-0.677*** (0.213)	-4.847*** (1.802)	0.027* (0.015)	0.094 (0.060)	-0.155 (0.194)
SE: LGA	0.022	0.211	1.860	0.015	0.061	0.193
SE: Comm-Round	0.022	0.211	1.461	0.011	0.046	0.157
SE: Conley (50km)	0.027	0.198	1.340	0.012	0.045	0.127
SE: Conley (100km)	0.028	0.141	1.394	0.011	0.042	0.161
Wild p-val: State	0.368	0.013	0.009	0.094	0.166	0.539
Observations	11951	11951	11144	11951	11951	11951

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. Each column represents an outcome; see the main analysis tables for explanations of the variables. Monetary values are in 2016 PPP USD. We focus on the main outcomes with significant estimates effects under different specifications in the robustness checks. Each regression follows the main specification, but we show how the standard error changes under different clustering approaches. The main regressions cluster SEs at the level of the community of residence in 2012, at which the flood exposure treatment is assigned. We then show SEs clustered at the LGA level, two-way clustered SEs at the community by round level, Conley SEs allowing for spatial correlation within 50 and 100 km, and p -values from a wild bootstrap of state-level clustered SEs. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A7: Sensitivity of estimated impacts of 2012 community flood incidence to alternative specifications

	(1)	(2)	(3)	(4)	(5)	(6)
	Main	Add rain controls	No controls	LGA-round FE	Zone-round FE	Round FE
Any food insecurity in last 12 months	0.05** (0.02)	0.05** (0.02)	0.04** (0.02)	0.10* (0.05)	0.06** (0.03)	0.09*** (0.03)
Daily HH consumption per capita (USD)	-0.68*** (0.21)	-0.67*** (0.21)	-0.69*** (0.22)	-1.76** (0.71)	-0.63*** (0.23)	-0.69*** (0.22)
Total value of HH farm production (USD 100s)	-4.85*** (1.80)	-4.90*** (1.81)	-4.63*** (1.74)	-3.10 (5.35)	-4.84** (1.93)	-5.88*** (1.84)
Total non-farm HH enterprise income (USD 100s)	0.55 (3.28)	0.56 (3.28)	0.06 (3.24)	7.02 (10.13)	-1.20 (3.33)	-0.58 (3.40)
Total wage employment income (USD 100s)	0.19 (6.56)	0.19 (6.57)	-0.38 (6.53)	10.83 (22.19)	-0.71 (6.02)	-0.25 (6.00)
Any HH farm activity	0.03* (0.01)	0.03* (0.01)	0.05*** (0.01)	0.10* (0.05)	0.02 (0.01)	0.03** (0.01)
Any HH non-farm enterprise activity	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.06** (0.02)	0.02 (0.02)	0.01 (0.02)
Any wage employment activity	0.00 (0.02)	0.00 (0.02)	0.01 (0.02)	0.00 (0.05)	0.02 (0.02)	0.02 (0.02)
Count of HH members wkg. in HH farm	0.08 (0.10)	0.08 (0.10)	0.11 (0.09)	0.39** (0.18)	0.12 (0.10)	0.14 (0.10)
Count of HH members wkg. in HH non-farm enterprise	0.00 (0.06)	0.00 (0.06)	0.01 (0.06)	0.07 (0.12)	-0.00 (0.06)	0.01 (0.06)
Count of HH members wkg. in wage employment	0.09 (0.06)	0.09 (0.06)	0.12** (0.06)	-0.01 (0.17)	0.13** (0.06)	0.15** (0.06)
Total area planted (ha)	-0.16 (0.19)	-0.15 (0.19)	-0.07 (0.20)	-0.19 (0.23)	-0.43* (0.26)	-0.38 (0.26)
Total family crop labor hours in past year	143.62 (133.94)	142.43 (133.17)	98.00 (140.14)	299.62 (245.23)	-0.45 (199.39)	222.22 (213.05)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. Each row represents an outcome; see the main analysis tables for explanations of the variables. Monetary values are in 2016 PPP USD. The columns show average effects of flood exposure across all post-flood survey rounds. Flood exposure is defined as at least two households in the community reporting being affected by floods in 2012, and is recentered around predicted community incidence. Estimates therefore represent the effect of residing in a community that was exposed to floods 2012 for a given predicted propensity of incidence. Each column indicates how the specification is changed relative to the main specification, presented in column (1), which includes household, state-by-round, and baseline characteristic-by-round fixed effects. In column (2) we add controls for the deviation in annual total precipitation relative to the historical mean at the community location. In column (3) we drop the baseline characteristic-by-round fixed effects, where we controlled for characteristics that differed significantly by flood status in [Table A2](#). Columns (4)-(6) replace the state-by-round FE with different forms of round FE. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A8: Sensitivity of estimated impacts of 2012 community flood incidence to alternative samples

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Main	All EAs	W/in 50 km of any 2012 flood report	Main, non-0 fluvial flood hazard	Main, drop 20km donut	Main, drop LGA donut	Main, only ag HHs
Any food insecurity in last 12 months	0.05** (0.02)	0.05** (0.02)	0.05** (0.02)	0.09*** (0.03)	0.03 (0.02)	0.05* (0.02)	0.05* (0.02)
Daily HH consumption per capita (USD)	-0.68*** (0.21)	-0.64*** (0.21)	-0.64*** (0.21)	-0.80* (0.41)	-0.62*** (0.23)	-0.53** (0.21)	-0.56*** (0.20)
Total value of HH farm production (USD 100s)	-4.85*** (1.80)	-4.74*** (1.78)	-4.88*** (1.78)	-7.72*** (2.61)	-5.48*** (2.00)	-5.26*** (1.83)	-5.74*** (1.98)
Total non-farm HH enterprise income (USD 100s)	0.55 (3.28)	1.07 (3.22)	1.02 (3.21)	-5.52 (5.43)	0.52 (3.46)	-1.18 (3.38)	4.52 (3.30)
Total wage employment income (USD 100s)	0.19 (6.56)	0.73 (6.56)	0.75 (6.57)	-0.79 (13.57)	0.22 (6.48)	0.10 (7.31)	1.38 (5.37)
Any HH farm activity	0.03* (0.01)	0.02 (0.02)	0.02 (0.02)	0.05** (0.02)	0.02 (0.02)	0.02 (0.01)	0.02* (0.01)
Any HH non-farm enterprise activity	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.03 (0.03)	0.00 (0.02)	0.01 (0.02)	0.01 (0.02)
Any wage employment activity	0.00 (0.02)	0.00 (0.02)	0.00 (0.02)	-0.01 (0.03)	0.03 (0.02)	0.00 (0.02)	0.01 (0.02)
Count of HH members wkg. in HH farm	0.08 (0.10)	0.06 (0.10)	0.05 (0.10)	-0.08 (0.17)	0.06 (0.12)	0.07 (0.10)	0.07 (0.11)
Count of HH members wkg. in HH non-farm enterprise	0.00 (0.06)	-0.01 (0.06)	-0.01 (0.06)	0.07 (0.10)	-0.02 (0.07)	-0.01 (0.06)	0.01 (0.07)
Count of HH members wkg. in wage employment	0.09 (0.06)	0.09 (0.06)	0.09 (0.06)	0.07 (0.09)	0.15** (0.06)	0.11* (0.06)	0.12* (0.07)
Total area planted (ha)	-0.16 (0.19)	-0.18 (0.19)	-0.22 (0.19)	0.20 (0.25)	-0.24 (0.22)	-0.16 (0.19)	-0.12 (0.21)
Total family crop labor hours in past year	143.62 (133.94)	114.48 (132.67)	107.86 (133.28)	-12.21 (167.05)	166.77 (143.02)	105.54 (147.49)	128.22 (146.55)
Observations	11951	12117	11947	4530	9704	11131	10182

Note: This table repeats [Table A8](#) but varies the sample instead of the specifications. Each column indicates how the sample is changed relative to the main specification, presented in column (1), which restricts the sample to communities within the common support of average 100-year flood depth within 5 km after stratifying by the number of years with any survey flood report from 2007-2011. In column (2) the community (EA) sample restrictions are removed. Column (3) replaces the main sample restriction with dropping any community more than 50 km from any community where floods were reported in 2012. Column (4) adds a sample restriction keeping only communities with non-0 average fluvial flood risk within 5 km according to the JRC 2024. Columns (5) and (6) add sample restrictions dropping communities which may be most affected by flood exposure spillovers. Column (5) drops communities less than 20 km from any community where a flood was reported in 2012. Column (6) drops non-flooded communities in the same LGA as another sample community where a flood was reported in 2012. Column (7) adds a sample restriction keeping only households engaged in agricultural activity at some point during the four survey rounds. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A9: Sensitivity to alternative ways of controlling for non-random component of 2012 flood incidence

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Recentering	No shock recentering					
	Main (SE)	Prediction-round controls (SE)	Risk/rain-round controls (SE)	No added controls (SE)	No baseline controls (SE)	No baseline controls (SE)	No baseline controls (SE)
Any food insecurity in last 12 months	0.05** (0.02)	0.04* (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.03 (0.02)	0.04* (0.02)
Daily HH consumption per capita (USD)	-0.68*** (0.21)	-0.74*** (0.22)	-0.86*** (0.22)	-0.82*** (0.22)	-0.82*** (0.22)	-0.71*** (0.22)	-0.74*** (0.22)
Total value of HH farm production (USD 100s)	-4.85*** (1.80)	-4.95*** (1.82)	-4.69*** (1.74)	-4.66*** (1.75)	-4.69*** (1.77)	-4.72*** (1.60)	-4.82*** (1.76)
Total non-farm HH enterprise income (USD 100s)	0.55 (3.28)	0.78 (3.21)	1.34 (2.94)	1.31 (2.84)	1.56 (2.84)	0.70 (2.79)	1.08 (3.12)
Total wage employment income (USD 100s)	0.19 (6.56)	1.51 (6.31)	1.94 (5.74)	4.48 (5.47)	4.37 (5.62)	4.20 (5.01)	1.88 (6.36)
Any HH farm activity	0.03* (0.01)	0.04** (0.01)	0.06*** (0.02)	0.06*** (0.02)	0.05*** (0.01)	0.04*** (0.01)	0.05*** (0.01)
Any HH non-farm enterprise activity	0.02 (0.02)	0.02 (0.02)	0.01 (0.02)	0.01 (0.02)	0.01 (0.02)	0.01 (0.02)	0.02 (0.02)
Any wage employment activity	0.00 (0.02)	0.01 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.01 (0.02)
Count of HH members wkg. in HH farm	0.08 (0.10)	0.12 (0.10)	0.21** (0.09)	0.21** (0.09)	0.19** (0.09)	0.11 (0.09)	0.12 (0.10)
Count of HH members wkg. in HH non-farm enterprise	0.00 (0.06)	0.00 (0.06)	0.01 (0.05)	0.01 (0.05)	0.00 (0.05)	-0.01 (0.05)	-0.01 (0.06)
Count of HH members wkg. in wage employment	0.09 (0.06)	0.11* (0.06)	0.13** (0.06)	0.14** (0.06)	0.14** (0.06)	0.13** (0.05)	0.12** (0.06)
Total area planted (ha)	-0.16 (0.19)	-0.19 (0.19)	-0.27 (0.18)	-0.24 (0.18)	-0.27 (0.19)	-0.47*** (0.18)	-0.15 (0.20)
Total family crop labor hours in past year	143.62 (133.94)	184.83 (134.88)	234.84* (124.50)	268.28** (133.02)	292.98** (138.10)	499.72*** (134.89)	129.97 (138.16)
Sample	Main	Main	Main	Main	Main	All EAs	Propensity common support

Note: This table repeats [Table A8](#) but varies how we control for the non-random component of 2012 flood incidence. Each column indicates how the specification is changed relative to the main specification, presented in column (1), where the 2012 survey flood exposure measure is recentered around its predicted value. Columns (2)-(7) show effects of the binary 2012 survey flood measure with no recentering and varying controls and sample restrictions. Columns (2)-(5) keep the main sample restrictions: communities within the common support of average flood risk stratified by the number of years of survey-reported flooding from 2007-2011, and within the common support of the predicted flood exposure measure by flood incidence. Column (2) adds predicted 2012 survey flood exposure-by-round fixed effects to the main baseline characteristic-by-round controls. Column (3) adds average 100-year flood depth within 5 km-by-round and community 2012 rainfall deviation-by-round fixed effects. Column (4) includes only the main baseline characteristic-by-round controls. Column (5)-(7) drops those controls, and vary the sample of communities. Column (6) includes all communities (EAs). Column (7) drops the balancing on flood risk but restricts the sample to the common support of predicted flood exposure.

Table A10: Separating effects of flood exposure and estimated flood risk

Impact of:	Flood Shock (1)	Flood Propensity (2)
Any food insecurity in last 12 months	0.05** (0.02)	-0.00 (0.04)
Daily HH consumption per capita (USD)	-0.68*** (0.21)	-0.52 (0.83)
Total value of HH farm production (USD 100s)	-4.85*** (1.80)	2.43 (5.25)
Total non-farm HH enterprise income (USD 100s)	0.55 (3.28)	-7.81 (11.12)
Total wage employment income (USD 100s)	0.19 (6.56)	3.60 (23.86)
Any HH farm activity	0.03* (0.01)	0.11*** (0.04)
Any HH non-farm enterprise activity	0.02 (0.02)	0.00 (0.05)
Any wage employment activity	0.00 (0.02)	0.10 (0.06)
Count of HH members wkg. in HH farm	0.08 (0.10)	0.50** (0.22)
Count of HH members wkg. in HH non-farm enterprise	0.00 (0.06)	-0.05 (0.17)
Count of HH members wkg. in wage employment	0.09 (0.06)	0.11 (0.17)
Any member left HH during/since last round	0.01 (0.02)	-0.04 (0.06)
Total area planted (ha)	-0.16 (0.19)	-0.93* (0.57)
Total family crop labor hours in past year	143.62 (133.94)	232.55 (397.37)
Sample	Main	Non-flooded EAs

Note: This table compares estimated effects of 2012 flood exposure following the main specification (column 1) to effects of 2012 flood propensity among non-flooded communities (column 2). The purpose of the comparison is to determine to what extent flood propensity alone may be associated with different trends over time after the major 2012 floods. In addition to dropping flooded communities from the sample, the specification in column 2 follows the main specification but replaces the recentered 2012 flood shock $\times$ post term with a 2012 estimated flood propensity $\times$ post term. Since the recentering of the flood shock in column 1 directly accounts for this flood propensity, the estimated effects control for any differential trends post-2012 by flood propensity, isolating impacts of actual flood exposure. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A11: Impacts of 2012 community flood incidence on crop yields and sales prices

	N	Control Mean (SD)	2012 flood recentered × Post (SE)	2012 flood recentered × 2013 round (SE)	2012 flood recentered × 2016 round (SE)	2012 flood recentered × 2019 round (SE)
Yield by area planted of cereals (kgs/ha) (household)	4478	2235.54 [4338.81]	-962.19** (426.07)	-1046.75* (553.73)	-907.38** (424.99)	-743.92 (605.83)
Yield by area planted of maize (kgs/ha) (household)	3142	1617.43 [3781.63]	-765.74* (416.98)	-973.99** (463.39)	-542.79 (503.37)	-1025.30* (590.59)
Yield by area planted of sorghum (kgs/ha) (household)	3110	1587.31 [2692.04]	3.70 (322.59)	-20.17 (522.30)	-80.65 (392.37)	747.45 (516.72)
Yield by area planted of millet (kgs/ha) (household)	1956	1268.53 [1638.98]	-196.89 (337.91)	-27.09 (425.85)	-406.96 (416.24)	808.75* (434.26)
Yield by area planted of cowpea (kgs/ha) (household)	2162	592.10 [1244.33]	256.61 (206.25)	91.62 (202.58)	450.91 (309.85)	-61.70 (467.79)
Yield by area planted of yam (kgs/ha) (household)	2207	8570.67 [18294.26]	-2999.88 (2435.71)	-4014.95 (3459.40)	-2311.77 (3303.06)	45.63 (3005.38)
Yield by area planted of cassava (kgs/ha) (household)	3222	1575.39 [4144.40]	-1725.52** (709.52)	-146.08 (719.66)	-3402.77*** (1178.67)	-1306.59 (1180.29)
Average cereal price (USD/kg)	4464	0.55 [0.92]	-0.27*** (0.08)	-0.27* (0.15)	-0.31*** (0.11)	-0.11 (0.15)
Average maize price (USD/kg)	2619	0.26 [2.03]	-0.40** (0.19)	-0.52* (0.27)	-0.37** (0.18)	-0.04 (0.21)
Average sorghum price (USD/kg)	3038	0.95 [0.70]	-0.11* (0.07)	-0.05 (0.04)	-0.20 (0.13)	-0.04 (0.18)
Average millet price (USD/kg)	1921	0.12 [0.48]	-0.23** (0.11)	-0.08 (0.05)	-0.39* (0.21)	-0.15 (0.11)
Average cowpea price (USD/kg)	2042	1.82 [2.06]	-0.41 (0.32)	-0.46 (0.36)	-0.33 (0.28)	-1.06* (0.60)
Average yam price (USD/kg)	2068	0.69 [0.48]	-0.01 (0.03)	-0.01 (0.05)	0.02 (0.06)	-0.22 (0.15)
Average cassava price (USD/kg)	958	0.68 [1.12]	-0.27 (0.69)	-0.58 (0.97)	-0.09 (0.52)	0.12 (0.44)

Note: This table presents effects of 2012 community flood exposure on household outcomes in subsequent survey rounds, following Table 2. Each row represents an outcome. We report results for the 6 crops with the highest average area planted in the sample. The results for cereals aggregate information for maize, sorghum, millet, and rice, weighted by the area planted to each crop for households growing multiple cereals. Yields are only defined for households with non-0 area planted for a particular crop. Prices are the average sales price reported by the household, or the median sales price in the community if they did not sell any output. Monetary values are in 2016 PPP USD. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A12: Impacts of 2012 community flood incidence on household crop production value among cereal producers

	(1)	(2)	(3)
Exposed to 2012 flood shock (recentered)	-8.13*** (2.78)	-8.23*** (2.78)	-5.66** (2.82)
Average cereal price (USD/kg)		0.08 (0.50)	
Yield by area planted of cereals (kgs/ha)			0.00*** (0.00)
Observations	4478	4464	4478
Baseline mean, non-flooded	31.13	31.21	31.13

Note: This table presents average effects of 2012 community flood exposure on total crop production value in 2016 USD PPP in subsequent survey rounds, following Table 7. The sample includes all households that cultivated cereal crops. The first column is the main specification for the impacts of flood exposure restricted to this sample. The next two columns introduce controls for household average cereal price and average cereal yield. Cereal price and yield are defined in Table 7/ * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A13: Impacts of 2012 community flood incidence on market food prices and agricultural labor wages

	N	Control Mean (SD)	2012 flood recentered × Post (SE)	2012 flood recentered × 2013 round (SE)	2012 flood recentered × 2016 round (SE)	2012 flood recentered × 2019 round (SE)
HH avg. PP daily wage earned by wage wkrs (USD)	1819	28.48 [36.25]	-6.57* (3.55)	-6.71 (4.84)	-5.57 (4.14)	-13.66 (10.92)
HH avg. PH daily wage earned by wage wkrs (USD)	1908	31.44 [42.62]	4.05 (5.19)	2.56 (5.37)	5.70 (5.99)	2.69 (5.75)
HH avg. daily wage paid to hired ag labor (USD)	4604	11.74 [33.51]	1.51 (1.53)	1.37 (1.76)	1.73 (1.46)	0.90 (1.67)
Comm. avg. daily wage for men's ag labor (USD)	9717	17.05 [11.16]	-0.34 (1.36)	-1.10 (1.74)	0.32 (1.29)	0.96 (2.18)
Comm. price shelled maize (USD/kg)	8355	0.01 [1.07]	-0.19 (0.18)	-0.00 (0.24)	-0.39** (0.18)	0.00 (.)
Comm. price local rice (USD/kg)	8737	0.01 [1.12]	-0.10 (0.15)	0.05 (0.20)	-0.25* (0.15)	0.00 (.)
Comm. price bread (USD/kg)	10143	0.00 [1.01]	-0.32** (0.15)	-0.17 (0.16)	-0.49*** (0.17)	0.00 (.)
Comm. price yam roots (USD/kg)	9128	-0.02 [0.94]	-0.11 (0.15)	-0.10 (0.18)	-0.12 (0.15)	0.00 (.)
Comm. price palm oil (USD/litre)	10129	0.01 [0.98]	0.06 (0.08)	-0.02 (0.11)	0.15 (0.10)	0.00 (.)
Comm. price groundnut oil (USD/litre)	9601	0.01 [0.97]	0.05 (0.09)	0.01 (0.09)	0.11 (0.12)	0.00 (.)
Comm. price banana (USD/kg)	7613	0.03 [1.02]	0.02 (0.16)	-0.05 (0.17)	0.10 (0.21)	0.00 (.)
Comm. price chicken (USD/kg)	8590	-0.06 [0.97]	0.05 (0.11)	-0.02 (0.10)	0.14 (0.16)	0.00 (.)
Comm. price sugar (USD/kg)	8323	-0.02 [0.94]	0.24 (0.18)	0.12 (0.21)	0.33* (0.18)	0.00 (.)

Note: This table presents effects of 2012 community flood exposure on household outcomes in subsequent survey rounds, following Table 2. We do not present dynamic analyses of effects by survey round as the collection of community price data in 2018-19 followed a different format than in the previous rounds so we are constrained to the first three survey rounds. Each row represents an outcome. Data for all outcomes preceded by "Comm." are from the community survey. Market prices for commonly-reported food goods are from the post-harvest survey for the nearest market to the community. The units for goods are different in different rounds so we convert all prices to standard deviations relative to the non-flooded community mean in each survey round after converting values to USD and winsorizing. The average daily wage for men's agricultural labor is the mean wage across different types of reported activities from the post-planting survey. The household average daily wage is the average wage paid for hired labor across activities for households that hired any farm labor, weighted by amount of days provided. Monetary values are in 2016 PPP USD. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A14: Heterogeneity in impacts of 2012 community flood incidence by baseline household and community characteristics

	Consumption		HH agriculture		Wage work		HH NFE	
	(1) Daily HH cons./cap.	(2) Any food insec.	(3) Value prod.	(4) Count mems.	(5) Total inc.	(6) Count mems.	(7) Total inc.	(8) Count mems.
Flood $\times$ Post $\times$ > 5 baseline HH members	1.14*** (0.37)	0.08** (0.04)	-5.84* (3.11)	-0.07 (0.18)	-10.83 (12.07)	0.12 (0.10)	5.29 (6.75)	0.03 (0.11)
Flood $\times$ Post $\times$ Above med. baseline HH assets value	0.09 (0.38)	0.00 (0.03)	-4.18 (3.33)	0.14 (0.18)	12.23 (13.01)	0.16 (0.11)	11.31* (6.38)	0.05 (0.11)
Flood $\times$ Post $\times$ Below med. dist. to major market	0.01 (0.46)	-0.12*** (0.04)	6.68* (3.43)	-0.11 (0.21)	2.74 (11.73)	0.18 (0.12)	-3.17 (5.99)	-0.08 (0.12)
Flood $\times$ Post $\times$ Below med. dist. to major road	-0.30 (0.55)	-0.03 (0.05)	1.80 (3.10)	-0.50** (0.19)	22.64* (13.29)	-0.13 (0.14)	0.91 (5.90)	-0.10 (0.11)
Flood $\times$ Post $\times$ Below med. dist. to >20k pop. town	-0.18 (0.49)	-0.04 (0.05)	0.17 (3.30)	-0.26 (0.18)	8.27 (12.69)	-0.11 (0.12)	-6.04 (6.36)	-0.15 (0.12)
Flood $\times$ Post $\times$ Below med. dist. to river/water	-0.58 (0.44)	0.02 (0.05)	2.29 (3.98)	0.02 (0.21)	14.08 (12.90)	-0.43*** (0.13)	-5.99 (6.97)	0.10 (0.11)
Flood $\times$ Post $\times$ Non-zero fluvial flood risk	-0.24 (0.46)	0.07* (0.04)	-4.76 (3.25)	-0.25 (0.20)	-1.75 (13.67)	-0.08 (0.12)	-6.03 (6.06)	0.10 (0.12)
Flood $\times$ Post $\times$ Comm. flooded in last 5 yrs	-0.15 (0.43)	0.06 (0.05)	-15.05*** (3.92)	0.35 (0.27)	-10.93 (14.03)	0.11 (0.16)	5.08 (8.20)	-0.11 (0.14)

Note: This table presents tests of heterogeneity in average effects of 2012 community flood exposure on household outcomes in subsequent survey rounds by baseline household and community characteristics. Each column represents an outcome. Column 1 shows daily consumption per capita and column 2 shows whether any food insecurity was reported over the previous 12 months. For each category of livelihood activities (HH agriculture, HH NFE, and wage work), we show effects on income or production value from that activity and count of household members engaged. The results for each column in each row are from separate triple-differences regressions fully interacting the indicated baseline household or community characteristic with the Post and Flood dummies, state-by-round fixed effects, and all controls from the main specification. We show only the estimated coefficient for the interaction term for simplicity, reflecting the estimated difference in effects for households/communities with a given characteristic. Monetary values are in 2016 PPP USD. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A15: Impacts of 2012 community flood incidence by baseline household engagement in NFE

	(1) Daily HH cons./cap.	(2) Any food insec.	(3) Value prod.	(4) Count mems.	(5) Total inc.	(6) Count mems.	(7) Total inc.	(8) Count mems.
Flood $\times$ Post, H=0	-1.45*** (0.49)	0.06 (0.04)	-1.69 (2.93)	-0.10 (0.15)	9.14 (8.23)	0.04 (0.09)	-1.32 (1.25)	0.11 (0.08)
Flood $\times$ Post, H=1	-0.51** (0.21)	0.04* (0.02)	-5.91*** (1.78)	0.12 (0.10)	0.66 (7.39)	0.11 (0.07)	3.37 (3.99)	-0.01 (0.06)
p , equality of effects	0.07	0.66	0.15	0.15	0.43	0.60	0.24	0.17

Note: This table presents average effects of 2012 community flood exposure on household outcomes in subsequent survey rounds, by baseline household engagement in NFE (76% of HHs). Each column represents an outcome. Column 1 shows daily consumption per capita and column 2 shows whether any food insecurity was reported over the previous 12 months. For each category of livelihood activities (HH agriculture, HH NFE, and wage work), we show effects on income or production value from that activity and count of household members engaged. The results for each column are from a single triple-differences regression fully interacting the baseline household characteristic H with the Post and Flood dummies, state-by-round fixed effects, and all controls from the main specification. We calculate the average effect in the group with $H = 1$ by taking the sum of the $Flood \times Post$ and $Flood \times Post \times H$ coefficients and estimate SEs using the *xtlincom* function in Stata. We also report the p -value for the triple interaction term, which tests for equality of effects between the two groups. Monetary values are in 2016 PPP USD. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

C Survey Data

The main data source for the analysis is Nigeria’s General Household Survey Panel (GHSP). The GHSP, part of the World Bank’s Living Standards Measurement Study - Integrated Surveys on Agriculture (LSMS-ISA), is a nationally-representative panel survey including 5,000 households conducted by the Nigeria National Bureau of Statistics. Before the first survey round, 500 enumeration areas (communities) were randomly sampled after stratifying by state. Ten households were then randomly sampled in each enumeration area from a census of households.

Four survey rounds were conducted between 2010 and 2019. Households are tracked over time, including if they move to a new location, but individual household members are not tracked. In 2018-19 the panel sample was partially refreshed, with 1,425 households from the original panel retained and 3,551 new panel households added to the sample. The GHSP data are publicly available from the World Bank’s Microdata Catalog.

The main analysis sample includes 15,356 observations across the four survey rounds from 4,750 unique households. All households are observed in both the 2010-11 and 2012-13 rounds, 4,497 households are observed in the 2015-16 round, and 1,430 households are observed in the 2018-19 round where we do not keep newly-sampled households in the partially-refreshed panel.

Each survey round includes both a post-planting and a post-harvest survey. Post-planting surveys took place in September-October 2010, September-October 2012, August-September 2015, and July-August 2018. Post-harvest surveys took place in February-April of 2011, 2013, and 2016 and in January-February 2019. Some information, such as individual labor supply, is recorded in each survey while other modules are only included in either the post-planting or post-harvest surveys.

Basic cleaning decisions include replacing missing values with 0 where appropriate (i.e., for income in an activity the household was not engaged in) and replacing impossible values (such as more than 24 hours per day) with missing values. We winsorize continuous variables by replacing values above the 99th percentile with the value at the 99th percentile. All variables representing monetary values are converted to 2016 PPP USD.

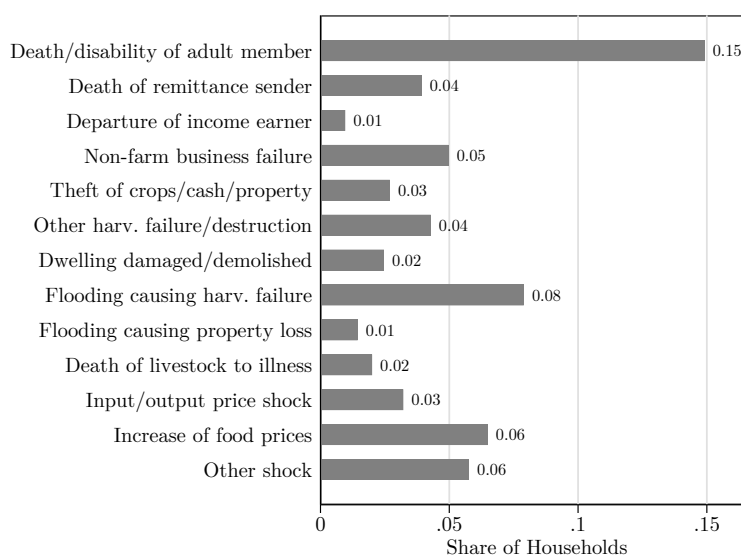
For certain variables—notably the measures of total household income by activity—we modify code developed by the Evans School Policy Analysis & Research Group (EPAR) to construct variables in consistent ways accounting for differences in the survey instruments across rounds. The code is available on GitHub: <https://github.com/EvansSchoolPolicyAnalysisAndResearch/LSMS-Agricultural-Indicators-Code>.

Differences across survey flood measures

Several survey questions ask about flood exposure, each of which captures specific ways in which floods can affect households. In the post-planting agricultural survey, households can list flooding as the main cause of loss of stored crops since the beginning of the new year for each cultivated crop. Such losses are reported by 13% of households for 2012 but flood-related losses are only

reported by 1% In the post-harvest agricultural survey, households can list flooding as a reason for not harvesting a particular crop. Failure to harvest is reported by 24% of households, but again floods are only listed as a reason by 3%. In both the post-planting and post-harvest household surveys, households can list flooding as a cause of household food insecurity in the past 12 months. Food insecurity is reported by 24% of households at post-planting and by 20% at post-harvest, and in both periods 3% list floods as a cause of food insecurity. Finally, in the post-harvest household survey shocks module, households are asked about various shocks they have been affected by over the past 5 years, the years in which they occurred, and the consequences of the shocks. Two shocks relate to floods: flooding that caused harvest failure is reported by 7% of households in 2012 and loss of property due to flood is reported by 1%. Floods are the second most commonly reported shock for 2012 (Figure A6).

Figure A6: Prevalence of shocks reported by households over the 12 months before the 2013 post-harvest survey round

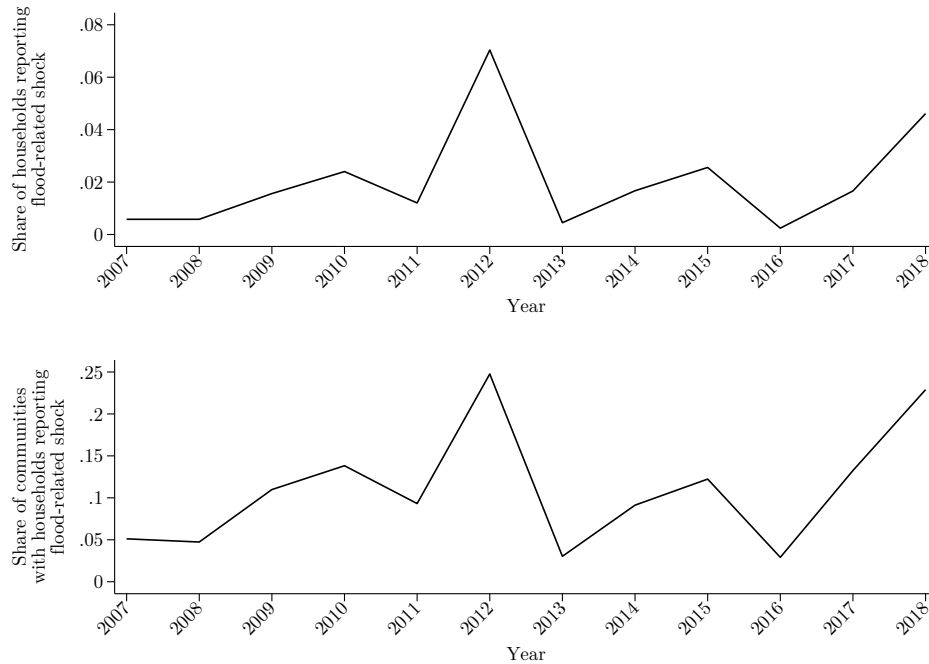


Note: Data are from the 2013 post-harvest household survey shocks module.

We can use the fact that household shocks are reported for multiple years to show how the prevalence of flood shocks was much greater in 2012 than in any other year from 2007-2018 (Figure A7). The fact that the share of households reporting flood shocks is much lower than the share of communities where such shocks are reported highlights how not every household is affected in communities where flooding occurs. This result is supported by the community questionnaire, which asks informants to report important events that made people worse off in the community, including floods, and the year in which they occurred. Floods are reported in 20% of communities in 2012, and on average these are reported to have affected 48% of community households (the median is 50%).

Across all flood questions, 12.4% of households report being affected by floods in some way

Figure A7: Survey flood shock reports by year

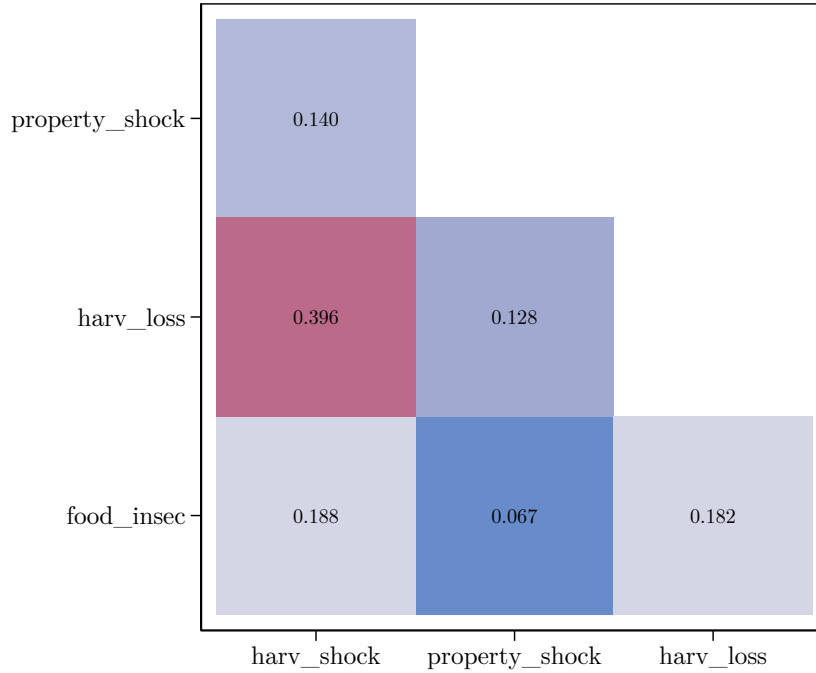


Note: Annual flood reports are from the Nigeria GHSP. In each survey round, households are asked to recall whether they experienced specific economic shocks over the last few years, including flooding that caused harvest failure and loss of property due to flood. These data were collected in the post-harvest survey waves in 2009, 2013, 2016, and 2019. This is a more restrictive measure of flood exposure than the main survey-based measure we use in this paper which incorporates reports of floods in the survey's food security and crop production modules, but there is no recall element for those questions. The top panel shows the share of households reporting any flood shock by year. The bottom panel shows the share of communities with any household flood shock report by year.

in 2012. [Figure A8](#) shows a heatmap of correlations between these measures. We aggregate the agricultural survey questions as 'harv_loss' and the food insecurity questions as 'food_insec', in both cases coded as 1 if the household reports a flood in either the post-planting or post-harvest wave. We compare these against the two flood questions from the household shock module. While the two harvest-related measures are highly-correlated, correlations between the other measures are weaker. This emphasizes that each measure is capturing different ways in which floods may affect households, and that they primarily affect different households rather than the same household being affected in multiple ways. The figure also indicates that household flooding reported in the shocks module—a common way survey-based flood measures are constructed—may miss some floods that did not cause the specific types of losses or damages that are asked about.

How do these household flood reports compare to what is reported in the GHSP community surveys? Twenty percent of households reside in communities where flooding was reported in 2012 in the community survey. The survey and community measures are correlated: in communities with a community survey flood report, 25% of households report a flood, compared to 7% in communities with no survey flood report. But this mismatch leads to 41% of households that report being affected by floods in 2012 residing in communities with no community survey flood

Figure A8: Correlations between survey measures of household-level 2012 flood incidence



Note: This heatmap shows pairwise correlations between different measures of flood incidence in 2012 at the level of households in the Nigeria GHSP. 'harv_shock' indicates a report of a flood that caused harvest failure in the household shocks module. 'property_shock' indicates a report of a flood that caused property loss in the household shocks module. 'harv_loss' indicates a report of loss of stored crops or failure to harvest in the post-planting and post-harvest household crop production modules. 'food_insec' indicates a report of floods as a cause of household food insecurity in either the post-planting or post-harvest food security modules.

report. Households reporting floods not captured in the community survey may be driven by lack of complete information error by the group of community survey informants, or by error in the household surveys. But even in communities with at least two (of ten sample households) separate sample households reports of flood in the shocks module, some community surveys do not report flooding. This suggests important measurement errors in the community survey and motivates the reliance on household reports to define survey-based community flood exposure for the main analyses rather than community reports. On the other hand, 25% of households are in communities with a community flood report but no survey flood report. A potential reason for this is floods affecting community households other than those included in the GHSP sample, or affecting households in ways not captured in the household survey questions. We therefore test sensitivity of the results to including community reports in the survey flood definition.

D Flood detection with MODIS satellite imagery

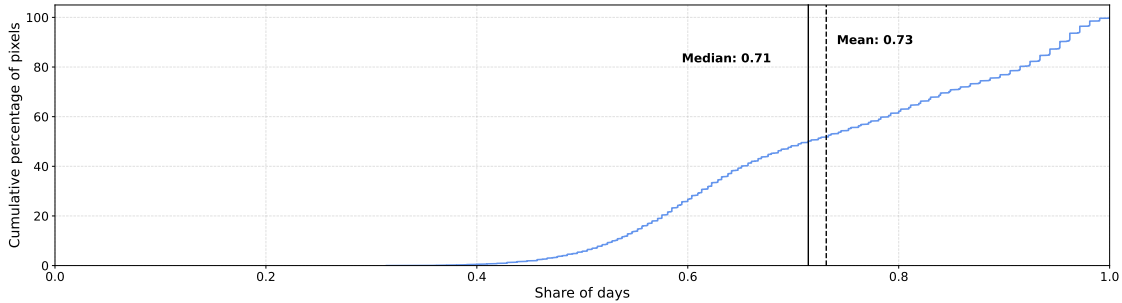
This appendix illustrates reasons why remote detection of flooding using MODIS satellite imagery, as used in this paper via the NASA NRT Global Flood Mapping Product, may feel to detect floods reported in surveys. First, optical satellite imagery is constrained by cloud cover which can limit the ability to observe changes in surface water. Second, MODIS has relatively coarse spatial resolution, limiting its ability to detect narrow or localized water features. Third, the satellite-based community indicator is constructed using a binary intersection rule: a community is classified as flooded whenever the any 2012 flooding is including in the NASA NRT Global Flood Mapping Product within a 5 km buffer. Floods detected within this buffer may not affect survey households for a variety of reasons.

D.1 Cloud cover and observability

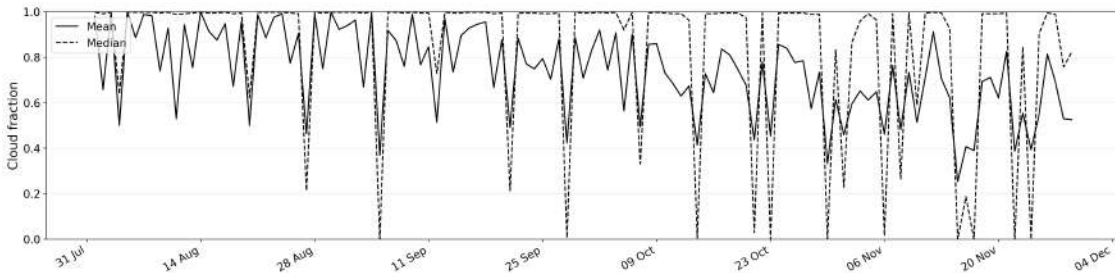
Optical satellite imagery cannot detect floods when the Earth’s surface is covered by clouds. [Figure A9](#) documents cloud conditions during the 2012 flood season in Nigeria, between August and November. Panel A shows that the median MODIS pixel (250 m) in Nigeria was cloudy on roughly 71% of days, and the mean was approximately 73%. This indicates a broad national pattern of frequent cloud cover during the months in which flooding was most likely to occur.

Figure A9: Cloud cover in MODIS satellite imagery in Nigeria, August-November 2012

(a) Distribution of share of cloudy days across all Nigeria pixels



(b) Daily mean and median cloud fraction within 5 km buffers around GHSP communities



Note: Data are from the MOD09GA collection on Google Earth Engine.

Panel B focuses on the empirical setting of the paper by plotting the daily mean and median cloud fraction within 5 km buffers around GHSP communities. Cloud cover remains high for much of the flood season, with means around 80-90% from August-September and 50-80% from October-

November. The median is often close to one, implying that on most dates at least half of the surveyed communities were entirely cloud-covered. Although the cloud fraction occasionally drops sharply, these clearer observation windows are brief and irregular.

Taken together, these patterns indicate that cloud cover during the period when most floods occurred in 2012 is a major constraint to remote sensing of flooding using MODIS imagery. The constraint implies that the absence of flooding in the NASA database cannot be interpreted as evidence that no flooding occurred on the ground, and likely explains a large share of the failure of the database to identify flooding in communities with survey-reported floods.

D.2 Spatial resolution and water detection

Another limitation of MODIS imagery is its 250 m resolution. Detection of surface water is based on reflection of different bandwidths of light, formalized in the Normalized Difference Water Index (NDWI). Pixels of mixed water and other surfaces will have lower NDWI values than pixels with a large share of water cover, and the water cover is therefore less likely to be identified. We illustrate how the MODIS data resolution limits its ability to detect water in comparison with Landsat 7 satellite imagery, which has a 30 m resolution but is collected less frequently.

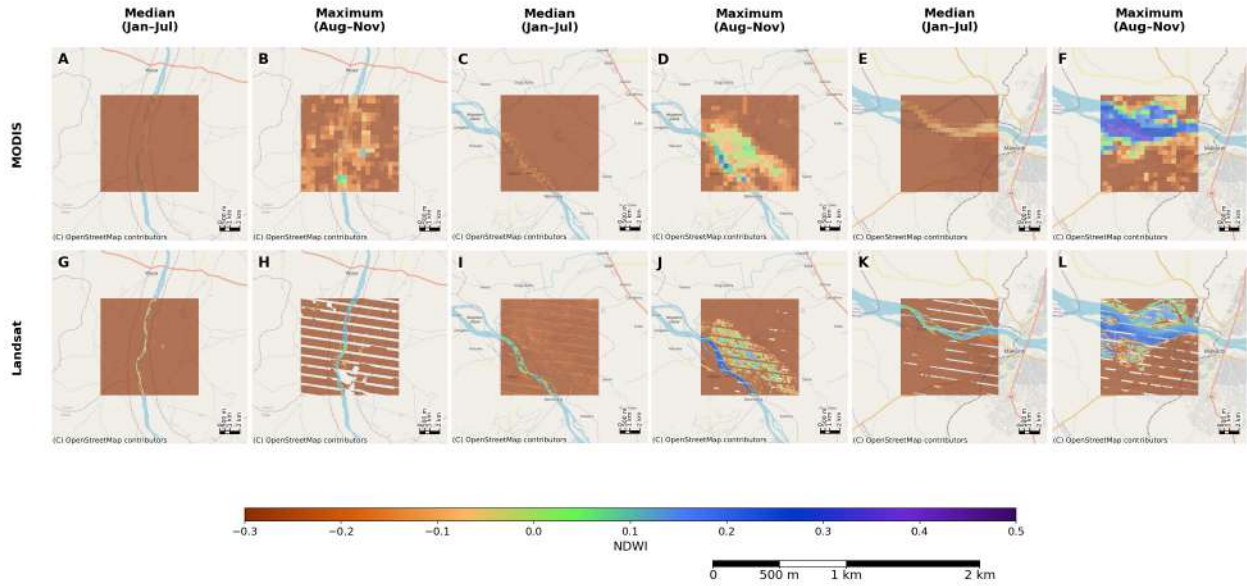
For both sources, we calculate the McFeeters NDWI (McFeeters, 1996), which is designed to enhance open surface water, rather than the Gao NDWI (Gao, 1996), which is more closely related to vegetation moisture. NDWI values above 0 or 0.1 are typically taken to likely represent surface water, with higher values indicating greater certainty. During the non-flood period (January–July 2012), we use median NDWI to represent typical surface-water conditions. During the flood period (August–November 2012), we use maximum NDWI to give the imagery the best possible chance to reveal changes in surface water.

Figure A10 shows three cases illustrating how the ability of 250 m resolution MODIS imagery to detect water depends on the spatial scale of the underlying feature, depicted using data from OpenStreetMap. In the first case, MODIS detects no meaningful variation in NDWI above the narrow river in the non-flood season (Panel A), and the signal remains unclear in the flood season (B). In the second case, with a slightly larger river, MODIS identifies a different NDWI signal during the non-flood period but it is not above typical thresholds for classifying water (C). It does identify some clear river water cells in the flood season, as well as some cells with potential flooding (D). In the third case, with a river over 500 m in width, MODIS imagery still does not clearly identify cells with water where the river lies during the non-flood season (E), though it does identify river cells and likely river overflows during the flood season (F).

In all cases, the 30 m resolution Landsat 7 imagery (bottom row) clearly identifies water cells above the rivers in the non-flood season, as well as expansions and overflows of these rivers during the flood season. In all cases, the higher resolution Landsat imagery provides sharper and more precise information on river channel structure and likely flood overflows.

The fact that MODIS imagery fails to detect even large rivers during certain periods clearly shows its limitations for detecting floods. Floods must spread over a large share of the 250 m

Figure A10: Progression in water detection across MODIS and Landsat imagery.



Note: Data are from the 250 m MODIS MOD09GA (top) and 30 m Landsat LE07/C02/T1.L2 (bottom) collections on Google Earth Engine. All data are from 2012. We present three cases with rivers of different widths to illustrate how the ability of MODIS to detect surface water depends on the spatial scale of the underlying feature. For each case, we visualize median NDWI during the January-July non-flood period and maximum NDWI during the August-November flood period for both MODIS and Landsat. NDWI values above 0 are more likely to indicate the presence of surface water.

pixel area and at a sufficient depth to be detectable using NDWI-based algorithms. Floods that spread unevenly over space – likely in uneven terrain – or that are relatively shallow are unlikely to be detected. MODIS imagery is therefore likely capture only a subset of more severe floods or floods in particular geographic conditions, which can further help explain the large number of GHSP communities with 2012 floods not identified in the NASA database.

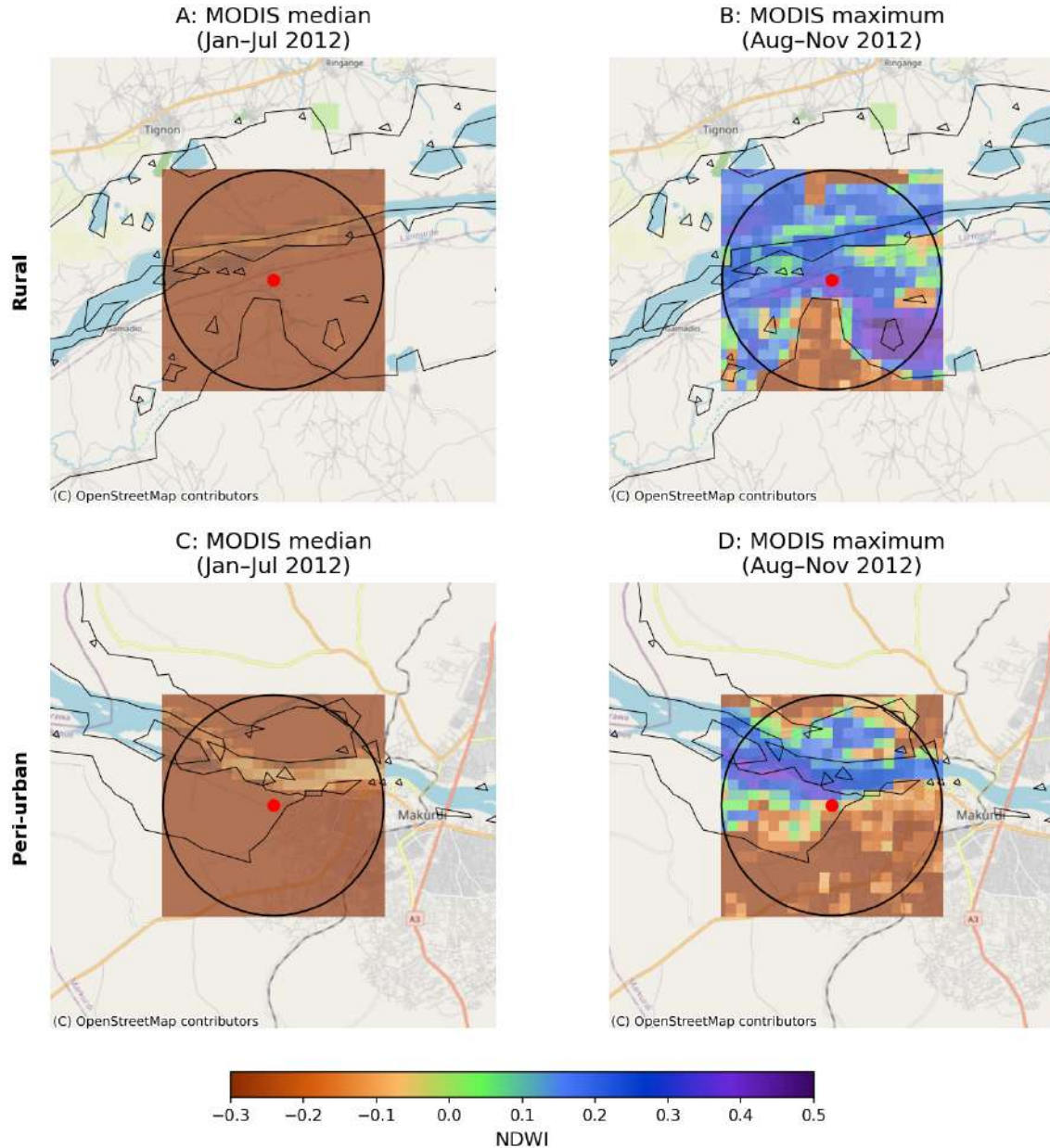
D.3 Illustrating survey vs. satellite flood detection

To illustrate how survey-based and satellite-based flood measures can agree or disagree at the community level, we present visual evidence from selected community examples. These examples are not intended to validate one source against the other mechanically. Rather, they show how the spatial distribution of water, uncertainty about the true location of the community within the buffer, and the construction of the MODIS-based indicator help interpret the relationship between the two measures.

Figure A11 presents two cases of agreement between the survey and satellite flood measures we use, considering rural and peri-urban communities near rivers. In both cases, we find much higher NDWI values in MODIS data during the flooding period than during the non-flooding period, with clear alignment with the MODIS-based flood detection boundaries. The figures appear to show clear cases of riverine flooding in large parts of the 5 km buffer around the communities.

Figure A12 presents examples of disagreement between the two flood measures. Panel a) shows two communities where flooding is detected within 5 km of the community centroid by the NASA

Figure A11: Examples of survey-satellite agreement in rural and urban settings



Note: The figure visualizes changes in NDWI and flood detection in a 5 km buffer around a rural GHSP community (top row) and an urban community (bottom). The jittered community centroids are indicated by red dots, and the 5 km buffers by black circles. The left panels show median MODIS NDWI in the non-flood period (January–July 2012), while the right panels show maximum MODIS NDWI in the flood period (August–November 2012). NDWI values above 0 are more likely to indicate the presence of surface water. Black polygons denote the MODIS-based NASA NRT Global Flood Mapping layer used for the satellite-based flood measure.

NRT GLObal Flood Mapping product, but is not reported in any household or community surveys.³¹ In the top row, changes in NDWI during the flooding season suggest a broader increase in local surface water in what appears to be a hydrologically complex setting characterized by marshes, so the satellite flood identification may reflect real water expansion that was not experienced as

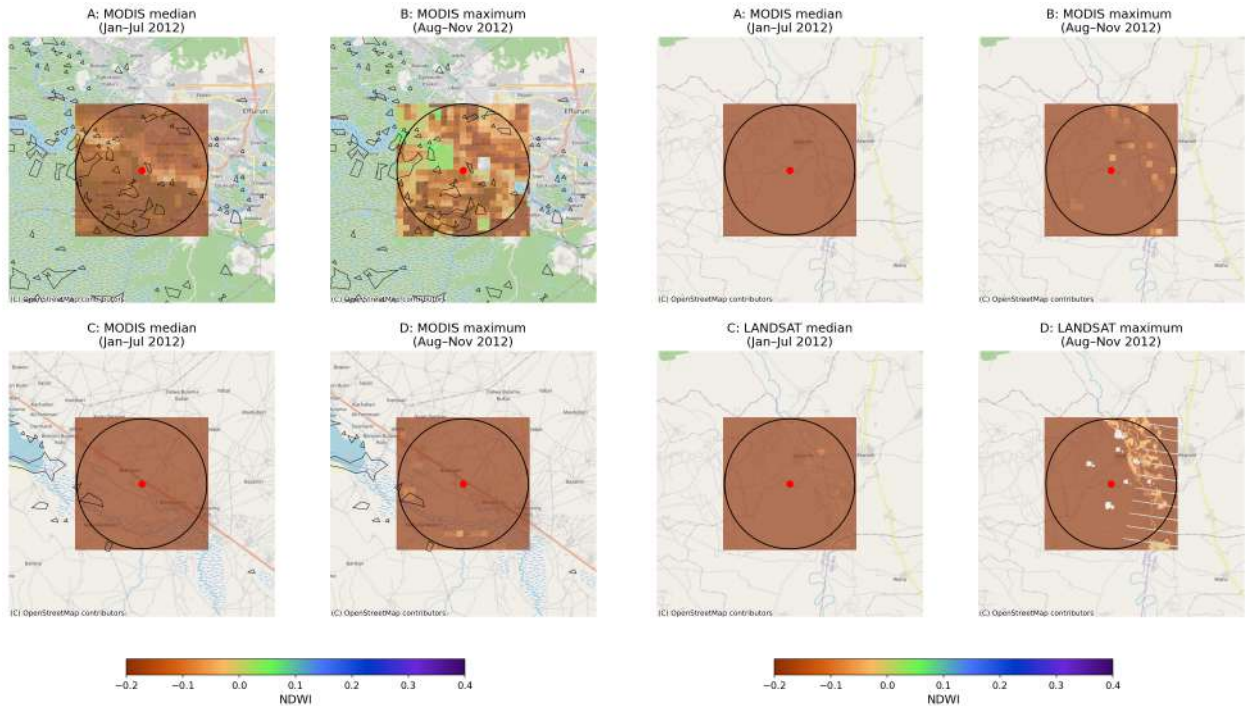
³¹In 17 of 73 communities with satellite-only flooding, there is no flood according to the main survey-based measure requiring at least 2 household flood reports, but there is at least one household or community survey flood report.

a flood. In the bottom row, visible change in NDWI within the buffer is much weaker, yet the community is still classified as flooded because the NASA flood layer intersects the 5 km buffer. The mismatch may represent either a limitation of the buffer based matching process for the NASA flood layer – required due to uncertainty about exact community locations – or a limitation or error in the NASA flood layer itself.

Figure A12: Examples of survey-satellite disagreement

(a) Satellite-only cases

(b) Survey-only case



Note: The figure visualizes changes in NDWI and flood detection in a 5 km buffer around a rural GHSP community (top row) and an urban community (bottom). The jittered community centroids are indicated by red dots, and the 5 km buffers by black circles. For each pair of images, the left panel shows median MODIS NDWI in the non-flood period (January–July 2012), while the right panel shows maximum MODIS NDWI in the flood period (August–November 2012). NDWI values above 0 are more likely to indicate the presence of surface water. Black polygons denote the MODIS-based NASA NRT Global Flood Mapping layer used for the satellite-based flood measure. Panel A presents two communities classified as flooded by the MODIS-based indicator but with no survey flood reports. Panel B presents a community with at least two survey flood reports no MODIS-based flooding detected within 5 km. For comparison, we also show changes in NDWI using higher-resolution Landsat which suggests a clearer pattern of riverbed expansion.

Panel b) illustrates the case of a community with multiple survey flood reports but no flood detection within 5 km in the NASA database. Changes in NDWI captured by MODIS suggest only weak differences during the flood season, but Landsat data suggests a broader pattern of localized riverbed expansion, some of which may have been experienced as flooding. This is the type of setting in which spatial resolution matters most: localized flooding may be meaningful for households while remaining only weakly visible in MODIS.